

Systematic benchmarking of evaluation paradigms, safety boundaries, and clinical reasoning gaps for multimodal large language models in rehabilitation

Ping Ye^{a,b,1}, Yu Li^{a,1}, Mengjian Qu^{a,c}, Xuan Zhang^a, Jing Liu^a, Tao Zhu^{b,*}, Jun Zhou^{a,*}

^aThe First Affiliated Hospital, Department of Rehabilitation, Hengyang Medical School, University of South China, Hengyang, Hunan, 421001, China

^bSchool of Computer Science, University of South China, Hengyang, Hunan, 421001, China

^cDepartment of Rehabilitation Sciences, The Hong Kong Polytechnic University, Kowloon, Hong Kong SAR, China

Abstract

Background: Motor rehabilitation and clinical biomechanical assessment rely on expert judgement to evaluate movement quality, pathological compensation, and spatiotemporal motor dynamics. Existing automated approaches mainly depend on conventional computer vision methods, such as skeletal keypoint-based analysis, but they provide limited clinical semantic interpretation and mechanistic reasoning. With the emergence of multimodal large language models (MLLMs), movement assessment may shift from geometric pose tracking toward end-to-end visual clinical reasoning. However, the ability of current MLLMs to identify fine-grained kinematic abnormalities and suppress visually unsupported hallucinations in rehabilitation videos remains insufficiently evaluated.

Objective: This study aimed to develop a multidimensional video benchmark, *ActionEval-MedBench*, to systematically evaluate the clinical reasoning capability, safety boundaries, and failure modes of MLLMs in motor rehabilitation and clinical biomechanics.

Methods: *ActionEval-MedBench* included 522 rigorously screened and de-identified videos, comprising 328 clinical upper-limb rehabilitation videos and 194 general functional biomechanics videos. Each video was paired with six multi-select multiple-choice questions, yielding 3,132 video–MCQ assessment items. We designed a six-dimensional ActionEval protocol covering semantic action recognition, kinematic feature alignment, pathological compensation identification, spatiotemporal dynamics analysis, comprehensive movement quality and clinical decision-making, and visual grounding for anti-hallucination assessment. Expert consensus labels were established through a dual-track Delphi-style procedure. Twenty-six leading MLLMs were evaluated under standardized zero-shot prompts and structured response-parsing rules. The primary outcome was the cohort-weighted ActionEval composite score, while the D6 false-positive hallucination rate (FP_{D6}) was used as an independent safety endpoint.

Results: Among 81,432 expected dimension-level response opportunities, 81,006 responses were successfully returned, whereas 426 were unavailable because of missing outputs or system-level failures. Among the returned responses, 80,931 were parsable and valid, while 75 were invalid or unparsable, yielding an overall valid-response rate of 99.38%. Gemini-3.1-Pro-Preview, Doubao-Seed-2.0-Lite-260428, Gemini-3.5-Flash, Claude-Opus-4-7, GLM-5V-Turbo, and GPT-5.5 formed a numerically high-performing cluster, with weighted ActionEval scores ranging from 66.17 to 65.46. The Friedman test showed significant overall performance differences among the 26 models ($\chi^2 = 2671.34$, $P < 0.001$), whereas Holm–Bonferroni-adjusted pairwise comparisons within the high-performing cluster were not statistically significant (all $P > 0.05$). Dimension-wise analysis revealed a marked semantic-to-clinical reasoning gap: models achieved strong performance in D1 semantic action recognition, with a mean F1-score of 91.1%, but declined substantially in higher-order clinical dimensions involving pathological compensation, spatiotemporal dynamics, and clinical decision-making, with mean F1-scores below 50% across D3–D5. Performance was lower in the clinical rehabilitation cohort than in the general functional biomechanics cohort, with an average drop of 9.44 percentage points; this difference was interpreted as a combined domain-shift burden between standardized functional videos and real-world fine-grained clinical rehabilitation videos rather than the isolated effect of pathology. In D6 negative-probe testing, nine models produced visually unsupported affirmative selections, with the highest FP_{D6} reaching 6.51%.

Conclusions: *ActionEval-MedBench* quantifies the cognitive boundaries and safety risks of current MLLMs in dynamic rehabilitation video understanding. Although MLLMs show promise for structured preliminary screening and clinical semantic reasoning, their limitations in detecting subtle pathological compensation and modeling spatiotemporal dynamics preclude their use as independent diagnostic tools. Future digital rehabilitation AI should follow a human-in-the-loop paradigm and incorporate domain-specific causal reasoning, kinematic-chain modeling, and rigorous visual-grounding safety probes.

Keywords: Multimodal Large Language Models, Motor Rehabilitation, Clinical Biomechanics, Video Understanding, Pathological Compensation, Visual Grounding, Clinical Reasoning, Benchmarking

1. Introduction

Accurate, continuous, and clinically interpretable assessment of motor function is fundamental to rehabilitation decision-making [1, 2]. For patients with movement impairments caused by neurological injury, orthopedic trauma, or sports-related injury, clinicians must evaluate not only whether a movement can be completed, but also how it is performed, including joint trajectories, postural control, compensatory strategies, spatiotemporal rhythm, and potential safety risks [3]. Standardized instruments such as the Fugl-Meyer Assessment provide important clinical references [4, 5], but their application remains dependent on in-person observation and professional experience, making frequent assessment resource-intensive and susceptible to inter-rater variability. Scalable and clinically sensitive automated assessment has therefore become increasingly important with the expansion of telerehabilitation and digital health [6].

Wearable sensors and conventional computer vision methods based on skeletal keypoint detection can quantify joint angles, movement velocity, and trajectory deviation [7, 8]. However, these measurements do not independently establish whether an abnormal pattern reflects weakness, pain-related protection, impaired motor control, or pathological compensation. Multimodal large language models (MLLMs), which integrate visual inputs with natural-language and medical knowledge, offer a potential route from motion perception to clinically meaningful interpretation [9, 10, 11]. Nevertheless, successful recognition of a macroscopic action, such as arm elevation or a push-up, does not demonstrate the ability to detect subtle shoulder hiking, trunk compensation, abnormal movement rhythm, or other clinically important deviations.

Existing multimodal video benchmarks predominantly evaluate general-domain action recognition, object perception, and event understanding [12, 13, 14, 15]. These tasks may not capture the patient-specific, task-dependent, and temporally dynamic features of rehabilitation movement. Moreover, safety-sensitive medical applications require explicit assessment of whether model outputs remain grounded in visible evidence [16]. Unsupported assertions about movement abnormalities, unsafe postures, or environmental objects may affect subsequent clinical interpretation. A rehabilitation-oriented benchmark should therefore evaluate both higher-order movement reasoning and baseline visual-evidence consistency through independent negative-evidence probes [17, 18].

To address these gaps, we developed *ActionEval-MedBench*, a multidimensional video benchmark for motor rehabilitation and clinical biomechanics. It comprises 522 de-identified videos: 328 clinical upper-limb rehabilitation videos containing fine-grained pathological motor features and 194 functional biomechanics videos selected from the public MM-Fit dataset [19]. Each video was paired with six clinically designed multi-label multiple-choice questions, yielding 3,132 assessment items. The resulting ActionEval-6D protocol evaluates

semantic action recognition (D1), kinematic feature alignment (D2), pathological compensation and erroneous movement identification (D3), spatiotemporal dynamics (D4), comprehensive movement quality and clinical decision-making (D5), and visual-evidence consistency using negative probes (D6).

Using a standardized zero-shot protocol, we evaluated 26 contemporary MLLMs and examined their progression from basic action recognition to higher-order rehabilitation reasoning. We also compared API-hosted and locally deployable models in privacy-sensitive settings and explored whether vision-specialized architectures exhibited advantages over general-purpose multimodal models. Model performance was assessed using set-based F1-score, exact match, dimension- and cohort-weighted composite scores, non-parametric statistical comparisons, and qualitative failure-mode analysis.

The contributions of this study are fourfold. First, *ActionEval-MedBench* provides a clinically grounded and auditable benchmark that combines fine-grained neuromotor rehabilitation with functional biomechanics. Second, the ActionEval-6D protocol quantifies the performance gap between macroscopic action recognition and reasoning about pathological compensation, temporal abnormalities, and movement quality. Third, D6 and its false-positive endpoint (FP_{D6}) provide an independent safety audit of visually unsupported scene- or object-level selections; this endpoint is interpreted as a baseline visual-grounding indicator rather than a comprehensive measure of clinical hallucination. Fourth, the study provides deployment-relevant evidence concerning the comparative capabilities of hosted and locally deployable models while identifying recurrent failure modes that remain relevant to clinical supervision.

Rather than evaluating whether MLLMs can replace rehabilitation professionals, this study defines their current capability boundaries and safety limitations. The findings are intended to support the development of human-centered rehabilitation AI in which models assist with structured observation and preliminary assessment while clinicians retain responsibility for evidence verification, clinical interpretation, and final decision-making [20, 21].

2. Methods

This study designed and implemented *ActionEval-MedBench*, a multidimensional benchmarking framework for video understanding in motor rehabilitation and clinical biomechanics. The benchmark was developed to systematically evaluate the higher-order clinical kinematic reasoning capabilities of current multimodal large language models (MLLMs) in dynamic rehabilitation videos. Specifically, it assessed whether models could move beyond basic visual semantic recognition and progress toward clinically relevant interpretation, including kinematic feature alignment, pathological compensation identification, spatiotemporal dynamics analysis, comprehensive movement quality assessment, clinical decision-making, and visual-evidence consistency verification.

*Corresponding authors.

E-mail addresses: tzhu@usc.edu.cn (T. Zhu), zhoujun8005@163.com (J. Zhou)

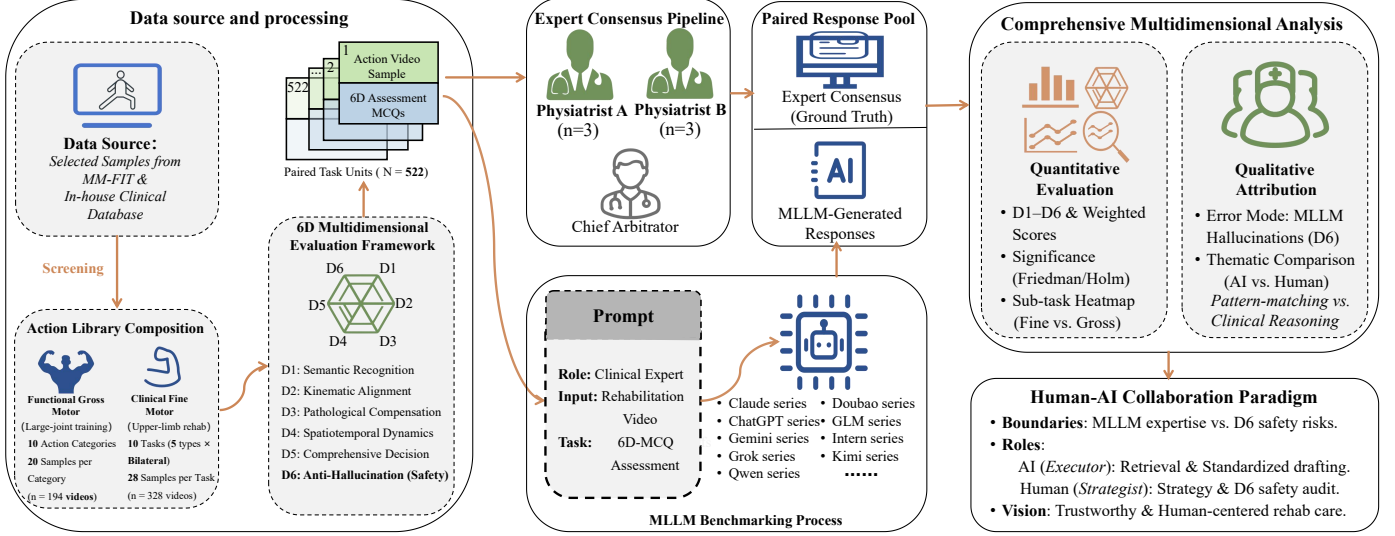


Fig. 1. **Methodological workflow of the ActionEval-MedBench study.** The workflow comprised four phases: (1) **Data preparation:** selection, screening, and de-identification of videos from the MM-FIT dataset and an in-house clinical database, followed by construction of functional gross-motor and clinical fine-motor rehabilitation cohorts; (2) **Expert consensus:** establishment of ground-truth labels through a dual-track expert-consensus pipeline and chief-expert arbitration; (3) **MLLM benchmarking:** zero-shot evaluation of 26 multimodal large language models using standardized prompts and video-based ActionEval-6D multiple-choice tasks; and (4) **Multidimensional analysis:** quantitative evaluation across D1–D6, weighted-score calculation, statistical comparison, and qualitative failure-mode attribution to characterize model capability boundaries and inform human–AI collaboration in rehabilitation medicine.

The study adopted a design paradigm combining expert-consensus ground truth with cross-model zero-shot benchmarking. The overall workflow consisted of four stages. First, video data were rigorously screened, de-identified, and organized into two complementary cohorts. Second, the ActionEval-6D video–MCQ assessment task library was constructed, and Delphi-style expert consensus labels were established as the reference standard. Third, 26 contemporary MLLMs were evaluated under standardized prompts and unified interface constraints in a zero-shot setting. Fourth, model performance was assessed through multidimensional quantitative analysis, false-positive clinical hallucination quantification using the D6 safety dimension, and qualitative failure-mode attribution for high-conflict cases. The complete methodological workflow is shown in Figure 1.

2.1 Data sources and cohort construction

This study adopted a dual-cohort parallel design to comprehensively evaluate the performance of multimodal large language models (MLLMs) in both basic movement perception and higher-order rehabilitation-related pathological reasoning. The dataset consisted of two complementary video cohorts: a clinical fine-grained neuromotor rehabilitation cohort and a general functional biomechanics cohort. Overall, the dataset contained 522 video recordings contributed by 48 participants, including 28 participants in the clinical rehabilitation cohort and 20 participants in the functional biomechanics cohort. The category-level composition and assessment focus of the two cohorts are summarized in Table 1.

Clinical fine-grained neuromotor rehabilitation cohort.

The clinical cohort included 328 videos contributed by 28 participants and collected from real-world rehabilitation physical therapy settings.

The videos involved patients with fine upper-limb motor impairments caused by neurological conditions such as post-stroke hemiplegia and other neuromotor injuries. Typical upper-limb rehabilitation tasks included forearm pronation–supination, arm raising, shoulder-touching and backward-reaching movements, finger tapping or fine finger-opposition tasks, and fist clenching [22]. The selected action categories are summarized in Table 1.

An initial pool of 340 candidate clinical video clips was assessed at the video level according to four prespecified eligibility criteria: complete execution of the target rehabilitation task, adequate visualization of task-relevant body regions, sufficient image quality for movement assessment, and successful de-identification of direct identifiers and potentially recognizable facial information in accordance with the approved protocol. Twelve clips that did not meet one or more eligibility criteria were excluded, resulting in a final clinical analytical cohort of 328 videos.

The final clinical videos had a mean duration of 5.005 s. All 328 videos were encoded at 30 frames/s with a vertical resolution of 640×960 pixels, corresponding to approximately 150 source frames per clip.

The clinical rehabilitation video cohort was reviewed and approved by the Ethics Committee of the First Affiliated Hospital of the University of South China (Approval No. 2021KS-KF-13-01), which waived the requirement for written informed consent to participate because this study involved retrospective analysis of pre-existing de-identified videos and did not involve any clinical intervention. All clinical videos were de-identified

Table 1
Summary of movement categories in the two video cohorts.

Cohort	Videos	Participants	Action labels	Representative movement categories	Assessment focus
Clinical fine-grained neuromotor rehabilitation cohort	328	28	10 task labels	Forearm pronation–supination, arm raising, shoulder-touching followed by extension, finger tapping, and fist clenching, assessed on left- and right-sided movements.	Fine-grained upper-limb motor control, side-to-side comparison, pathological compensation, and impaired movement quality.
General functional biomechanics cohort	194	20	10 action categories	Squat, push-up, dumbbell shoulder press, lunge, standing dumbbell row, sit-up, dumbbell triceps extension, biceps curl, seated dumbbell lateral raise, and jumping jack.	Gross-motor kinetic-chain control, standard functional biomechanics, and baseline non-pathological movement interpretation.

Notes: The clinical cohort contained 10 task labels corresponding to five upper-limb rehabilitation movement types assessed bilaterally. The functional biomechanics cohort contained 10 representative gross-motor action categories selected from the MM-Fit dataset. The 522 video recordings were contributed by 48 participants, including 28 participants in the clinical rehabilitation cohort and 20 participants in the functional biomechanics cohort. Participants contributed a median of 10 recordings each (range, , 8–30), and multiple task- or side-specific recordings could originate from the same participant. The table summarizes movement categories at the cohort level and does not imply equal sample numbers for every action label.

before analysis in accordance with the approved protocol. Direct identifiers were removed before data transmission or analysis.

Direct participant identifiers and detailed participant-level demographic and clinical information were not retained in the deeply de-identified analytical dataset. However, a non-identifying recording-to-participant grouping variable was available exclusively for statistical clustering. This variable indicated which recordings originated from the same participant but contained no information that could be used to identify the participant or characterize their demographic or clinical status. It was used only during statistical analysis and was not included in any model-visible input.

The 328 clinical rehabilitation recordings originated from 28 participants and should not be interpreted as representing 328 independent patients. Multiple task- or side-specific recordings, including left- and right-limb recordings, could originate from the same participant. Apart from cohort membership and the anonymous grouping information required for clustered analysis, no participant-level characteristics were available. Age, sex, individual diagnosis, disease duration, treatment history, and standardized impairment- or disease-severity measures could not be reconstructed. The screening process was therefore documented primarily using video-level eligibility criteria and the final number of retained recordings rather than as a conventional participant-level enrollment flow.

An important design feature of this cohort was its intra-subject healthy-versus-affected control structure. Because neurological impairments such as stroke commonly present with asymmetric unilateral motor deficits, movements performed by the unaffected limb provided an internal reference for relatively normal physiological trajectories, whereas movements performed by the affected limb reflected pathological compensation or impaired motor control. Since paired movements from the same patient were recorded under comparable physical environments, camera viewpoints, object scales, and macroscopic task demands, this intra-subject control design reduced potential confounding from low-level visual factors such as move-

ment amplitude, background variation, and recording conditions.

General functional biomechanics cohort.

The functional cohort included 194 videos contributed by 20 participants and selected from the publicly available MM-Fit multimodal fitness dataset [19].

We screened and curated video clips covering 10 representative gross-motor kinetic-chain tasks, including squats, lunges, push-ups, dumbbell shoulder presses, dumbbell rows, sit-ups, triceps extensions, biceps curls, lateral raises, and jumping jacks. The selected action categories are summarized in Table 1. This cohort served as a reference for normal biomechanical performance and was designed to evaluate the baseline ability of MLLMs to interpret standard, highly visible human movement patterns in non-pathological settings.

The 194 functional biomechanics videos had a mean duration of 15.022 s, a median duration of 15.000 s, and a range of 5.000–134.800 s. All videos were encoded at 30 frames/s with a horizontal resolution of 1280×720 pixels. Most functional clips were approximately 15 s long, although the duration range included several shorter clips and one 134.8-s outlier.

The source-video characteristics reported above should be distinguished from the lower effective temporal-sampling rates and processed frame counts used by individual model-inference interfaces, as detailed in Section 2.3 and Supplementary Table B.14.

2.2 ActionEval-6D evaluation framework and expert consensus labelling

To avoid the limitation that a single accuracy metric may obscure important weaknesses of MLLMs in complex clinical reasoning, we developed the ActionEval-6D evaluation framework. This framework decomposes rehabilitation video understanding into six independent but clinically related dimensions. For each dynamic video, six structured multi-select multiple-choice questions (MCQs) were designed, yielding 3,132 assessment items from 522 videos. The framework was intended to

transform the complex task of rehabilitation movement understanding into a systematic evaluation of clinically relevant capabilities. The definitions of the six dimensions are summarized in Table 2.

In this progressive evaluation system, D1 was designed to measure the baseline capability of visual semantic perception. D2 to D5 gradually increased the requirement for higher-order clinical kinematic reasoning and rehabilitation-related pathological interpretation, whereas D6 was treated as an independent safety dimension for medical AI. Importantly, D6 was not intended to capture all possible forms of clinical hallucination. Instead, it was specifically designed as a set of negative evidence probes targeting scene-level or contextual visual grounding. Errors involving action misclassification, fabricated postural features, or over-extrapolation of pathological compensation were not incorporated into D6, but were instead reflected by performance degradation in D1–D5 and further examined in the qualitative failure-mode analysis.

Each MCQ item was associated with an adjudicated ground-truth option set $\mathcal{G} \subseteq \{A, B, C, D\}$. Depending on the item design, $\mathcal{G}$ could be empty, representing the expert-confirmed absence of all queried findings, or could contain one or more clinically validated options. Options not included in the ground-truth set served as plausible distractors. The multi-select format enabled partial-credit evaluation, while dimension-specific random-subset and select-all baselines were calculated to quantify the remaining influence of ground-truth label cardinality and option-selection strategies.

The gold-standard labels for all 3,132 assessment items were established using a dual-track consensus and arbitration pipeline to minimize subjectivity in clinical evaluation and generate a robust reference standard [23, 24].

Intra-track independent annotation. An expert panel of six clinicians with substantial experience in rehabilitation medicine was randomly divided into two independent working groups. The two groups independently reviewed the videos and completed the assessment items under mutually blinded conditions. They were also blinded to all model outputs. Expert judgements were based on video evidence and relevant rehabilitation assessment principles.

Consistency check. For each item, the consensus answers independently generated by Group A and Group B through within-group majority voting were compared. If the two group-level consensus answers were identical, the answer was directly accepted as the gold-standard label for that item.

Chief-expert arbitration. Items with disagreement between the two groups, such as borderline judgements regarding pathological compensation, were submitted to a chief arbitrator with senior clinical qualifications. The chief arbitrator performed a final video review and adjudication, producing the final expert-consensus label used as the reference standard for model evaluation.

2.3 Model evaluation and standardized zero-shot inference

A total of 26 contemporary multimodal large language mod-

els (MLLMs) were included in this study, covering major proprietary and open-weight model families, including Claude, Gemini, GPT, Grok, GLM, Doubao, Kimi, Qwen, MiniMax, Intern/InternVL, Llama, and Step. To objectively evaluate the out-of-the-box capability and generalization potential of these models when processing clinical rehabilitation videos, all models were tested in a zero-shot setting without any task-specific fine-tuning, prompt optimization based on the test set, or additional training on *ActionEval-MedBench*.

To support predefined subgroup analyses and examine the deployment potential of different technical routes in real-world medical settings, all included models were categorized along two key dimensions. First, according to model accessibility and clinical deployment feasibility, models were classified as API-only/hosted models or open-weight/local-deployable models. This classification was intended to examine whether locally deployable models could provide clinically useful reasoning performance in privacy-sensitive rehabilitation scenarios where patient videos may need to remain within institutional networks [25]. Second, according to architectural design and development focus, models were categorized as general-purpose multimodal models, vision- or video-specialized models, or other specialized models. This classification was used to evaluate whether stronger visual encoding or video-specific architectural design was associated with improved higher-order clinical kinematic reasoning, such as identifying complex pathological compensations.

During the evaluation phase, all models followed a unified task-level inference protocol with standardized questions, prompts, output requirements, and scoring rules. Media preprocessing followed the prespecified model-family-specific route supported by the corresponding provider or, for models without native video-file support, the study-side client implementation.

All 26 models were evaluated through programmatic API endpoints; no provider-operated web interface or local/self-hosted inference was used during benchmark execution. The API-only/hosted versus open-weight/local-deployable classification described model availability and deployment potential rather than the inference access channel used in this study. Thus, models categorized as open-weight/local-deployable were also evaluated through API endpoints to maintain a consistent programmatic inference workflow. Model-family-specific API access routes, visual-input representations, and temporal-sampling configurations are reported in Supplementary Table B.14.

Each request contained a de-identified visual input—either a complete video clip or a chronologically ordered sequence of contact-sheet images—together with its corresponding MCQ item. For the clinical cohort, all visual inputs underwent the approved de-identification procedure and removal of direct identifiers before model inference; no direct patient identifiers were submitted to external API services.

Local source filenames, directory paths, preassigned action indices, and ground-truth action names were used exclusively for study-side data indexing and were not included in the model-visible prompt, question stem, answer options, attachment captions, uploaded-file metadata, or any other textual in-

Table 2

Definition of the ActionEval-6D multidimensional clinical evaluation framework.

Dimension	Concept	Primary assessment objective
D1	Semantic action recognition	Identifies the macroscopic movement task or rehabilitation exercise being performed in the video.
D2	Kinematic feature alignment	Assesses the consistency between key joint trajectories, movement amplitude, trunk posture, and task-specific rehabilitation or biomechanical standards.
D3	Pathological compensation and erroneous movement identification	Detects compensatory strategies, abnormal postures, erroneous kinetic-chain transmission, and pathological movement patterns during task execution.
D4	Spatiotemporal dynamics analysis	Analyzes movement control features across time, including rhythm, speed variation, pauses, tremor, and smoothness during eccentric or concentric phases.
D5	Comprehensive movement quality and clinical decision-making	Integrates multidimensional movement phenotypes to infer functional limitation, potential training risk, and clinically relevant intervention or prescription recommendations.
D6	Visual grounding and anti-hallucination	Evaluates whether the model remains grounded in visual evidence and avoids fabricating scene, object, or contextual information that is absent from the video.

put during inference. Before model submission, each video was represented using an opaque sample identifier that contained no action-category information. Consequently, none of the 522 model-visible video inputs contained a filename-derived or action-name cue (0/522), and the complete benchmark constituted the cue-free evaluation set. Models were required to determine the target movement from the visible video evidence and select the most appropriate action from the predefined action list. The prompt used role-based instruction to ask the model to act as a senior expert in clinical rehabilitation and biomechanics [26]. Four inference constraints were imposed: (1) **visual grounding**, requiring the answer to be based only on dynamic movement phenotypes visible in the video; (2) **prior suppression**, discouraging the model from relying solely on medical keywords in the question stem or semantic priors from the answer options; (3) **structured output**, requiring the model to return a unique and explicit option set unless an empty set was permitted by the task rule; and (4) **anti-hallucination**, prohibiting unsupported assertions about environmental context, objects, scenes, or findings not visible in the video.

To maximize reproducibility and reduce stochastic variation during benchmarking, the generation temperature was set to 0.0 for all models whose programmatic API allowed explicit parameter control. This deterministic setting was selected because the benchmark required objective comparison against expert-consensus labels rather than creative response generation. For APIs that did not expose a configurable temperature parameter, the provider’s default inference configuration was used. These cases were recorded as API-level constraints, and no additional prompt tuning, repeated sampling, or manual output selection was performed. Thus, all models were evaluated under the lowest-stochasticity configuration available through their respective programmatic API routes. Regarding the visual-input pipeline, the source-video characteristics differed between the

two cohorts. The 328 clinical rehabilitation clips had a mean duration of 5.005 s and were uniformly encoded at 30 frames/s with a vertical resolution of 640×960 pixels. The clips contained 150–184 source frames, with approximately 150 frames per clip on average. The 194 functional biomechanics clips had a mean duration of 15.022 s, a median duration of 15.000 s, and a range of 5.000–134.800 s. All functional clips were encoded at 30 frames/s with a horizontal resolution of 1280×720 pixels, corresponding to a median of approximately 450 source frames and a range of approximately 150–4,044 source frames.

Models whose official inference interfaces accepted video files received the complete de-identified clips, after which video decoding and temporal preprocessing were performed by the corresponding provider-side pipeline.

For the nine models across five model families that did not support native video-file input—Claude, GPT, Grok, Llama, and MiniMax—visual inputs were generated using the study-side `make_contact_sheets()` pipeline. For a source video containing N frames, the pipeline uniformly selected

$$k = \min(N, K_{\max})$$

frames across the complete temporal axis, where $K_{\max}$ was the model-family-specific maximum frame budget. Each sampled frame was resized to 320×180 pixels. Up to 12 temporally ordered frames were arranged into one 4×3 contact-sheet image and encoded as JPEG with a quality factor of 85. The resulting contact sheets were submitted to the model in chronological order. The effective sampling rate was duration-adaptive and was calculated as k/T frames/s, where T denotes the source-video duration in seconds. No question-dependent frame selection, key-frame optimization, or retrospective manual adjustment was performed.

For the Claude family, GPT-5.5, Grok-4.2, and the Llama-4 family, $K_{\max} = 96$. Because all 522 benchmark videos con-

tained at least 150 source frames, every video reached the 96-frame limit. Consequently, each request contained exactly 96 uniformly sampled frames represented by eight contact-sheet images. The effective sampling rate was approximately 19.2 frames/s for a typical 5-s clinical clip and 6.4 frames/s for a 15-s functional clip. Across the functional cohort, the duration-adaptive effective sampling rate ranged from approximately 0.71 frames/s for the 134.8-s outlier to 19.2 frames/s for the 5-s clip.

For the MiniMax-M2.7 family, $K_{\max} = 240$. All clinical clips contained fewer than 240 source frames; therefore, their complete frame sequences were retained, yielding 150–184 frames and 13–16 contact-sheet images per request at the original temporal density of 30 frames/s. In the functional cohort, 193 of 194 clips reached the 240-frame limit and were represented by 20 contact-sheet images, whereas one 5-s sit-up clip contained 150 frames and was represented by 13 contact sheets. For capped functional clips, the effective sampling rate was $240/T$ frames/s, corresponding to 16.0 frames/s for a 15-s clip and approximately 1.78 frames/s for the 134.8-s outlier. The uncapped 5-s clip retained its original temporal density of 30 frames/s.

The contact-sheet inputs were treated as surrogate visual representations rather than as being fully equivalent to temporally continuous native-video input. Although the selected frames preserved chronological ordering and covered the complete clip duration, fixed-count temporal sampling, resizing each frame to 320×180 pixels, and combining multiple frames into contact-sheet images may reduce the visibility of subtle joint-level abnormalities and short-duration temporal events. The potential implications for D3 pathological-compensation identification and D4 spatiotemporal-dynamics analysis are discussed in Section 4.7.

For native-video APIs, the effective temporal-sampling configuration was provider-specific rather than identical across model families. At the time of evaluation, the Gemini video interface documented a default sampling rate of 1 frame/s, corresponding to approximately 5 processed frames for a typical clinical clip and a median of 15 processed frames for a functional clip (range: approximately 5–135 frames). The DashScope-based Qwen and Kimi video interfaces documented a default sampling rate of 2 frames/s, corresponding to approximately 10 processed frames for a typical clinical clip and a median of 30 processed frames for a functional clip (range: approximately 10–270 frames), unless a provider-specific maximum-frame limit was reached [27, 28, 29]. When the number of frames calculated from the documented sampling rate exceeded the provider-defined maximum, the corresponding interface performed uniform temporal downsampling within that limit. For other proprietary video interfaces, the officially documented default media-processing route available during the evaluation period was retained without manual modification. Model- or API-family-specific input routes, sampling configurations, and processed-frame counts are summarized in Supplementary Table B.14.

For proprietary native-video APIs that did not return the realized frame indices or internal preprocessing logs, the exact timestamps and final number of frames processed in an in-

dividual request could not be independently recovered. Accordingly, their reported processed-frame counts represent expected values calculated from the source-clip duration and documented API sampling rate rather than provider-returned processing logs. These expected values may be reduced when a provider-specific maximum-frame limit is applied. In contrast, the realized frame and contact-sheet counts for the five non-native-video client families were fully recoverable from the original inference code and were verified across all 522 benchmark videos.

To ensure engineering stability during large-scale automated evaluation, a predefined limited retry mechanism was triggered when temporary network interruption, API timeout, or server-side failure occurred. After the retry threshold was reached, a successfully returned but invalid or unparsable response was classified as a returned invalid response and assigned an item-level score of zero under the primary benchmark scoring rule. By contrast, a response opportunity for which no model output was returned because of a missing output or system-level failure was classified as unavailable and excluded from the corresponding model-specific primary point estimate. Alternative valid-only and complete-denominator response-handling strategies were evaluated in sensitivity analyses, as described in Sections 2.4 and 2.5.

2.4 Response parsing, scoring metrics, and D6 safety quantification

Response parsing and usability control. Because the benchmark used multi-label MCQs, model-generated natural-language responses were converted into structured option subsets using a predefined rule-based parser. The parser extracted all valid option letters from A to D and mapped them into standardized prediction sets. For example, responses such as “A and B” or “A, B” were both parsed as the prediction set $\{A, B\}$.

A parsable valid response was defined as a successfully returned option subset $\mathcal{P} \subseteq \{A, B, C, D\}$. For items whose task rule permitted the absence of all queried findings, an explicitly returned empty set ($\mathcal{P} = \emptyset$) was retained as a valid response. An explicit empty set was distinguished from a blank or missing response. Returned responses containing refusals, unparsable text, illegal option symbols, or structurally unusable content were classified as invalid or unparsable responses, whereas response opportunities affected by missing outputs or system-level failures were recorded separately as unavailable.

For primary benchmark scoring, the 81,006 successfully returned dimension-level responses constituted the returned-response scoring pool. The 80,931 parsable valid responses were scored according to their predicted option sets, whereas the 75 returned invalid or unparsable responses were assigned an item-level score of zero. The 426 response opportunities unavailable because of missing outputs or system-level failures were excluded from model-specific primary point estimates and recorded separately.

Two alternative response-handling strategies were used to evaluate robustness. In the valid-only sensitivity analysis, the

75 returned invalid or unparseable responses were excluded rather than assigned zero, leaving 80,931 parsable valid responses. In the more conservative complete-denominator sensitivity analysis, all 81,432 expected response opportunities were retained, and both the 75 returned invalid or unparseable responses and the 426 unavailable responses were assigned a score of zero.

Primary and secondary scoring metrics. We used a two-level metric system to evaluate MLLM performance, with reasoning effectiveness as the primary focus and clinical strictness as a complementary criterion. The primary metric was the item-level set-based F1-score, which was used to quantify clinical pathological reasoning performance in the main analysis. In the multi-label setting, the F1-score provides a harmonic balance between precision and recall, thereby capturing partial recognition of complex clinical signs and avoiding the all-or-nothing limitation of exact-match scoring. Therefore, all weighted composite scores, including S_{total} , and subsequent multidimensional visualizations were based on the item-level set-F1 score.

Unless otherwise specified, the leaderboard, dimension-specific F1-scores, cohort-specific scores, clinical reasoning gaps, and weighted composite scores followed the primary returned-response scoring rule described above: parsable valid responses were scored according to their predicted option sets, returned invalid or unparseable responses were assigned zero, and unavailable response opportunities were excluded from model-specific point estimates.

Human-reference and chance-level baselines. To contextualize model performance, two chance-level baselines were calculated separately for each ActionEval dimension. For the random-subset baseline, each option in the four-option universe was independently selected with a probability of 0.5, equivalent to assigning equal probability to all 16 possible option subsets, including the empty set. The expected item-level set-F1 was calculated by exact enumeration rather than Monte Carlo simulation. For the select-all baseline, all four options were selected for every item. Both baselines used the same item-level set-F1 definition and empty-set conventions as the primary model evaluation.

The empirical human ceiling was calculated using the three Group A experts' independent pre-arbitration annotations. For each expert, the annotated option set was compared with the final adjudicated ground-truth set, and dimension-specific item-level set-F1 was macro-averaged across all available items. The primary human-ceiling estimate was defined as the arithmetic mean of the three individual-expert F1-scores. The F1-score of the Group A three-expert majority-vote label was additionally reported as a consensus reference. Because the final adjudicated ground truth incorporated evidence from the expert-panel process, these results should be interpreted as empirical human-reference agreement rather than as a completely independent theoretical upper bound.

Set-F1 decomposition metrics. To characterize the error structure underlying the item-level set-F1 score, we additionally calculated the mean predicted-set size, item-level precision, and item-level recall for every model and ActionEval dimension. For item i , let $\mathcal{P}_{mi}$ denote the option set predicted by

model m , and let $\mathcal{G}_i$ denote the adjudicated expert ground-truth option set. Item-level predicted-set size, precision, and recall were defined as:

$$\text{Predicted-set size}_{mi} = |\mathcal{P}_{mi}|, \quad (1)$$

$$\text{Precision}_{mi} = \frac{|\mathcal{P}_{mi} \cap \mathcal{G}_i|}{|\mathcal{P}_{mi}|}, \quad (2)$$

$$\text{Recall}_{mi} = \frac{|\mathcal{P}_{mi} \cap \mathcal{G}_i|}{|\mathcal{G}_i|}. \quad (3)$$

When both the predicted and ground-truth sets were empty, precision, recall, and set-F1 were assigned a value of 1. When exactly one of the two sets was empty, precision, recall, and set-F1 were assigned a value of 0. For two non-empty sets, complete overlap received a score of 1, no overlap received a score of 0, and partial overlap received the corresponding proportional set-F1 score.

For each model m and dimension d , mean predicted-set size, precision, and recall were calculated by macro-averaging the corresponding item-level values across parsable valid responses:

$$\overline{|\mathcal{P}|}_{md} = \frac{1}{N_{md}} \sum_{i=1}^{N_{md}} |\mathcal{P}_{mi}|, \quad (4)$$

$$\overline{\text{Precision}}_{md} = \frac{1}{N_{md}} \sum_{i=1}^{N_{md}} \text{Precision}_{mi}, \quad (5)$$

$$\overline{\text{Recall}}_{md} = \frac{1}{N_{md}} \sum_{i=1}^{N_{md}} \text{Recall}_{mi}.$$

Here, N_{md} denotes the number of parsable valid responses for model m in dimension d . Mean ground-truth-set size was additionally reported to facilitate interpretation of option under-selection or over-selection. Dimension-level set-F1 remained the macro-average of the item-level set-F1 values and was not recalculated as the harmonic mean of the macro-averaged precision and recall.

The decomposition metrics were calculated conditionally among parsable valid responses because predicted-set size, precision, and recall could not be meaningfully assigned to missing or unparseable outputs. Returned invalid or unparseable responses and unavailable response opportunities were therefore excluded from the decomposition analysis. Consequently, the results reported in Supplementary Table C.19 constitute a valid-response-only diagnostic analysis and should be distinguished from the primary benchmark scores, in which returned invalid or unparseable responses were assigned zero. The decomposition analysis did not alter the primary leaderboard or model rankings.

Because 292 of the 522 D3 items had an empty ground-truth set, the all-item D3 averages combined negative items with items containing expert-labelled pathological-compensation

findings. To more directly evaluate the omission of pathological signs, we additionally performed a diagnostic analysis restricted to the 230 D3 items containing at least one expert-labelled pathological-compensation option ($|\mathcal{G}_i| > 0$). Within this positive-ground-truth subset, predicted-set size, precision, recall, set-F1, and the proportion of responses with an empty predicted set were recalculated. This diagnostic analysis was descriptive and did not affect the primary model scores or rankings. Complete model- and dimension-level decomposition results are reported in Supplementary Table C.19.

Post hoc D3 stratification by adjudicated gold-label status. Because the item-level pre-arbitration Group A and Group B consensus labels were not retained in the archived analytical dataset, the exact split between the 404 D3 items with inter-group agreement and the 118 D3 items requiring arbitration could not be reconstructed. Aggregate arbitration counts reported in Table 4 were therefore not used to infer or synthetically assign item-level arbitration status.

To address the related scientific question of whether poor D3 performance persisted among items with a determinate reference interpretation, we performed a reviewer-requested post hoc stratification using the archived adjudicated ground-truth set and its accompanying rationale. Items with either an explicit expert-labelled pathological-compensation finding or a rationale-confirmed absence of pathological compensation were classified as having a determinate, rationale-supported gold label ($n = 429$). This subset comprised 230 items containing at least one expert-labelled compensation option and 199 items with a rationale-confirmed absence. The remaining 93 items were classified as internally unresolved under the archived gold-set/rationale classification. The term “determinate” in this analysis does not denote six-rater unanimity and should not be interpreted as a reconstruction of the 404/118 inter-group agreement/arbitration split reported in Table 4.

For each model, item-level set-F1 was macro-averaged separately within the determinate and internally unresolved subsets using the primary returned-response scoring rule. Parsable valid responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned an item-level score of zero, and unavailable response opportunities were excluded from the corresponding model-specific subset estimate. The two subset estimates were compared using a two-sided paired Wilcoxon signed-rank test across the 26 model-level F1 pairs. Additional descriptive analyses separately examined the 230 positive-compensation items and the 199 rationale-confirmed-absence items. This post hoc diagnostic analysis used the same response-handling rule as the primary dimension-level analysis and did not affect the primary leaderboard, composite scores, or model rankings. Aggregate results are reported in Table 6, and the complete model-level results are provided in Supplementary Table C.18.

As a secondary metric, we independently reported exact-match (EM) accuracy. EM required the model-predicted option set $\mathcal{P}$ to be identical to the expert-consensus ground-truth set $\mathcal{G}$:

$$\mathcal{P} = \mathcal{G}.$$

Although EM was not used to calculate the weighted compos-

ite score, it provided a stringent reference for evaluating output determinacy and clinical unambiguity.

Multidimensional and cross-cohort weighted scores. The primary outcome of this study was the cohort-weighted ActionEval composite score, denoted as S_{total} . The calculation consisted of two steps. First, for each model, the dimension-weighted score across D1–D6 was calculated as:

$$S_{\text{dimension}} = 0.10S_{D1} + 0.10S_{D2} + 0.30S_{D3} + 0.10S_{D4} + 0.25S_{D5} + 0.15S_{D6}. \quad (6)$$

The dimension weights were specified a priori, before inspection of model-performance results, through consensus-based clinical-priority judgement by the multidisciplinary rehabilitation expert panel. The weighting scheme was designed to reflect the relative clinical consequences and decision relevance of the six dimensions rather than their statistical difficulty. D1, D2, and D4 were each assigned 10% as foundational perceptual and kinematic capabilities; D3 received 30% because distinguishing true motor recovery from pathological compensation is central to rehabilitation assessment [3]; D5 received 25% because it integrates movement quality, functional limitation, safety risk, and intervention-relevant judgement, consistent with the multidimensional conceptualization of functioning and disability in the International Classification of Functioning, Disability and Health [30]; and D6 received 15% as an independent visual-grounding safety component.

This approach is consistent with transparent multicriteria evaluation principles, under which criterion weights represent explicitly stated clinical value judgements and should be accompanied by uncertainty or sensitivity analyses [31]. The weights were not fitted to model outputs or patient outcomes and should therefore be interpreted as an expert-defined benchmark utility function rather than as an externally validated clinical scoring scale.

Second, the final composite score was obtained through cross-cohort weighting:

$$S_{\text{total}} = 0.65S_{\text{clinical}} + 0.35S_{\text{functional}}. \quad (7)$$

where S_{clinical} and $S_{\text{functional}}$ denote the mean scores of the clinical fine-grained rehabilitation cohort and the functional biomechanics cohort, respectively.

The cross-cohort weights were also specified before inspection of model-performance results. The clinical fine-grained rehabilitation cohort received greater weight because it represented the primary clinical-use scenario of the benchmark and contained the more clinically consequential pathological-reasoning tasks, whereas the functional biomechanics cohort was retained as a complementary test of general movement understanding and cross-cohort generalization. These weights likewise represent the intended evaluation priorities of the benchmark rather than an empirically optimized clinical scoring rule.

Secondary outcomes included overall EM accuracy, dimension-specific F1-scores, cohort-specific scores, cross-cohort performance drop, and subgroup differences according to model accessibility and architectural specialization.

To evaluate whether the principal model-ranking conclusions depended on the prespecified dimension weights, the composite scores were recalculated under three alternative weighting schemes: (1) equal weighting, assigning one-sixth of the total weight to each dimension; (2) lower-D6 weighting, reducing the D6 weight to 5% and proportionally redistributing the remaining weight across D1–D5 while preserving their relative ratios; and (3) D6-excluded weighting, assigning zero weight to D6 and proportionally normalizing the weights of D1–D5. Rank-order robustness was assessed using Spearman’s rank correlation between the primary and alternative model rankings, together with top-five model overlap and the maximum absolute rank displacement. The complete results of these analyses are reported in Supplementary Table C.17.

D6 false-positive hallucination quantification. D6 was designed as an independent safety dimension consisting of scene- or object-level negative-evidence probes. In all D6 items, the queried visual element was confirmed by experts to be absent from the corresponding video. Therefore, conventional F1-score may produce a ceiling effect and obscure clinically relevant false-alarm risk. To address this issue, we defined the D6 false-positive hallucination rate (FP_{D6}) as the core safety endpoint. A false-positive visual-grounding event was recorded when the expert-consensus label indicated the absence of the queried visual element, but the model selected an affirmative option indicating its presence.

Let $N_{FP,D6}$ denote the number of D6 contextual or object-level false-positive visual-grounding events, and let N_{D6} denote the number of D6 negative probes included in the corresponding analysis. The D6 false-positive hallucination rate was calculated as:

$$FP_{D6} = \frac{N_{FP,D6}}{N_{D6}}. \quad (8)$$

For each model, the point estimate of FP_{D6} was reported together with an exact binomial or Wilson 95% confidence interval. Because the current D6 design specifically targeted scene- and object-level visual-evidence consistency, this metric should be interpreted as an estimate of baseline visual-grounding failure rather than as a comprehensive measure of all forms of clinical kinematic hallucination. Moreover, because D6 used a constrained select-all-that-apply format, these quantitative events represent visually unsupported affirmative selections rather than, by themselves, definitive evidence of hallucination in the stronger generative sense. Over-extrapolation involving action categories, pathological postures, or kinematic compensations was instead captured through D1–D5 performance and further examined in the qualitative failure-mode analysis.

2.5 Statistical analysis and qualitative failure-mode attribution

Descriptive statistics and non-parametric hypothesis testing. All quantitative analyses were conducted using a long-format dataset at the model–video–MCQ level. The theoretical

response dataset consisted of 81,432 model–video–MCQ response opportunities, corresponding to 26 MLLMs, 522 videos, and six ActionEval dimensions for each video.

The 522 video recordings were contributed by 48 participants, including 28 participants in the clinical rehabilitation cohort and 20 participants in the functional biomechanics cohort. Multiple task- or side-specific recordings, including left- and right-limb recordings, could originate from the same participant.

Of the 81,432 expected model–video–MCQ response opportunities, 81,006 responses were successfully returned and constituted the primary returned-response scoring pool. Among these, 80,931 responses were parsable and valid, whereas 75 returned responses were classified as invalid or unparsable and were assigned an item-level score of zero in the primary benchmark scoring. The remaining 426 response opportunities were unavailable because of missing outputs or system-level failures and were excluded from model-specific primary point estimates. The 80,931 parsable valid responses additionally constituted the valid-only subset used for Set-F1 decomposition and sensitivity analysis.

Unless otherwise specified, model-level point estimates, dimension-specific scores, cohort-specific scores, clinical reasoning gaps, and weighted composite scores followed this primary returned-response scoring rule. Parsable valid responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned zero, and unavailable response opportunities were excluded from model-specific point estimates. For matched non-parametric comparisons, only observations with scores available for the models involved in the corresponding comparison were included; returned invalid responses assigned zero were treated as observed scores, whereas unavailable responses were not treated as observed zeros.

To generate Figure 3, the model-level dimension-specific item-level set-F1 scores were arranged into a 26×6 matrix, with one row for each evaluated MLLM and one column for each ActionEval dimension (D1–D6). Each cell represented the mean item-level set-F1 score for one model–dimension combination and was expressed as a percentage. The heatmap used a fixed colour scale ranging from 0% to 100%. The plotted values followed the primary returned-response scoring rule described above. The heatmap was generated using Python version 3.10.0 (Python Software Foundation, <https://www.python.org/>), seaborn version 0.13.2 (<https://seaborn.pydata.org/>), and Matplotlib version 3.10.8 (<https://matplotlib.org/>).

To account for intra-participant correlation arising from multiple recordings contributed by the same person, 95% confidence intervals for model-level estimates were calculated using 2,000 participant-level cluster-robust bootstrap resamples [32]. Participants were sampled with replacement separately within the clinical and functional cohorts. Within each bootstrap replicate, all video recordings and associated ActionEval items contributed by a sampled participant were retained together, ensuring that repeated task-specific recordings and paired left- and right-limb recordings were not separated during resampling. Cohort-specific estimates were recalculated within each repli-

cate and subsequently combined using the prespecified 65% clinical and 35% functional cohort weights. This procedure accounted for both dependence among the six ActionEval items derived from the same video and dependence among multiple videos contributed by the same participant.

The participant-level bootstrap procedure was used to estimate confidence intervals and did not alter the model-level point estimates or the paired non-parametric hypothesis tests described below. Video-level aggregated analyses were retained as complementary sensitivity analyses for assessing whether the principal conclusions depended on the six-item matched structure within each video.

For overall cross-model performance comparison, the Friedman test was used to assess global differences among the 26 models on matched video-MCQ item-level scores [33], as the score distributions may not satisfy strict normality assumptions. If the global test indicated statistical significance, pairwise comparisons between models were further performed using the Wilcoxon signed-rank test on the corresponding matched observations. To control the inflation of type I error due to multiple comparisons, all pairwise P values were adjusted using the Holm-Bonferroni method. When multiple top-ranked models showed numerically small differences and no statistically significant pairwise differences after correction, they were collectively defined as the top-performing cluster.

Because statistical significance can be amplified in large matched datasets, the global Friedman result was interpreted jointly with Kendall’s coefficient of concordance (W), absolute score differences, multiplicity-adjusted pairwise comparisons, and participant-level bootstrap confidence intervals. A statistically significant global test was therefore not interpreted as evidence that all models were separated by clinically meaningful performance differences.

Capability decomposition metrics. To quantify the structural capability gap and cross-domain generalization loss of MLLMs in rehabilitation video understanding, two derived metrics were predefined. First, the clinical reasoning gap was used to measure the performance drop when models moved from low-level visual perception to higher-order clinical reasoning:

$$\text{Clinical reasoning gap} = S_{D1} - \frac{S_{D3} + S_{D4} + S_{D5}}{3}. \quad (9)$$

Second, cross-cohort performance drop was used to evaluate model generalization from standardized functional biomechanics videos to fine-grained clinical rehabilitation videos:

$$\text{Performance drop} = S_{\text{functional}} - S_{\text{clinical}}. \quad (10)$$

Because the two cohorts differed in movement scale, body region, participant status, recording context, and data source, this performance drop was interpreted as a combined cross-domain generalization burden rather than as the isolated causal effect of pathological compensation difficulty.

Subgroup comparisons were conducted using the Mann-Whitney U test, permutation tests, or bootstrap confidence in-

tervals for between-group differences, as appropriate. Effect sizes were reported where applicable.

For the subgroup analyses summarized in Table 10, the 10 two-sided Mann-Whitney U tests presented in the table were treated as a single family of hypotheses. The corresponding unadjusted P values were corrected using the Benjamini-Hochberg false-discovery-rate (BH-FDR) procedure. Statistical significance was determined using BH-FDR-adjusted P values at a two-sided threshold of 0.05, while unadjusted P values were retained for transparency. Holm-Bonferroni and Bonferroni corrections were additionally applied to the same family of 10 tests as sensitivity analyses; both procedures yielded the same substantive conclusion as the primary BH-FDR analysis.

Unless otherwise specified, a two-sided significance threshold of $P < 0.05$ was used for statistical tests not subject to a separately defined multiplicity-adjustment procedure.

Sensitivity analyses. To assess whether the main conclusions depended on specific scoring assumptions or preprocessing decisions, a series of sensitivity analyses were performed. These included alternative weighting schemes, recalculation after removing the D6 safety dimension, isolated analysis using only the clinical cohort, complete-case question-intersection analysis, video-level aggregated analysis, valid-only response scoring, complete-denominator invalid/missing-as-zero scoring, and alternative Holm-Bonferroni and Bonferroni multiplicity adjustments for the 10 subgroup comparisons reported in Table 10.

In the valid-only sensitivity analysis, the 75 returned invalid or unparsable responses were excluded rather than assigned zero, leaving 80,931 parsable valid responses. This analysis evaluated whether assigning zero to returned invalid responses affected the leaderboard, top-performing cluster, or principal dimension-level conclusions.

In the more conservative complete-denominator sensitivity analysis, all 81,432 expected response opportunities were retained. Both the 75 returned invalid or unparsable responses and the 426 response opportunities unavailable because of missing outputs or system-level failures were assigned zero. This analysis evaluated whether exclusion of unavailable response opportunities resulted in an overly optimistic estimate of model performance or materially affected model rankings.

The Set-F1 decomposition analysis was conducted separately among the 80,931 parsable valid responses. Because predicted-set size, precision, recall, and empty-prediction frequency could not be meaningfully defined for missing or unparsable outputs, invalid and unavailable responses were excluded from these decomposition estimates. The decomposition analysis was therefore treated as a valid-response-only diagnostic analysis and did not replace the primary leaderboard or dimension-specific F1 estimates.

Qualitative failure-mode analysis. The qualitative analysis consisted of two sequential stages. First, a deterministic rule-based retrieval module screened the structured quantitative results and identified candidate error records according to prespecified criteria, including frequently misclassified items, errors shared by the top-performing model cluster, marked D1-to-D3/D5 performance discrepancies, and D6 false-positive

visual-grounding events. This automated module functioned only as a case-retrieval and data-organization tool; it did not generate qualitative codes, assign failure-mode labels, or interpret the clinical mechanisms underlying model errors.

Second, clinical and algorithmic reviewers manually examined the retrieved cases, including the original video, question and answer options, expert-consensus label, model-selected option set, and accompanying free-text rationale. Using a Grounded Theory-informed constant-comparison approach, the reviewers iteratively generated preliminary codes, compared recurrent error patterns across cases and models, and consolidated related codes into higher-order failure-mode categories through discussion and consensus. The resulting thematic categories were used only for qualitative interpretation and did not affect the quantitative scores or model rankings.

3. Results

3.1 Dataset composition, response quality, and expert consensus standard

The ActionEval-MedBench dataset included 522 deeply de-identified dynamic movement video recordings contributed by 48 participants. The clinical fine-grained rehabilitation cohort comprised 328 recordings from 28 participants, whereas the functional biomechanics cohort comprised 194 recordings from 20 participants. Participants contributed a median of 10 recordings each (range, 8–30), and multiple task- or side-specific recordings could originate from the same participant.

Through zero-shot inference on 3,132 MCQ assessment items across 26 evaluated MLLMs, a total of 81,432 expected dimension-level response opportunities were generated. Of these, 81,006 responses were successfully returned, whereas 426 response opportunities were unavailable because of missing outputs or system-level failures, corresponding to an overall response-yield rate of 99.48%. Among the returned responses, 80,931 were parsable and valid, while 75 were classified as invalid or unparsable. Under the primary returned-response scoring rule, the 80,931 parsable valid responses were scored according to their predicted option sets, and the 75 returned invalid or unparsable responses were assigned an item-level score of zero. The 426 unavailable response opportunities were excluded from model-specific primary point estimates. The overall valid-response rate was 99.38% relative to the complete expected response set. The high response yield and parsability of the returned outputs support the engineering feasibility of using structured select-all-that-apply MCQs for large-scale MLLM evaluation.

The dataset characteristics and the ActionEval-6D evaluation framework are summarized in Table 3.

To establish the reference standard, we implemented a dual-track independent consensus and arbitration pipeline based on a 3+3 expert-panel structure. As shown in Table 4, the comparison between the group-level consensus labels independently generated by Group A and Group B yielded a pooled Cohen’s

κ of 0.807. This pooled estimate was interpreted descriptively and in conjunction with the dimension-specific agreement coefficients and arbitration rates, rather than as evidence of uniformly high reliability across all six dimensions. The spontaneous pipeline consensus rate was 86.88%, corresponding to 2,721 items with complete agreement between the two independent groups. Only 13.12% of items (411/3,132) required chief-expert arbitration. These results indicate substantial pooled inter-group agreement while also revealing dimension-dependent heterogeneity in annotation reliability and the need for arbitration in clinically ambiguous items.

The pooled Cohen’s κ of 0.807 should be interpreted cautiously because it aggregates all six ActionEval dimensions and is partly elevated by the perfect agreement observed in D6 ($\kappa = 1.000$). D6 consisted of comparatively constrained visual-grounding negative probes and showed no pre-arbitration discrepancies in the present dataset. Consequently, the pooled estimate does not imply uniformly high reliability across ActionEval-6D. The dimension-specific κ estimates and arbitration rates provide more informative measures of annotation difficulty, particularly for D3–D5, where agreement was lower and arbitration was more frequently required.

The dimension-specific agreement pattern further reflected the different levels of clinical complexity across ActionEval-6D tasks. For relatively objective perceptual and visual-grounding tasks, D6 achieved perfect agreement ($\kappa = 1.000$), and D1 showed very high agreement ($\kappa = 0.896$), with arbitration rates of 0.00% and 4.02%, respectively. In contrast, dimensions requiring higher-order judgement showed lower agreement and higher arbitration rates. D3 pathological compensation identification ($\kappa = 0.650$), D4 spatiotemporal dynamics analysis ($\kappa = 0.761$), and D5 comprehensive decision quality ($\kappa = 0.706$) required arbitration in 22.61%, 19.35%, and 18.39% of items, respectively.

This distribution suggests that clinical movement-pathology reasoning is intrinsically more uncertain and expertise-dependent than basic visual recognition or scene-level visual-grounding verification. The higher arbitration rates in D3–D5 also support the need for a dual-track blinded consensus procedure when constructing gold-standard labels for fine-grained rehabilitation video understanding.

3.2 Quantitative performance evaluation

Based on the cross-cohort weighted composite score (S_{total}), the 26 evaluated MLLMs showed a clear performance gradient on ActionEval-MedBench. The global non-parametric test based on matched video–MCQ item-level F1-scores showed statistically significant overall performance differences among models (Friedman test, $\chi^2 = 2671.34$, $P < 0.001$; Kendall’s $W = 0.039$). However, Kendall’s $W = 0.039$ indicated a small overall effect magnitude. Therefore, this global significance result should not be interpreted as evidence that all models were separated by clinically meaningful performance differences. Instead, the leaderboard was interpreted jointly using statistical significance, absolute score differences, effect magnitude, and

cluster-level stability. The overall leaderboard and statistical testing results are summarized in Table 5 and Figure 2.

Participant-level cluster-robust bootstrapping produced wider confidence intervals than the original video-level re-sampling procedure, reflecting dependence among multiple recordings contributed by the same participant. Among the 10 models reported in Table 5, the participant-level confidence intervals were 1.21–1.36 times as wide as the corresponding video-level intervals. The revised bootstrap procedure did not alter the model-level point estimates or leaderboard order because it affected uncertainty estimation rather than score calculation. Substantial overlap remained among the confidence intervals of the leading models, reinforcing the need for cautious interpretation of small numerical differences near the top of the leaderboard. Confidence-interval overlap was interpreted descriptively and was not used as a substitute for the prespecified paired hypothesis tests.

At the top of the leaderboard, the six highest-ranking models—Gemini-3.1-Pro-Preview, Doubao-Seed-2.0-Lite-260428, Gemini-3.5-Flash, Claude-Opus-4-7, GLM-5V-Turbo, and GPT-5.5—showed highly clustered weighted ActionEval scores, ranging from 66.17 to 65.46. The absolute score range within this group was only 0.71 percentage points. Pairwise Wilcoxon signed-rank tests with Holm–Bonferroni correction showed no statistically significant differences within this top-ranked group across all 15 pairwise comparisons ($P = 1.000$ for all comparisons). Therefore, these six models were defined as a statistically indistinguishable top-performing cluster rather than as a strict single-model hierarchy.

The mid-ranking models, beginning with Kimi-K2.6, showed relatively stable overall performance but remained numerically below the top-performing cluster. However, given the small effect magnitude in the global rank test, the participant-level uncertainty intervals, and the narrow absolute score differences near the top of the leaderboard, small differences between adjacent models should be interpreted cautiously. Models ranked lower in the leaderboard showed larger performance reductions, with errors concentrated mainly in the higher-order clinical dimensions D3–D5. This pattern suggests that clinically meaningful distinctions among contemporary MLLMs in rehabilitation video understanding may be better reflected by dimension-level reasoning profiles, particularly the gap between D1 semantic action recognition and D3–D5 higher-order clinical reasoning, than by small differences in adjacent leaderboard ranks.

3.3 Capability decomposition: quantifying the semantic-to-clinical reasoning gap

To deeply investigate the performance differences of the 26 state-of-the-art multimodal large language models across rehabilitation assessment tasks of different clinical depths, we systematically decomposed model capability using the ActionEval-6D multidimensional framework. The dimension-specific performance is summarized in Table 8.

First, the dimension-wise capability heatmap (Figure 3) re-

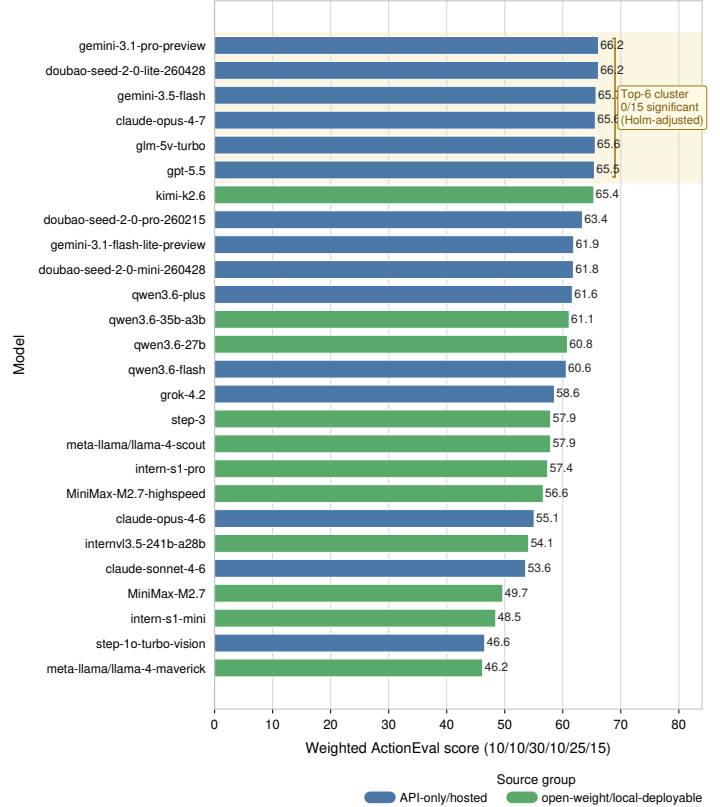


Fig. 2
Overall benchmark leaderboard. Cross-cohort weighted ActionEval composite scores of the 26 evaluated MLLMs. Blue bars indicate API-only or hosted models, and green bars indicate open-weight or locally deployable models. The shaded area marks the statistically indistinguishable top-performing cluster after Holm–Bonferroni correction.

vealed a systematic perception-to-reasoning performance cliff. Under the primary returned-response scoring rule, D1 semantic action recognition represented the strongest relative capability across the 26 evaluated models, with a cross-model mean F1-score of 91.1%. This indicates that current MLLMs generally performed well in recognizing the macroscopic movement category represented in the video, although the high absolute D1 score should also be interpreted in relation to the dimension-specific chance and select-all baselines reported in Table 7.

However, when the task shifted from macroscopic action recognition to higher-order clinical kinematic interpretation, model performance declined substantially. Under the same primary returned-response scoring rule, the cross-model mean F1-score decreased to 60.4% in D2 kinematic feature alignment. In the clinically demanding dimensions, the corresponding mean F1-scores were 45.2% for D3 pathological compensation identification, 49.2% for D4 spatiotemporal dynamics analysis, and 43.1% for D5 comprehensive movement quality and clinical decision-making. By contrast, D6 visual grounding and anti-hallucination verification reached a mean F1-score of 98.7%, although this high value reflected the ceiling effect of the constrained negative-probe design and was therefore interpreted jointly with the dedicated FP_{D6} endpoint.

The valid-response-only set-F1 decomposition further clari-

fied the error structure underlying the lower performance in the higher-order clinical dimensions (Supplementary Table C.19). Across the 26 evaluated models, the mean of the model-level item-macro recall estimates was 46.39% in D3 and 43.00% in D5. In D5, recall was lower than precision in all 26 models, and the mean predicted-set size was 0.976 compared with a mean expert ground-truth-set size of 1.008, indicating a small but consistent tendency to omit expert-labelled findings.

The all-item D3 decomposition required more careful interpretation because 292 of the 522 D3 items had an empty ground-truth set. Across all D3 items, the cross-model mean precision and recall were 45.61% and 46.39%, respectively, while the mean predicted-set size was 0.676 compared with a mean ground-truth-set size of 0.556. Therefore, the complete D3 item set did not show a simple pattern of recall being systematically lower than precision.

When the diagnostic analysis was restricted to the 230 D3 items containing at least one expert-labelled pathological-compensation option, cross-model mean recall was only 36.48%. Models selected an average of 0.742 options per item, compared with 1.261 expert-labelled options, and 45.39% of responses contained no selected compensation option. This omission-dominant pattern indicates that models frequently failed to identify expert-labelled pathological-compensation signs, providing quantitative support for the FM-2 pathological-compensation blind-spot pattern.

A reviewer-requested post hoc stratification based on the adjudicated gold set and its accompanying rationale further showed that the low D3 score was not attributable solely to internally unresolved items (Table 6). Across the 26 models, the 429 D3 items with a determinate, rationale-supported gold label yielded a mean F1-score of 43.0%, which was 12.5 percentage points lower than the 55.5% observed for the 93 items classified as internally unresolved under the archived gold-set/rationale classification. The difference was statistically significant in a two-sided paired Wilcoxon signed-rank test across the 26 model-level subset estimates ($P = 0.029$).

More importantly, when the analysis was restricted to the 230 determinate D3 items containing at least one expert-labelled pathological-compensation finding, the mean detection F1-score was only 33.8%, which was 11.4 percentage points below the all-item D3 mean of 45.2%. Thus, the D3 performance deficit persisted, and became more pronounced, when evaluation focused specifically on explicit pathological-compensation findings. The comparatively higher score for the internally unresolved subset should not be interpreted as indicating that those items were intrinsically easier, because it was partly influenced by correct empty-set predictions under the predefined scoring convention.

To further precisely quantify and diagnose this capability gap, we introduced the derived metric of clinical reasoning gap. The results are shown in Figure 4 and Table 8.

In the perception-versus-clinical-reasoning scatter plot (Figure 4A), the diagonal dashed line ($y = x$) represents perfect equivalence between low-level perception, namely D1, and higher-order reasoning, namely the mean score of D3–D5, indicating the absence of a capability gap. All 26 evaluated models

fell markedly below the equivalence line in the lower-right region of the plot. Although their D1 semantic action-recognition scores clustered in the relatively high range of 80%–95%, the mean D3–D5 clinical reasoning scores were concentrated in the lower range of 20%–55%. This systematic spatial displacement quantitatively demonstrates the technical bottleneck that models can “see the action” but cannot adequately “understand the pathology.” The high absolute D1 values should nevertheless be interpreted in conjunction with the dimension-specific chance and select-all baselines because label cardinality contributed partly to the observed score level.

In the clinical reasoning gap bar chart (Figure 4B) and Table 8, we quantitatively ranked the width of the performance cliff for each model. The results showed that the median reasoning gap across the 26 models was as high as 43.57 percentage points (pp). Intern-S1-Pro exhibited the smallest reasoning gap in the entire evaluation, at 30.51 pp. However, this was mainly attributable to its relatively low D1 perception score of 78.88%, which mathematically narrowed the performance cliff, rather than indicating that it had the strongest reasoning capability.

The chance-level and empirical human-reference benchmarks are summarized in Table 7. The random-subset and select-all baselines varied substantially across dimensions because of differences in ground-truth set size and label composition. In the clinically demanding dimensions, the mean individual-expert F1 was 78.74% for D3, 83.14% for D4, and 81.93% for D5, whereas the corresponding dimension-wise best-observed model scores were 58.30%, 64.17%, and 52.55%, respectively. Thus, the dimension-wise best model scores remained 20.44, 18.97, and 29.38 percentage points below the empirical human ceiling in D3, D4, and D5. By contrast, the human–model gaps were substantially smaller in D1 and D6, consistent with the relative saturation of basic action recognition and constrained negative-probe visual grounding. Human-reference scores were not aggregated into a composite score because cohort-specific expert estimates required to reproduce the model leaderboard’s 65% clinical and 35% functional weighting procedure were unavailable.

In contrast, even among the first-tier models, Gemini-3.1-Pro-Preview, which ranked first overall, still showed a clinical reasoning gap as large as 39.07 pp, with a D1 score of 94.25% but a higher-order reasoning mean of only 55.18%. The worst-performing model in this respect, Meta-Llama/Llama-4-Maverick, showed an even larger gap of 66.76 pp, with a D1 score of 92.57% but a reasoning mean that dropped to 25.81%. These quantitative findings indicate to rehabilitation clinicians that relying solely on macroscopic recognition accuracy to evaluate the reliability of large models may lead to serious safety misjudgement. Even when a model appears to understand the video, with D1 scores above 90%, it may still have an average cognitive blind spot exceeding 40% when providing clinical suggestions, as reflected by D3–D5 scores below 50%. This reveals a potential clinical safety risk driven by automation bias: clinicians may overtrust immature high-risk decision recommendations because of the model’s strong performance on comparatively simple recognition tasks.

To intuitively present the trajectory of capability collapse when models transition from perception to higher-order reasoning, we plotted the D1–D6 radar profile (Figure 5). This figure overlays the top eight high-performing models, shown as colored solid lines, with the overall average performance of the 26 models, shown as a red dashed shaded profile. The radar polygons exhibited a clear asymmetric centripetal contraction. In the two lower-difficulty wings, D1 and D6, all solid and dashed profiles largely overlapped and remained close to the saturated outer ring of 90%–100%. However, in the high-difficulty core region, D3, D4, and D5, the radar profiles contracted markedly inward, forming a sharp centripetal depression.

The consistent collapse of the radar polygons visually demonstrates that the performance differences and discriminative power of large models are determined primarily by their reasoning depth in the high-difficulty clinical dimensions D3–D5, rather than by low-level perception. For example, compared with the average model represented by the red dashed profile, the overall top-performing model Gemini-3.1-Pro-Preview showed the most prominent outward expansion in D3 pathological compensation identification, exceeding the mean by 12.47 pp. In contrast, in the D6 anti-hallucination safety test, because all models showed high certainty on negative probes, its advantage was only 1.25 pp. This multidimensional performance profile provides strong architectural guidance for future multimodal AI algorithms, suggesting that greater emphasis should be placed on temporal kinetic-chain modeling of pathophysiological features.

The composite model rankings were robust to alternative dimension-weighting assumptions (Supplementary Table C.17). Spearman’s rank correlations with the primary ranking were 0.964 under equal weighting and 0.999 under both lower-D6 and D6-excluded weighting. The corresponding top-five overlaps were 4/5, 5/5, and 5/5, while the maximum absolute rank displacements were 5, 1, and 1 positions, respectively. These results indicate that the principal comparative conclusions were not materially dependent on the prespecified weighting scheme, although equal weighting produced limited rank changes among individual models.

3.4 Cohort stratification: generalization from standardized functional movements to fine-grained clinical rehabilitation videos

To evaluate the ecological validity and cross-cohort generalization ability of MLLMs in real clinical environments, we systematically divided the dataset into two parallel subcohorts. Figure 6A and Figure 6B show the independent performance leaderboards for the functional biomechanics cohort, which mainly consisted of large-joint and highly visible gross-motor movements, and the clinical fine-grained rehabilitation cohort, which consisted of real-world patient-specific rehabilitation videos, respectively.

Overall, the two cohorts showed a polarized pattern of task difficulty from the algorithmic perspective. In the functional biomechanics cohort (Figure 6A), gross-motor actions with

clear visual boundaries were relatively favorable for the models, and seven models achieved scores above 70%. Among them, Doubao-Seed-2.0-Lite-260428 demonstrated the strongest performance in general large-joint trajectory tracking and physical-rule interpretation, ranking first in this cohort with a score of 75.72%.

In contrast, in the clinical fine-grained rehabilitation cohort (Figure 6B), when models faced real-world rehabilitation videos characterized by low-amplitude movements, patient-specific variability, and subtle clinical movement features, all models showed lower performance, and no model exceeded the 65% threshold. GPT-5.5 ranked first in the clinical cohort, with a weighted score of 63.07%.

Second, the polarized evaluation pattern was visualized using the slope chart of functional-to-clinical performance shift (Figure 7). When the evaluation domain shifted from the functional biomechanics cohort to the clinical rehabilitation cohort, the performance lines of most models showed a downward shift toward the clinical side. Statistical testing showed that the mean clinical performance drop across the 26 models reached 9.44 percentage points, with the mean functional cohort score being 64.68% compared with a mean clinical cohort score of 55.25% ($P < 0.001$).

It is important to note that this cross-cohort performance drop should not be interpreted as the isolated causal effect of pathological compensation or neurological impairment. The two cohorts differed not only in clinical status, but also in body region, movement scale, visual target size, data source, recording context, and task characteristics. Therefore, the 9.44 percentage-point drop is better interpreted as a combined domain-shift burden when transferring from standardized, high-visibility functional biomechanics videos to real-world fine-grained clinical rehabilitation videos.

Table 9 provides a quantitative summary of representative models with the most severe performance drops. Among the models with relatively high overall rankings, Gemini-3.1-Flash-Lite-Preview showed the largest cross-cohort performance loss, with a performance drop of 16.92 pp. This was followed by MiniMax-M2.7-Highspeed, with a drop of 15.66 pp; Kimi-K2.6, with a drop of 14.73 pp; and the overall top-ranked Gemini-3.1-Pro-Preview, with a drop of 14.53 pp.

These findings indicate that even high-performing MLLMs remain sensitive to distribution shifts between standardized functional movement videos and real-world fine-grained clinical rehabilitation videos. However, because cohort membership is confounded with movement scale, body region, data source, and participant status, these results should be interpreted as evidence of cross-domain generalization difficulty rather than as proof that pathological compensation alone caused the observed performance degradation.

In sharp contrast, at the bottom of Figure 7, a small number of low-performing models, such as Step-1o-Turbo-Vision, with an overall score of 46.56%, showed relatively flat or even slightly upward trajectories from the functional to clinical cohort. For example, Step-1o-Turbo-Vision scored 45.31% in the functional cohort and 47.23% in the clinical cohort, resulting in a negative performance drop of −1.92 pp. This seemingly

abnormal flat trajectory does not indicate stronger adaptability to clinical videos. Rather, it likely reflects a statistical floor effect: when a model already performs poorly across both cohorts, small differences between functional and clinical scores may reflect random fluctuation rather than meaningful domain adaptation. Therefore, the flat or slightly upward trajectories observed in several low-performing models should be interpreted cautiously and should not be used as evidence of superior clinical generalization.

3.5 Visual faithfulness (D6), safety boundary, and subgroup analysis

Analysis of visual faithfulness (D6). This study rigorously examined the anti-hallucination capability of the evaluated models under visually misleading conditions through the independent safety dimension D6, namely video-evidence consistency and anti-hallucination verification. Although as many as 17 models achieved excellent F1-scores of $\geq 99\%$ under the conventional D6 metric, showing an obvious ceiling effect, the true safety boundary among models became more apparent when we applied a more sensitive safety measure, namely the false-positive hallucination rate (FP_{D6}) (Figure 8).

The results showed that nine models produced at least one visually unsupported affirmative selection under the constrained D6 negative-probe format. *intern-s1-mini* showed the highest operational D6 false-positive visual-grounding rate, reaching 6.51% (95% CI: 4.70–8.96%). The other eight models, including *MiniMax-M2.7* and *Grok-4.2*, also exhibited varying degrees of false alarms.

To exclude the potential *ceiling-effect bias* of D6, whose overall scores were close to full marks, on the overall ranking, we further performed a sensitivity re-analysis after removing D6 from the weighting formula entirely (see Supplementary Appendix C). The results showed that even when the D6 safety dimension, which accounted for 15% of the total weight, was completely excluded from the weighted formula, the relative leading arrangement of the top-performing model cluster (Top-6) remained highly stable, and the *clinical reasoning cliff* from D1 to D3–D5 remained statistically significant. This robustness evidence indicates that the capability discontinuity revealed in this study was not an artifact of D6 weighting or a ceiling-effect bias, but reflected a broader difficulty of current multimodal models in higher-order clinical movement reasoning.

Subgroup differences by technical route and deployment attributes. To provide decision support for future private deployment in real medical network environments, we conducted structured subgroup analyses of the evaluated models (descriptive metrics in Figure 9, significance testing in Table 10).

1. API-only/hosted models vs. open-weight/local-deployable models.

API-hosted models attained a numerically higher global weighted composite score than open-weight/local-deployable models (61.19% vs. 55.96%; $U = 130.00$; unadjusted $P =$

0.0147). However, this difference did not remain statistically significant after correction across the 10 comparisons reported in Table 10 (BH-FDR-adjusted $P = 0.0735$). The only deployment-based difference that remained statistically significant after correction was conventional D6 anti-hallucination F1 (99.59% vs. 97.44%; $U = 135.50$; unadjusted $P = 0.0045$; BH-FDR-adjusted $P = 0.0450$). By contrast, the dedicated FP_{D6} hallucination rate did not differ significantly between the two deployment groups after correction (0.14% vs. 0.92%; $U = 52.00$; unadjusted $P = 0.0661$; BH-FDR-adjusted $P = 0.1653$). No statistically significant differences were detected in D3 pathological compensation (BH-FDR-adjusted $P = 0.1736$) or D5 clinical decision-making (BH-FDR-adjusted $P = 0.1653$). These findings leave open the potential feasibility of locally deployable models for selected core reasoning tasks in privacy-constrained medical environments, although the absence of statistically significant differences should not be interpreted as evidence of equivalence. Overall, the corrected results indicate a descriptive advantage for hosted models in the composite score and a metric-specific difference in conventional D6 F1, rather than broad evidence of superior clinical reasoning or hallucination safety.

2. General-purpose multimodal models vs. vision-specialized models. In the specialization-based subgroup analysis, general-purpose multimodal models showed numerically higher weighted composite scores and higher mean performance in the core clinical reasoning dimensions D3–D5, whereas vision-specialized models showed a descriptively lower false-positive hallucination rate in D6. However, because the number of vision-specialized models was small ($n = 4$) and most between-group comparisons did not reach statistical significance, this finding should be interpreted as exploratory rather than confirmatory.

This exploratory pattern is consistent with, but does not prove, the hypothesis that visual encoding alone may be insufficient for higher-order rehabilitation reasoning. Stronger visual feature alignment may help improve basic visual grounding and reduce scene- or object-level hallucinations, but such an advantage did not translate into statistically significant superiority in pathological compensation identification or comprehensive clinical decision-making in the present benchmark. The numerical advantage of general-purpose multimodal models may also partly reflect language priors, medical-text exposure, or MCQ option-pattern matching, which cannot be fully excluded in the current study design.

Stability of model reasoning across multi-scale samples and downside risk. Beyond mean scores, the stability of model performance across 522 highly diverse real-world video samples is equally important for practical clinical use. Therefore, we further conducted a video-level robustness analysis. The boxplot of video-level score distributions (Figure 10) showed that the top-performing model cluster (Top-6) generally achieved higher median scores and extended further into the upper performance range across the 522 video recordings contributed by 48 participants. However, the top-performing models did not exhibit uniformly narrower score distributions. They showed

slightly larger or comparable standard deviations (mean SD, 25.61 vs. 23.32 for the middle- and lower-ranking models) and larger interquartile ranges (49.75% vs. 38.36%). This broader dispersion primarily reflected their higher upper tails, with scores extending into the 90%–100% range, rather than a uniformly more concentrated distribution. Accordingly, standard deviations and interquartile ranges alone were not interpreted as evidence of superior performance stability. Lower-tail stability and downside risk were evaluated separately using the cumulative distribution functions and low-score rates described below.

To characterize lower-tail performance, we further introduced the cumulative distribution function (CDF; Figure 11) and calculated the probability that the video-level weighted score fell below 50 points. The 50-point value was used as an exploratory low-score reference cutoff. Its selection was heuristic rather than outcome-derived and should not be interpreted as an externally validated clinical usability threshold.

The CDF trajectories (Figure 11) showed clear performance stratification. The curves of the top-performing cluster were generally shifted toward the lower right and exhibited steeper slopes.

When evaluated against this exploratory 50-point reference cutoff, the cumulative proportion of low-score cases in the top-performing cluster was relatively small, with a median low-score rate of 31.4%. This suggests that, relative to the other evaluated models, the top-performing models were less likely to exhibit severe performance degradation on complex or ambiguous videos. In contrast, a lower-performing model, Step-1o-Turbo-Vision, showed a low-score rate of 42.2%. These proportions provide a descriptive comparison of lower-tail performance rather than evidence that models scoring above 50 points are clinically usable. Taken together, these distributions indicate that mean performance alone may not fully characterize model stability across heterogeneous rehabilitation videos. Lower-tail performance may therefore serve as a complementary descriptive consideration, rather than a validated clinical safety criterion, when comparing models.

3.6 Qualitative thematic analysis: attribution of clinical reasoning failure mechanisms

The automated retrieval module identified 1,232 candidate high-conflict error records using the prespecified criteria described in Section 2.5, including frequently misclassified items, errors shared by the top-performing model cluster, marked D1-to-D3/D5 performance discrepancies, and D6 false-positive visual-grounding events. The module served only to retrieve and organize candidate cases; it did not generate qualitative codes, assign failure-mode labels, or determine the final thematic categories.

Subsequent human-led, Grounded Theory-informed coding by clinical and algorithmic reviewers identified five recurrent and clinically interpretable failure modes: perception–reasoning disconnection (FM-1), pathological compensation blind spots (FM-2), language priors overriding visual evidence

(FM-3), temporal integration breakdown (FM-4), and kinetic-chain reasoning deficiency (FM-5). These categories describe recurrent error patterns observed during the transition from basic visual perception to higher-order clinical reasoning and should not be interpreted as direct evidence of the models’ internal cognitive processes.

FM-1: Perception–reasoning disconnection. The model accurately matched the macroscopic semantic category of the movement, but failed to capture subtle abnormal joint compensation.

FM-2: Pathological compensation blind spots. The model misclassified clinically typical compensatory erroneous movements as normal variations in healthy motor execution.

This qualitative pattern was consistent with the quantitative set-F1 decomposition. Among the 230 D3 items containing at least one expert-labelled pathological-compensation option, the cross-model mean recall was only 36.48%. Models selected an average of 0.742 options per item, compared with 1.261 options in the expert ground-truth sets, and 45.39% of responses contained no selected compensation option (Supplementary Table C.19). This omission-dominant response profile indicates that models frequently failed to identify expert-labelled compensatory signs and is behaviorally consistent with the FM-2 normalization-bias pattern. However, these quantitative findings should not be interpreted as direct evidence of the models’ internal decision mechanisms.

FM-3: Language priors overriding visual evidence. Driven by patterns learned from text-based training data, such as formulaic expressions from standard rehabilitation guidelines, the model generated diagnostic or scene descriptions that contradicted the visual evidence in the corresponding video. This pattern was frequently observed in D6 false-positive visual-grounding events, although the qualitative analysis could not directly establish the model’s internal causal mechanism.

FM-4: Temporal integration breakdown. The model failed to integrate movement evidence consistently across temporally ordered frames and consequently missed dynamic abnormalities that depended strongly on temporal information, including tremor, pauses, and freezing of gait.

FM-5: Kinetic-chain reasoning deficiency. The model responses frequently failed to reflect the core-to-distal force-transmission relationships used by rehabilitation clinicians when interpreting coordinated movement. Instead, the responses often appeared to rely on superficial matching between medical terminology in the question and semantically related answer options.

4. Discussion

4.1 Principal findings

This study developed and applied *ActionEval-MedBench*, a multidimensional video benchmarking framework for evaluating multimodal large language models (MLLMs) in motor rehabilitation and clinical biomechanics. By integrating 522

de-identified rehabilitation and functional biomechanics videos with 3,132 clinically designed multi-select MCQ items, we systematically audited 26 contemporary MLLMs across six clinically meaningful dimensions [34]. The central finding of this study is that current MLLMs demonstrate a clear dissociation between strong low-level visual semantic recognition and limited higher-order clinical movement reasoning.

On the one hand, most models achieved high performance in D1 semantic action recognition and relatively strong performance in D6 visual-grounding negative probes, suggesting that current MLLMs can often identify macroscopic action categories and avoid certain scene-level false-positive hallucinations. On the other hand, their performance declined substantially in D3 pathological compensation identification, D4 spatiotemporal dynamics analysis, and D5 comprehensive clinical decision-making. This pattern indicates that the current generation of MLLMs can often recognize what action is being performed, but they remain limited in judging how the action is performed and what its biomechanical or clinical significance may be.

This finding directly addresses the central research question of this study: whether general multimodal video understanding can be translated into clinically reliable rehabilitation movement analysis [35]. Our results suggest that the answer is currently conditional rather than affirmative. MLLMs show considerable potential as structured screening and documentation tools, but they are not yet sufficiently reliable for independent clinical interpretation of subtle pathological compensation, abnormal kinetic-chain patterns, or dynamic motor-control deficits.

Our findings complement the recent multidimensional evaluation by Cao et al. [21], who compared rehabilitation plans generated by five LLMs with a standardized human-expert benchmark across 48 authentic clinical cases from six rehabilitation subspecialties. Their outputs were blindly evaluated by six senior physiatrists across clinical applicability and safety, scientific rigor, individualization, and clarity. Several top-performing LLMs received higher ratings than the standardized human benchmark for knowledge-intensive rehabilitation-plan generation, whereas the qualitative analysis identified persistent human strengths in strategic pathway design, clinical prioritization, and humanistic care.

The two studies evaluate complementary but distinct stages of the rehabilitation-AI pipeline. Cao et al. primarily examined downstream clinical synthesis: whether an LLM could formulate a rehabilitation plan after the relevant clinical case information had already been provided. In contrast, ActionEval-MedBench examines the upstream visual-perception and evidence-extraction stage: whether an MLLM can reliably identify action semantics, subtle pathological compensation, temporal motor abnormalities, and visually unsupported assertions directly from dynamic movement videos. Our results suggest that strong language-based plan-generation performance should not be assumed to imply reliable video-grounded clinical reasoning. A model may generate a fluent and apparently comprehensive rehabilitation plan while still failing to detect the compensatory or temporal abnormalities on which

that plan should be based.

Because the two studies differ substantially in input modality, task structure, evaluation unit, scoring scale, and clinical endpoint, their numerical results should not be compared directly. Instead, they jointly suggest that trustworthy rehabilitation AI requires competence at both stages: accurate extraction of clinically meaningful evidence from patient movement and safe synthesis of that evidence into individualized recommendations. This combined evidence further supports a human-AI collaboration model in which MLLMs assist with structured visual observation and preliminary plan drafting, while rehabilitation professionals retain responsibility for verifying the visual evidence, interpreting causal mechanisms, prioritizing clinical problems, and making final treatment decisions.

4.2 From action recognition to clinical movement reasoning

A major contribution of this study is the quantitative characterization of the semantic-to-clinical reasoning gap. Under the primary returned-response scoring rule, the evaluated models achieved a cross-model mean F1-score of 91.1% in D1 semantic action recognition, indicating that macroscopic movement categories were generally well captured. However, the corresponding mean F1-scores were only 45.2%, 49.2%, and 43.1% in D3, D4, and D5, respectively, and the median clinical reasoning gap reached 43.57 percentage points. This gap was observed even among top-performing models, including Gemini-3.1-Pro-Preview, which ranked first overall but still showed a 39.07 percentage-point gap between D1 and the mean of D3–D5.

This finding has important implications for the evaluation of medical multimodal AI. In rehabilitation, the clinically relevant question is not simply whether a model can identify that a patient is raising an arm, performing a squat, or completing a reaching task. Rather, the key question is whether the model can detect shoulder hiking, trunk compensation, abnormal rhythm, tremor, reduced smoothness, unsafe posture, or task-specific motor inefficiency. These features are often low-amplitude, temporally distributed, and patient-specific. They require integration of visual observation, biomechanical knowledge, and clinical reasoning.

The observed reasoning gap suggests that current MLLMs may rely heavily on semantic pattern recognition and textual priors rather than truly modeling the dynamic pathophysiological mechanisms underlying abnormal movement [36]. This interpretation is supported by the qualitative failure-mode analysis, which identified perception–reasoning disconnection, pathological compensation blind spots, language-prior hallucination, temporal integration breakdown, and kinetic-chain logic deficiency as recurring error mechanisms.

The set-F1 decomposition provided additional quantitative support for the FM-2 pathological-compensation blind-spot pattern (Supplementary Table C.19). Because 292 of the 522 D3 items had an empty ground-truth set, precision and recall calculated across all D3 items were similar (45.61% and 46.39%, respectively) and should not be interpreted as evi-

dence that recall was systematically lower than precision. However, among the 230 D3 items containing at least one expert-labelled pathological-compensation option, cross-model mean recall was only 36.48%. Models selected an average of 0.742 options per item, compared with 1.261 expert-labelled options, and 45.39% of responses contained no selected compensation option.

This omission-dominant response profile is behaviorally consistent with a normalization tendency in which clinically relevant compensatory features are omitted or treated as absent. Nevertheless, low recall and empty predictions do not independently establish the models' internal decision mechanisms. The interpretation of normalization bias is therefore supported jointly by the quantitative omission pattern and the manually reviewed FM-2 cases and should be understood as a behavioral failure pattern rather than direct evidence of an internal cognitive process.

Collectively, these findings suggest that future rehabilitation-oriented MLLMs should not merely improve visual feature extraction, but should incorporate domain-specific causal reasoning, temporal motor-control modeling, and biomechanical prior knowledge.

4.3 Visual grounding in safety-critical contexts

In the clinical deployment of medical AI, conventional performance metrics such as average accuracy or F1-score often fail to fully capture long-tail risks in safety-critical scenarios. This issue is particularly important in rehabilitation video analysis, where an apparently minor visual error may lead to an inappropriate functional judgement or unsafe exercise prescription. By introducing D6 as an independent safety dimension, this study constructed a negative-probe-based assessment of visual-evidence consistency for clinical video understanding.

Our findings show that although many models achieved near-ceiling D6 F1-scores, the D6 false-positive analysis revealed meaningful differences in visual-evidence consistency. Nine models produced affirmative selections for visual elements that experts confirmed were absent, and the highest FP_{D6} reached 6.51%. Because D6 was implemented using a constrained select-all-that-apply MCQ format, that is, a forced-choice response setting, these events are more precisely interpreted as visually unsupported affirmative selections or false-positive visual-grounding errors. They do not, by themselves, constitute definitive evidence of hallucination in the stronger generative sense unless the accompanying free-text rationale explicitly fabricates the absent feature. We therefore interpret FP_{D6} conservatively as an operational visual-grounding safety indicator. Stronger claims of factual hallucination are reserved for qualitative cases involving explicit textual fabrication, such as the FM-3 example in which the model described a nonexistent blue yoga mat and its material. Nevertheless, if such unsupported selections are propagated into narrative reports or clinical recommendations, they may still create unnecessary concern, inappropriate exercise modification, or unsafe rehabilitation decisions [37].

Therefore, FP_{D6} should be understood as an operational safety-audit indicator rather than as a direct measure of generative hallucination or an independently validated clinical safety boundary. Before MLLMs are used in rehabilitation workflows, especially in semi-automated screening or telerehabilitation scenarios, they should be required to demonstrate not only high average performance but also low D6 false-positive rates under negative visual evidence. Future medical multi-modal systems should move beyond an accuracy-centered evaluation paradigm and incorporate uncertainty estimation, abstention mechanisms, and escalation protocols [38]. When the visual evidence is ambiguous or insufficient, a clinically safe model should be able to withhold unsupported conclusions or request expert review.

At the same time, the low FP_{D6} observed in the top-performing cluster provides an important complementary insight. It suggests that leading MLLMs were not simply guessing MCQ answers from language priors in all cases, but were often anchored to visible video features. This supports the feasibility of using visual-grounding probes as a practical audit tool for differentiating visually grounded reasoning from text-driven hallucination. However, D6 in the present benchmark targeted scene- and object-level negative probes; it should not be overinterpreted as a complete measure of all clinical kinematic hallucinations. Higher-order hallucinations involving fabricated pathological compensation, movement rhythm, or kinetic-chain abnormalities require separate assessment through D1–D5 performance and qualitative failure-mode analysis.

4.4 Clinical implications for deployment: toward a human-in-the-loop paradigm

The subgroup analyses provide practical evidence for future deployment strategies in digital rehabilitation. As shown in Table 10, API-only or hosted models attained a numerically higher overall weighted score than open-weight or locally deployable models; however, this difference did not remain statistically significant after correction across the 10 subgroup comparisons (BH-FDR-adjusted $P = 0.0735$). The only deployment-based difference that remained statistically significant after correction was conventional D6 anti-hallucination F1 (BH-FDR-adjusted $P = 0.0450$). However, the corresponding difference in the dedicated FP_{D6} hallucination-rate endpoint was not statistically significant (BH-FDR-adjusted $P = 0.1653$), indicating that the D6 finding was metric-specific and should not be interpreted as evidence of broad safety superiority. No statistically significant differences were detected in the core higher-order reasoning dimensions most directly related to rehabilitation decision quality, including D3 pathological compensation (BH-FDR-adjusted $P = 0.1736$) and D5 clinical decision-making (BH-FDR-adjusted $P = 0.1653$).

These findings have important translational implications. In real clinical environments, patient rehabilitation videos are highly sensitive data. Hospitals and rehabilitation institutions may face strict privacy, cybersecurity, and regulatory con-

straints, such as HIPAA, GDPR, or equivalent local medical data governance requirements [39]. Under such conditions, locally deployable models may offer a practical and privacy-preserving route for clinical implementation. Although these models showed numerically lower overall performance in the present sample, the corrected analysis did not detect deployment-based differences in D3 or D5. This absence of statistically significant differences does not establish equivalence between hosted and locally deployable models, but it indicates that local deployment should not be dismissed as technically infeasible solely on the basis of deployment category. Further validation using larger model samples, clinically defined non-inferiority margins, and dedicated safety endpoints is required before deployment-route equivalence can be established.

Nevertheless, the results of this study do not support the use of current MLLMs as autonomous diagnostic tools. Instead, they support a human-in-the-loop collaboration paradigm [40]. In this framework, AI systems should act as preliminary screeners, structured observers, and documentation assistants. Their strengths in D1 and D2 can be used to classify large volumes of rehabilitation videos, extract visible movement patterns, summarize task execution, and generate standardized records for clinician review. Human experts, in contrast, should remain responsible for high-order clinical interpretation, especially in D3 and D5, where pathological mechanisms, functional limitation, safety risk, and individualized treatment planning must be integrated.

This division of roles is consistent with the broader direction of responsible medical AI. MLLMs may reduce the workload associated with repetitive visual observation and improve the standardization of preliminary assessments across institutions. However, rehabilitation physicians and therapists remain indispensable as clinical strategists. They must determine whether a movement abnormality reflects pain avoidance, muscle weakness, spasticity, motor planning deficits, poor coordination, or compensatory behavior. They must also decide whether a patient should progress, regress, modify, or stop a particular exercise. These decisions require contextual clinical knowledge, longitudinal understanding of patient recovery, and ethical responsibility that current MLLMs cannot independently provide.

4.5 Why vision-specialized models are not sufficient for clinical rehabilitation reasoning

Another important finding is that vision-specialized models did not consistently outperform general-purpose multimodal models in the overall weighted score or in the core clinical reasoning dimensions D3–D5. Their relative advantage was mainly observed in D6, where stricter visual feature alignment may have helped reduce false-positive hallucinations. This result suggests that stronger visual encoders alone are not sufficient to solve clinical rehabilitation reasoning.

Clinical movement analysis is not a pure perception problem. It requires causal interpretation of dynamic motor behavior. For example, a forward arm elevation movement may appear visually complete, but the clinical interpretation depends

on whether the movement is achieved through normal glenohumeral control, scapular hiking, trunk extension, compensatory momentum, or altered timing. These distinctions require integration of temporal dynamics, kinetic-chain reasoning, and rehabilitation-specific pathophysiological knowledge. A model that simply extracts more accurate visual features may still fail if it cannot map those features onto clinically meaningful mechanisms [41].

Therefore, future model development should focus on the integration of three components: high-resolution visual perception, temporal biomechanical modeling, and domain-specific clinical reasoning. Potential directions include incorporating rehabilitation-specific video pretraining, explicit modeling of joint and trunk coordination, structured reasoning over kinetic chains, and multimodal alignment with expert-annotated pathological compensation patterns. In addition, models should be encouraged to provide uncertainty-aware outputs rather than overconfident categorical judgements when evidence is limited.

4.6 Implications for telerehabilitation and scalable rehabilitation assessment

The results of this study also have implications for telerehabilitation. Remote rehabilitation systems increasingly rely on patient-recorded videos, home-based exercise monitoring, and asynchronous clinician review. In these settings, scalable video triage and structured movement assessment are highly valuable. MLLMs may help identify whether the correct exercise was attempted, whether the movement was completed, whether gross posture or task execution appears abnormal, and whether the video requires urgent clinician review.

However, the same remote setting also amplifies risk. Home videos may vary substantially in camera angle, lighting, occlusion, background, clothing, and task instruction quality. Fine-grained compensatory patterns may be difficult to observe, and patients may not have immediate professional supervision. The performance degradation observed in the clinical fine-grained rehabilitation cohort highlights that MLLMs are vulnerable when facing subtle, patient-specific abnormalities. Therefore, MLLM-assisted telerehabilitation should be deployed as a triage and documentation layer rather than as a fully automated assessment system.

A clinically realistic workflow may include three steps. First, the model performs structured preliminary analysis of uploaded rehabilitation videos, including action category, visible range, gross posture, and possible abnormalities. Second, the system flags uncertain, low-confidence, high-risk, or visually inconsistent cases for clinician review. Third, the rehabilitation specialist integrates model output with medical history, physical examination, patient-reported symptoms, and longitudinal recovery goals before making therapeutic decisions. Such a workflow may improve efficiency while preserving clinical accountability.

4.7 Limitations and Future Directions

Although *ActionEval-MedBench* provides a rigorous, standardized evaluation framework for multimodal rehabilitation AI, several critical limitations remain to be addressed in future research.

Format-Induced Clinical Gaps. The current MCQ-based format ensures objectivity, scalability, and reproducibility across large-scale benchmarking; however, it inherently fails to simulate the open-ended, multi-turn clinical reasoning that characterizes real-world rehabilitation. In clinical practice, physicians and physical therapists do not operate in a vacuum of static choices. They continuously formulate hypotheses, ask iterative follow-up questions, reassess movements from alternative viewpoints, perform manual palpation, evaluate resistance or pain levels, and dynamically integrate patient-reported symptoms. These interactive, multi-modal elements are not captured by the current closed-ended design, presenting an operational gap between bench performance and bedside utility.

Dataset Breadth and Bio-Kinematic Constraints. The video dataset primarily focuses on fine-grained upper-limb rehabilitation and general functional biomechanical tasks. While this scope effectively captures subtle clinical motor impairments, current general MLLMs and vision-specialized models frequently struggle to decode complex movements due to a lack of explicit **3D skeletal dynamic constraints**. Without pre-integrated kinematic-chain logic, models often fail to “see through” two-dimensional pixel representations to detect subtle **kinematic-chain restrictions** [42], such as hidden scapular elevation or trunk compensation. Future iterations must expand the asset repository to include complex gait analysis, balance disorders, spinal cord injury, pediatric rehabilitation, neurodegenerative diseases, and post-operative musculoskeletal syndromes to comprehensively validate external validity across broader rehabilitation subspecialties.

Limited participant-level clinical characterization.. Although a non-identifying recording-to-participant grouping variable was available for participant-level cluster-robust bootstrapping, detailed participant-level demographic and clinical information was not retained in the deeply de-identified retrospective dataset. The 522 recordings were contributed by 48 participants, including 28 participants in the clinical rehabilitation cohort and 20 participants in the functional biomechanics cohort. The 328 clinical rehabilitation recordings should therefore not be interpreted as representing 328 independent patients. Apart from cohort membership and the anonymous grouping information required to identify repeated recordings from the same participant, no data were available on age, sex, individual diagnosis, disease duration, treatment history, or standardized impairment- or disease-severity measures. Consequently, we could not evaluate whether model performance varied across demographic groups, diagnostic categories, or levels of clinical severity. This limitation restricts the assessment of cohort representativeness and participant-level subgroup generalizability and should be addressed in future prospectively constructed datasets with richer, privacy-preserving participant-level clinical characterization.

Environmental and Robustness Confounders. Although all video assets underwent rigorous de-identification and the ground-truth labels were established via a dual-track Delphi-style consensus, the data remains bounded by single-center recording conditions, camera perspectives, and clinical sources. Current benchmarks cannot fully isolate domain-shift effects caused by environmental variability. Future studies should deploy multi-center, multi-device frameworks to systematically stress-test model robustness against real-world confounders, including variable frame rates, sub-optimal lighting, **occlusion-driven compensatory alignments**, and **perspective distortions** induced by non-orthogonal camera angles.

Heterogeneous temporal-input preprocessing.. Although all source videos included in the final analytical dataset were encoded at 30 frames/s, the effective number, temporal density, and spatial representation of frames available to each model depended on its visual-input route. For native-video interfaces, video decoding and temporal sampling were managed by the corresponding provider, and some proprietary APIs did not expose the realized frame indices or internal preprocessing logs. Consequently, the exact temporal evidence processed by these models could not always be independently recovered. For the nine models evaluated through non-native-video routes, visual inputs were generated using a client-side contact-sheet pipeline with maximum frame budgets of either 96 or 240 frames. Although the selected frames were uniformly distributed across the complete clip duration, the fixed frame budgets produced duration-dependent effective sampling rates. In addition, each selected frame was resized to 320×180 pixels and combined with up to 11 other frames in a 4×3 contact-sheet image. This representation may reduce sensitivity to subtle joint displacement, brief tremor, short pauses, rapid movement transitions, and other temporally localized or spatially fine-grained abnormalities. Accordingly, observed differences in D4 spatiotemporal-dynamics performance, and potentially in fine-grained D3 pathological-compensation identification, may partly reflect variation in temporal sampling density, frame resolution, and input representation rather than model reasoning capability alone. The present benchmark should therefore be interpreted as comparing the end-to-end model systems available through their implemented input routes, rather than as isolating architectural differences under an identical temporal and spatial information budget. Future evaluations should standardize the effective temporal-information budget, record realized frame indices, and separately compare native-video, frame-sequence, and contact-sheet inference pipelines. Model-family-specific input routes and sampling configurations are reported in Supplementary Table B.14. Although all models were evaluated through programmatic APIs, some providers did not expose complete backend versioning, decoding settings, realized frame indices, or internal media-processing parameters. The benchmark should therefore be interpreted as a comparison of the end-to-end API systems available during the evaluation period rather than as an architecture-controlled comparison under fully identical inference conditions.

Expert-defined dimension weighting. The dimension weights constitute an expert-defined benchmark utility function rather than an externally validated clinical scoring scale. Although the weights were prespecified through multidisciplinary clinical-priority judgement and the main conclusions were examined under alternative weighting schemes, different clinical settings or stakeholder groups may assign different priorities to perceptual accuracy, compensation recognition, temporal reasoning, comprehensive clinical judgement, and hallucination safety. Future studies should validate or recalibrate these weights using broader stakeholder panels and prospectively collected clinical outcomes.

Interpretation and Granularity of D6 False Positives. Because D6 used a constrained select-all-that-apply MCQ format, its quantitative false-positive events represent visually unsupported affirmative selections under a constrained-response, or forced-choice, setting rather than spontaneous open-ended generation. Such selections should not automatically be equated with hallucinations in the stronger generative sense. Definitive claims of factual hallucination would require corroborating evidence that the model explicitly fabricated an absent feature in its free-text rationale, as illustrated by the qualitative FM-3 case. In addition, the current D6 probes primarily targeted scene- and object-level visual grounding and did not capture the full spectrum of clinically important errors involving pathological compensation, movement trajectories, or temporal motor abnormalities. Future benchmarks should combine constrained negative probes with systematic analysis of free-text rationales and higher-order kinematic evidence.

Heuristic 50-Point Reference Cutoff. The 50-point cutoff used to summarize lower-tail performance in Figures 10 and 11 was selected as an intuitive midpoint of the 0–100 scoring range and is therefore heuristic and necessarily arbitrary. It was not derived from patient outcomes, an expert-defined minimum acceptable performance level, prospective clinical validation, or regulatory criteria. Accordingly, the proportion of videos scoring below 50 points should be interpreted only as a descriptive summary of lower-tail performance rather than as evidence that a model is or is not clinically usable. Future studies should establish outcome-linked thresholds and examine the robustness of conclusions across multiple candidate cutoffs.

Temporal Longevity of Rankings. This study represents a capability snapshot within a rapidly shifting ecosystem of foundation model development. As medical LLMs and vision-language architectures evolve, individual model rankings will inevitably shift. To prevent benchmarking obsolescence, future work must establish a continuous, dynamic evaluation platform capable of tracking the relationship between architectural scaling, open-source progress, and clinical reasoning proficiency over time.

Translational and Intervention-Oriented Validation. Finally, future research must transition from passive benchmarking to intervention-oriented validation. Prospective clinical trials are

urgently needed to investigate whether MLLM integration into rehabilitation workflows delivers tangible clinical efficacy [43]. Specifically, future work should evaluate whether these models can safely reduce clinician documentation workload, improve inter-rater diagnostic consistency, accelerate clinical reasoning formulation, and enhance patient compliance without compromising patient safety or therapeutic outcomes.

4.8 Summary of the discussion

In summary, this study shows that current MLLMs have meaningful potential for structured preliminary rehabilitation video assessment, but their clinical use must be constrained by clearly defined safety boundaries. Their strengths lie in macroscopic action recognition, structured response generation, and preliminary visual screening. Their weaknesses lie in subtle pathological compensation detection, spatiotemporal motor-control interpretation, kinetic-chain reasoning, and clinically consequential decision-making. Therefore, the most appropriate near-term role of MLLMs in rehabilitation is not autonomous diagnosis, but expert-supervised assistance within a human-in-the-loop workflow.

The broader implication is that the next generation of rehabilitation AI should not be optimized solely for higher average benchmark scores. Instead, it should be designed around clinical interpretability, visual grounding, uncertainty awareness, patient safety, and collaborative integration with rehabilitation professionals [11]. *ActionEval-MedBench* provides a practical step toward this goal by transforming expert-dependent rehabilitation movement analysis into a quantifiable, auditable, and clinically interpretable evaluation task.

5. Conclusions

This study developed *ActionEval-MedBench*, a multidimensional video-based benchmark for systematically evaluating MLLMs in motor rehabilitation and clinical biomechanics. Across 26 contemporary models, the results revealed a clear gap between basic visual action recognition and higher-order clinical reasoning, particularly in pathological compensation identification, spatiotemporal dynamics analysis, and rehabilitation-relevant decision-making. Although current MLLMs show promise for structured preliminary screening and standardized movement description, their limited ability to model subtle kinematic abnormalities and clinical movement mechanisms precludes their use as independent diagnostic or therapeutic decision-making tools.

These findings highlight three priorities for trustworthy rehabilitation AI: stronger temporal and kinetic-chain modeling, mandatory visual-grounding safety checks using negative evidence probes, and human-in-the-loop deployment in which AI supports initial screening and structured reporting while clinicians retain responsibility for final interpretation and treatment decisions. *ActionEval-MedBench* provides a standardized empirical basis for auditing this transition from general video

recognition toward safe, interpretable, and clinically grounded rehabilitation reasoning.

Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CI	Confidence Interval
CV	Computer Vision
D1–D6	The six dimensions of the ActionEval framework
D1	Semantic Action Recognition
D2	Kinematic Feature Alignment
D3	Pathological Compensation Identification
D4	Spatiotemporal Dynamics Analysis
D5	Comprehensive Movement Quality and Clinical Decision-Making
D6	Visual Grounding and Anti-Hallucination Verification
EM	Exact Match
FM	Failure Mode
FP_{D6}	D6 False-Positive Hallucination Rate
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
MCQ	Multiple-Choice Question
MLLM	Multimodal Large Language Model
MLLMs	Multimodal Large Language Models
pp	Percentage Points

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki. The clinical rehabilitation video cohort was reviewed and approved by the Ethics Committee of the First Affiliated Hospital of the University of South China (Approval No. 2021KS-KF-13-01), which waived the requirement for written informed consent to participate because this study involved retrospective analysis of pre-existing de-identified rehabilitation videos and did not involve any clinical intervention.

All clinical videos were de-identified before analysis in accordance with the approved protocol. Direct identifiers were removed before data transmission or analysis. The approved protocol explicitly covered the transmission of the minimum necessary de-identified clinical movement video content to third-party commercial API providers for automated model inference. The functional biomechanics videos were obtained from the publicly available MM-Fit dataset and used in accordance with its terms of use.

All clinical experts participating in the consensus annotation and arbitration process were informed of the study purpose, annotation procedures, and data-usage policies before participation.

Consent for publication

No participant images, names, direct identifiers, or identifiable personal information are included in the main manuscript or supplementary materials.

Data and code availability

The custom code and reproducibility materials used for video preprocessing, visual-input preparation, model inference, response parsing, statistical analysis, and figure generation are available on GitHub at https://github.com/PingYe616/Benchmark_data and archived in Zenodo at <https://doi.org/10.5281/zenodo.22133038>. The archived materials include the source code, prompt templates, ActionEval-6D question set, response-parsing scripts, statistical-analysis scripts, and a README describing the software environment and execution instructions. Clinical rehabilitation videos are not publicly available because of privacy and ethics restrictions. The functional biomechanics videos are publicly available from the MM-Fit dataset [19].

Competing interests

The authors declare that they have no competing interests.

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Author contributions

PY and YL contributed equally to this work and should be considered co-first authors. PY, YL, and JZ conceived and designed the study. PY and YL developed the ActionEval-6D benchmarking protocol, designed the MCQ assessment items, and drafted the manuscript. YL and JL were responsible for software development, API-inference engineering, and automated response parsing. MQ and XZ participated in clinical video collection, de-identification, and data curation. MQ and XZ performed the independent dual-track Delphi consensus labeling, with JZ serving as the chief clinical arbitrator. JL and TZ performed the statistical analysis and visualized the data. TZ and JZ supervised the project, acquired the funding, and critically revised the manuscript for important intellectual content. All authors read and approved the final manuscript.

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Appendix A. ActionEval-6D multiple-choice question corpus

Appendix A.1. Examples of question-bank samples

This appendix presents representative cleaned sample structures from the ActionEval-6D benchmark. The following examples were selected from the evaluation pipeline and illustrate the annotation format and expert-consensus

ground-truth (GT) labels across different rehabilitation cohorts and ActionEval dimensions.

The human-readable cohort/action descriptions shown in the examples below were added only after model inference for supplementary reporting and should not be interpreted as identifiers or metadata presented to the evaluated models. During inference, videos were represented using opaque sample identifiers containing no action-category information. Local source filenames, directory paths, action indices, and ground-truth action names were used only for study-side data indexing and were not included in the model-visible prompt, attachment caption, uploaded-file metadata, or other textual input. Across the complete benchmark, 0 of the 522 model-visible video inputs contained a filename-derived or action-name cue.

Clinical rehabilitation sample example

Cohort/action type: Clinical / right-hand finger opposition

Sample identifier: clinical::sample_C001

Table A.12

Example ground-truth labels for a clinical rehabilitation sample.

Dimension	GT label
D1 (semantic action recognition)	A B C
D2 (auxiliary parameter extraction / kinematic alignment)	A
D3 (pathological compensation identification)	C
D4 (spatiotemporal dynamics analysis)	A C
D5 (clinical decision / movement quality)	B
D6 (safety consistency / visual grounding)	D

Functional biomechanics sample example

Cohort/action type: Functional / 01 Squat

Sample identifier: functional::sample_F001

Table A.13

Example ground-truth labels for a functional biomechanics sample.

Dimension	GT label
D1	A B C D
D2	A B C D
D3	A B D
D4	A B D
D5	D
D6	D

Appendix A.2. Multi-label scoring rules

To ensure standardized and reproducible model evaluation, this study used a predefined multi-label scoring framework. The scoring rules were as follows.

- Parsing rule.** Model-generated option strings were uniformly parsed into standardized option sets. For example, outputs such as A and B, A, B, or ["A", "B"] were all normalized as the option set {A, B}. For items whose task rule permitted the absence of all queried findings, an explicitly returned empty selection was normalized as the empty set $\emptyset$ and encoded internally using the dedicated EMPTY representation. Such an explicit empty set was retained as a valid prediction and was distinguished from a missing field, blank output, refusal, unparsable response, or system-level failure.
- Primary metric: item-level set-F1.** The primary metric was the item-level set-based F1-score. For each item, the model-predicted option set $\mathcal{P} \subseteq \{A, B, C, D\}$ was compared with the adjudicated expert-consensus

ground-truth set $\mathcal{G} \subseteq \{A, B, C, D\}$. Precision, recall, and set-F1 were calculated as:

$$\text{Precision} = \frac{|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{P}|}, \quad \text{Recall} = \frac{|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{G}|},$$

$$\text{Set-F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}},$$

where $\mathcal{P}$ denotes the predicted option set and $\mathcal{G}$ denotes the adjudicated expert-consensus ground-truth option set.

Because an empty set was a valid ground-truth or predicted response for applicable items, explicit conventions were used to avoid undefined denominators. When $\mathcal{P} = \mathcal{G} = \emptyset$, precision, recall, and set-F1 were assigned a value of 1. When exactly one of $\mathcal{P}$ and $\mathcal{G}$ was empty, precision, recall, and set-F1 were assigned a value of 0. For two non-empty sets, complete overlap received a score of 1, no overlap received a score of 0, and partial overlap received the corresponding set-F1 score defined above. These conventions were applied at the item level before dimension-level macro-averaging.

For primary benchmark scoring, the 81,006 successfully returned dimension-level responses constituted the returned-response scoring pool. The 80,931 parsable valid responses were scored according to their predicted option sets, whereas the 75 returned invalid or unparseable responses were assigned an item-level score of zero. The 426 response opportunities unavailable because of missing outputs or system-level failures were excluded from model-specific primary point estimates.

In the valid-only sensitivity analysis, the 75 returned invalid or unparseable responses were excluded rather than assigned zero, leaving 80,931 parsable valid responses. In the more conservative complete-denominator sensitivity analysis, all 81,432 expected response opportunities were retained, and both the 75 returned invalid or unparseable responses and the 426 unavailable responses were assigned zero.

3. **Secondary metric: exact match.** Exact match (EM) was retained as a conservative secondary metric to evaluate model performance under a strict all-or-nothing criterion. EM required the predicted option set to be identical to the adjudicated expert-consensus ground-truth set:

$$\mathcal{P} = \mathcal{G}.$$

Two explicitly empty sets were therefore treated as an exact match. Returned invalid or unparseable responses were assigned an EM score of zero under the primary benchmark scoring rule, whereas unavailable response opportunities were excluded from model-specific primary EM estimates. The same valid-only and complete-denominator sensitivity rules described above were applied when EM was recalculated under alternative response-handling assumptions.

4. **Weighted aggregation.** For each model, dimension-weighted scores were first calculated separately within the clinical rehabilitation and functional biomechanics cohorts using the prespecified D1–D6 weights:

$$S_{\text{dimension}} = 0.10S_{D1} + 0.10S_{D2} + 0.30S_{D3} + 0.10S_{D4} + 0.25S_{D5} + 0.15S_{D6}.$$

The resulting cohort-specific dimension-weighted scores were then combined using the prespecified 65% clinical and 35% functional weighting scheme:

$$S_{\text{total}} = 0.65S_{\text{clinical}} + 0.35S_{\text{functional}}.$$

This two-stage procedure yielded the final cohort-weighted ActionEval composite score. The same response-handling rule used for the corresponding dimension-level scores was retained during cohort-specific and final weighted aggregation.

5. **Safety assessment for D6.** D6 was independently defined as a safety endpoint. It quantified the false-positive visual-grounding rate under negative-ground-truth probes, aiming to evaluate evidence consistency and robustness in complex rehabilitation video scenarios.

A D6 false-positive event was operationally defined as the selection of an affirmative option indicating the presence of a scene, object, or contextual element that experts had confirmed to be absent. Such events were interpreted as visually unsupported affirmative selections under a constrained-response format rather than, by themselves, as definitive evidence of open-ended generative hallucination.

Appendix B. Zero-shot inference strategy and engineering implementation audit

Appendix B.1. Zero-shot inference policy

All evaluated vision-language models or multimodal large language models were operated in a zero-shot setting. No model was fine-tuned using benchmark videos, expert labels, or task-specific answer sets. The evaluation protocol required each model to process the input video and return structured reasoning results in a predefined JSON format.

Each input video was associated with its corresponding evaluation item only within the study-side data-management pipeline. During model inference, the video was presented using an opaque sample identifier containing no action-category information. Local source filenames, directory paths, preassigned action indices, ground-truth action names, and filename-derived action bindings were not included in the model-visible prompt, question stem, answer options, attachment captions, uploaded-file metadata, or other textual input. None of the 522 model-visible video inputs contained a filename-derived or action-name cue (0/522); consequently, the complete benchmark constituted the cue-free evaluation set. The model was therefore required to determine the target movement directly from the visible video evidence and select the most appropriate action from the predefined action list.

The evaluation covered six dimensions, D1–D6. All D1–D6 items used a select-all-that-apply multiple-choice format in which zero or more affirmative options could be correct, depending on the item. An explicitly empty selection was permitted when none of the queried findings was present. The prompt explicitly required the model to answer only on the basis of visible visual evidence, to avoid unsupported inference, and to return option codes rather than free-text diagnoses. D6 was designed as a negative-evidence visual-grounding test by querying objects, scenes, or contextual elements that experts had confirmed to be absent from the corresponding video, thereby evaluating visually unsupported affirmative selections under negative-ground-truth probes.

Prompt for Multimodal Large Language Models

PROMPT =

“You are an experienced neurorehabilitation and motor-function assessment expert. Your task is to evaluate the input movement video using the standardized ActionEval-6D questionnaire.

First, determine the target movement directly from the visible video evidence and select the most appropriate action from the predefined action list. Then answer only the D1–D6 questions for the selected movement.

D1–D6 use a select-all-that-apply format. Depending on the item, zero or more affirmative options may be supported by the visible evidence, and an explicitly empty selection is permitted when none of the queried findings is present. Select all and only the options supported by the visible evidence. All judgments must rely strictly on visible video evidence. Do not infer unobserved findings from prior knowledge, expected action patterns, or option semantics.

Return only a valid JSON object containing the video ID, recognized action ID, action name, recognition explanation, and D1–D6 answers with brief evidence.

Full prompt templates, the complete ActionEval-6D question set, and scoring scripts are available in the GitHub repository.”

Appendix B.2. Model-specific visual-input routes and temporal-sampling configurations

Because the evaluated models differed in their native support for dynamic video input, model-family-specific visual-input routes were documented to improve the reproducibility and interpretation of the benchmarking results. Models with native video-input capability received the complete de-identified video through the corresponding official interface, whereas models without native

video support received a chronologically ordered sequence of uniformly sampled frames. Provider-documented default temporal-sampling configurations were retained when available. When the provider did not expose its internal sampling rate or realized frame indices, these parameters were reported as not reported (NR) rather than inferred. The resulting input routes and available temporal-sampling information are summarized in Table B.14.

Appendix B.3. Data input processing and privacy protection

The benchmark contained two video cohorts: a functional fitness or biomechanics cohort and an upper-limb rehabilitation cohort. During inference, model inputs were identified only by opaque sample identifiers that contained no action-category information. Source filenames, folder names, preassigned action indices, and ground-truth action names were not included in the model-visible payload.

After completion of model inference, the returned outputs were linked to the study-side metadata through the opaque sample identifier for quality control, scoring, and audit. The resulting internal export files were stored by model identifier and benchmark subset in wide-table, long-table, and option-level CSV formats. These post-inference export fields could include cohort or folder, opaque video ID, internal filename, action index, action name, evaluation dimension, selected option, and reasoning text. The filename, action-index, and action-name fields were appended only after the model response had been generated and were never transmitted in the prompt, question text, attachment caption, uploaded-file metadata, or other model-visible content.

This study strictly followed data de-identification principles. No personal identifying information was used in the evaluation or scoring pipeline. Before result release and data sharing, all original folder names and filenames were replaced with de-identified sample IDs. The scoring and aggregation pipeline used only de-identified sample identifiers, action types, evaluation dimensions, model identifiers, parsed options, and expert gold-standard labels.

Appendix B.4. Response parsing and quality control

Model raw outputs were required to follow a structured JSON format, including video ID, recognized action index or action name, reasoning text, and D1–D6 evaluation blocks. The parsing pipeline used rule-based normalization to convert selected options into standardized CSV fields.

For items whose task rule permitted the absence of all queried findings, an explicitly returned empty selection was normalized as the empty set $\emptyset$ and encoded internally using the dedicated EMPTY representation. Such an explicit empty selection was retained as a parsable valid response and was distinguished from a blank or missing response field, refusal, structurally unusable output, parsing failure, or system-level failure. A successfully returned response containing a blank required field, refusal, illegal option value, unparsable text, or structurally unusable content was classified as returned invalid or unparsable. By contrast, a response opportunity for which no model output was returned after the predefined retry procedure was classified as unavailable. The model-generated reasoning text was retained for audit purposes.

During data preprocessing, strict quality-control auditing was performed. Audit scripts checked JSON readability, structural completeness, required-field availability, parsing-error placeholders, illegal option values, and blank responses. Across 26 models and 522 videos, 13,572 model–video JSON outputs were expected, with each output designed to contain six dimension-specific responses, yielding 81,432 model–video–dimension response opportunities. Of these, 81,006 dimension-level responses were successfully returned, whereas 426 response opportunities, corresponding to 71 model–video outputs, were unavailable because no usable model output was returned after the predefined retry procedure or because of system-level failures. The overall response-yield rate was therefore 99.48%. Among the 81,006 returned responses, 80,931 were parsable and valid, while 75 were classified as invalid or unparsable. This corresponded to a parsable-validity rate of 99.91% among returned responses and an overall valid-response rate of 99.38% relative to the complete expected response set. Quality-control outcomes were summarized at the dimension-response level to maintain consistency with the unit of analysis used in the main manuscript.

Under the primary returned-response scoring rule, the 81,006 successfully returned dimension-level responses constituted the returned-response scoring pool. The 80,931 parsable valid responses were scored according to their normalized predicted option sets, including explicitly returned empty sets where permitted by the task rule. The 75 returned invalid or unparsable responses

were assigned an item-level score of zero. The 426 response opportunities unavailable because of missing outputs or system-level failures were excluded from model-specific primary point estimates and recorded separately.

Two alternative response-handling strategies were used in sensitivity analyses. In the valid-only sensitivity analysis, the 75 returned invalid or unparsable responses were excluded rather than assigned zero, leaving 80,931 parsable valid responses. In the complete-denominator sensitivity analysis, all 81,432 expected response opportunities were retained, and both the 75 returned invalid or unparsable responses and the 426 unavailable responses were assigned an item-level score of zero.

D6 false-positive visual-grounding events were identified by comparing each parsable D6 option set with the negative expert-consensus reference label. If a model selected an affirmative option indicating the presence of a visual element, namely option A, B, or C, when the expert-consensus label indicated its absence, the response was recorded as a D6 false-positive event and included in the calculation of FP_{D6} . Because this classification was derived from a constrained select-all-that-apply response, it was interpreted as a visually unsupported affirmative selection rather than, by itself, as definitive evidence of fabrication in the accompanying free-text reasoning.

Appendix C. Robustness and sensitivity analyses

Appendix C.1. D6-excluded sensitivity analysis

To rule out the possibility that the video-evidence consistency or anti-hallucination dimension D6 introduced ceiling-effect bias, this study examined model stability after removing D6 from the composite weighting formula. Specifically, the 15% D6 weight in the original weighted evaluation system was removed, and the remaining D1–D5 dimensions were re-normalized to recalculate the model leaderboard and the composition of the top-performing model cluster.

The results showed that, after excluding D6, the top-performing model cluster remained highly stable and completely overlapped with the main analysis. The Spearman rank correlation between the main ranking and the D6-excluded ranking was 0.999, indicating that the leaderboard structure was minimally dependent on D6 weighting. In addition, the paired Wilcoxon test comparing D1 with the mean of D3–D5 remained highly significant:

$$P = 1.49 \times 10^{-8}.$$

These results indicate that the clinical reasoning gap identified in the main analysis was not driven by the D6 weighting scheme or the ceiling effect of D6. Instead, it reflected a real performance bottleneck when models transitioned from basic visual perception to higher-order clinical reasoning.

Appendix C.2. Complete model leaderboard and statistical testing

To ensure full transparency of the evaluation results, Table C.15 presents the complete performance of all 26 evaluated MLLMs on ActionEval-MedBench. Qwen3.6-27B was used as the reference model, and paired video-level Wilcoxon tests were used to calculate the statistical significance of each model relative to the reference model. Significance markers were defined as follows: *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$, and ns $P \geq 0.05$.

Appendix C.3. Summary of robustness analyses

To ensure the robustness of the main conclusions, this study performed multiple sensitivity analyses under different weighting schemes, data-processing strategies, and cohort restrictions. The results are summarized in Table C.16.

Overall, the sensitivity analyses showed that the main leaderboard, top-performing cluster, and semantic-to-clinical reasoning gap were highly consistent across alternative analytic settings. These findings indicate that the core conclusions of this study did not depend on a specific scoring parameter, weighting strategy, or response-handling rule, but reflected the robust capability boundaries of current MLLMs in rehabilitation clinical reasoning tasks.

Appendix C.4. Sensitivity of composite rankings to dimension weighting

To examine the dependence of the composite model rankings on the expert-defined benchmark utility function, model rankings were recalculated under equal weighting, lower-D6 weighting, and D6-excluded weighting. Rank stability was assessed using Spearman’s rank correlation with the primary ranking, top-five model overlap, and maximum absolute rank displacement. The results are summarized in Table C.17.



Fig. 3
Dimension-wise ActionEval-6D performance. The heatmap presents the model-level mean item-level set-F1 scores (%) of the 26 evaluated MLLMs across the six ActionEval dimensions. The underlying data were arranged into a 26×6 model-by-dimension matrix, with one row for each evaluated MLLM and one column for each ActionEval dimension. Each cell represents one model–dimension combination. The values followed the primary returned-response scoring rule: valid responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned an item-level score of zero, and unavailable response opportunities were excluded from model-specific point estimates. D1 represents semantic action recognition, D2 kinematic feature alignment, D3 pathological compensation identification, D4 spatiotemporal dynamics analysis, D5 comprehensive movement quality and clinical decision-making, and D6 visual grounding and anti-hallucination verification. Colors encode F1-scores from 0% (red) to 100% (green), and the thick vertical line separates D6 as an independent visual-grounding safety dimension. Models generally achieved high scores in D1 and D6, whereas performance declined substantially in the higher-order clinical reasoning dimensions D3–D5. The heatmap was generated using Python (version 3.10.0; Python Software Foundation, <https://www.python.org/>), seaborn (version 0.13.2; <https://seaborn.pydata.org/>), and Matplotlib (version 3.10.8; <https://matplotlib.org/>).

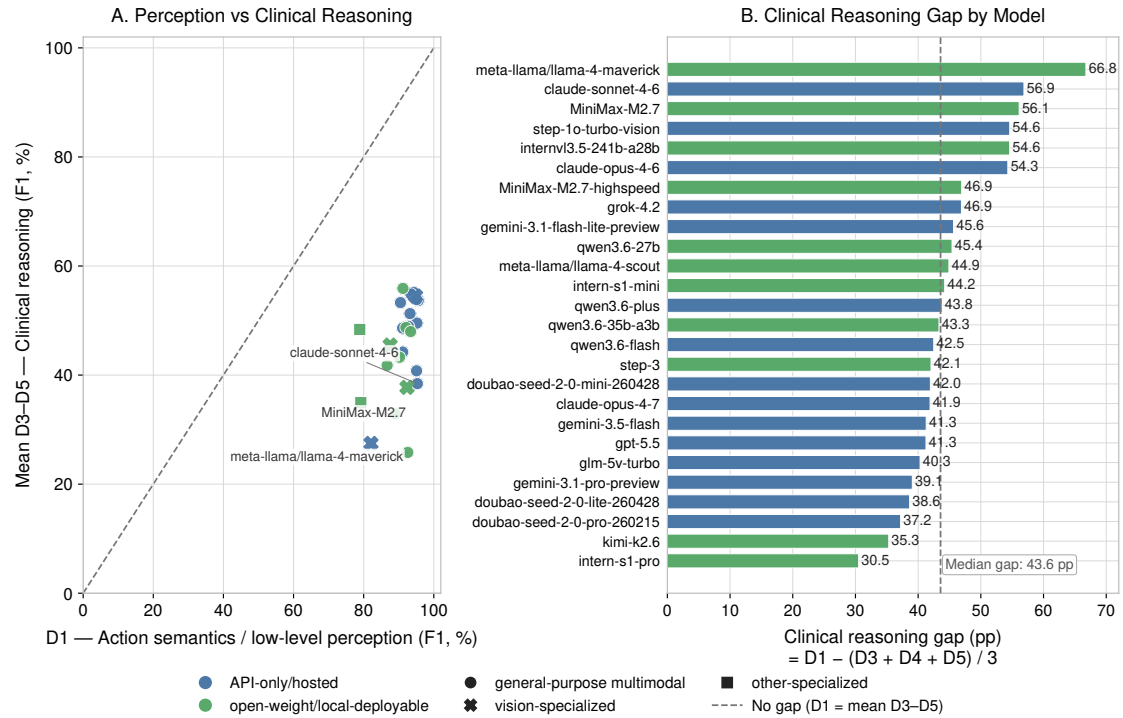


Fig. 4
Clinical reasoning gap: perception-to-reasoning performance cliff. **A** Scatter plot comparing D1 semantic action recognition with the mean score across D3–D5 clinical reasoning dimensions. The diagonal dashed line represents equal performance between low-level perception and higher-order clinical reasoning. All models fall below this line, indicating lower performance in pathological compensation identification, spatiotemporal dynamics analysis, and clinical decision-making than in basic action recognition. **B** Model-level clinical reasoning gap, defined as $S_{D1} - (S_{D3} + S_{D4} + S_{D5})/3$. The vertical dashed line indicates the median gap across all evaluated models. Blue bars denote API-only or hosted models, and green bars denote open-weight or locally deployable models.

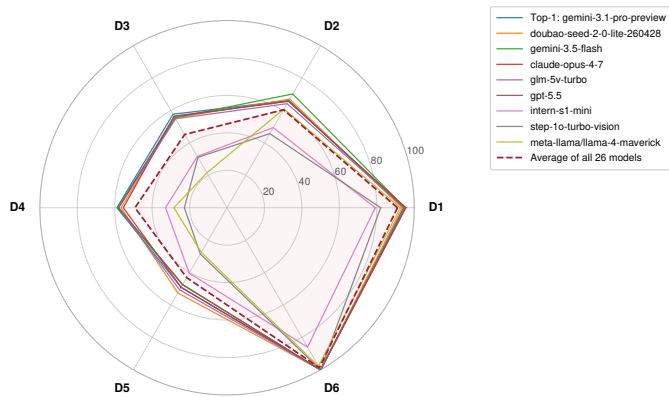


Fig. 5
D1–D6 radar profile: top models versus average. The radar plot compares the dimension-wise ActionEval-6D performance of representative high-performing and low-performing models against the average performance of all evaluated models. Model profiles remained close to the outer range in D1 semantic action recognition and D6 visual grounding, whereas D3–D5 showed marked inward contraction, indicating weaker performance in pathological compensation identification, spatiotemporal dynamics analysis, and comprehensive clinical decision-making.

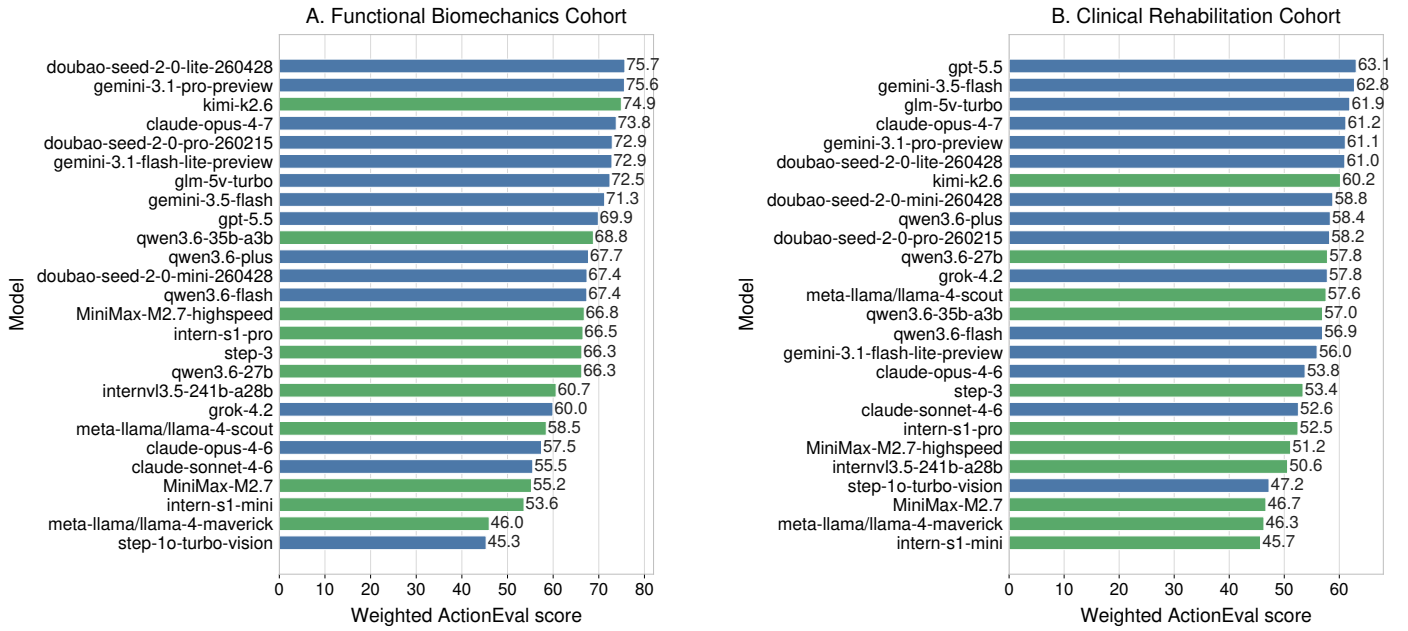


Fig. 6
Functional and clinical action benchmark leaderboards. (A) Independent leaderboard for the functional biomechanics cohort, which mainly consisted of large-joint, high-visibility gross-motor actions. (B) Independent leaderboard for the clinical fine-grained rehabilitation cohort, which included real-world patient-specific rehabilitation videos with subtle movement features. Blue bars denote API-only or hosted models, whereas green bars denote open-weight or locally deployable models.

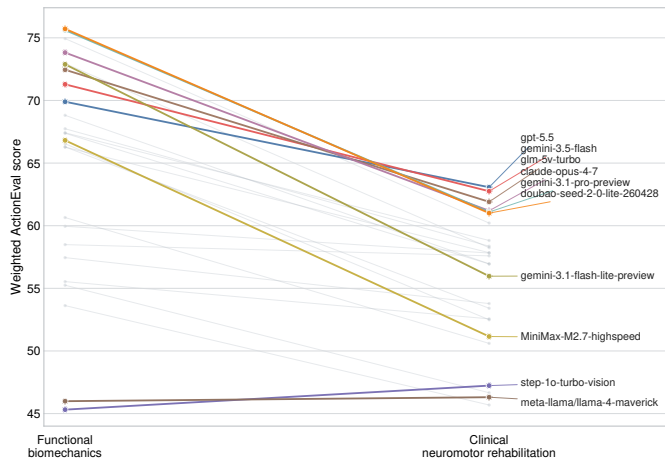


Fig. 7
Functional-to-clinical performance shift. The slope chart shows the change in weighted ActionEval scores when each model was evaluated on the functional biomechanics cohort and the clinical fine-grained neuromotor rehabilitation cohort. Most models showed a downward shift from functional to clinical videos, indicating performance degradation when moving from standardized, high-visibility functional actions to real-world fine-grained clinical rehabilitation videos.

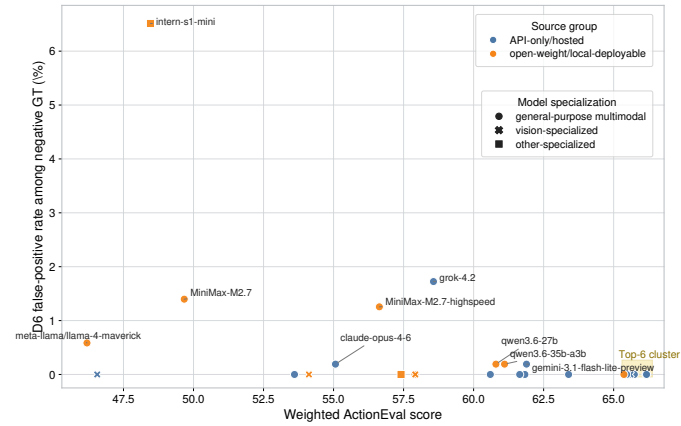


Fig. 8
Safety endpoint beyond the D6 ceiling effect. The scatter plot shows the relationship between overall weighted ActionEval score and the operational D6 false-positive visual-grounding rate (FP_{D6}) among negative-ground-truth probes. Although many models achieved near-ceiling D6 F1-scores, the false-positive visual-grounding analysis revealed additional safety differentiation. Blue dots denote API-only or hosted models, and orange dots denote open-weight or locally deployable models. Marker shape indicates model specialization category. The shaded region highlights the top-performing cluster with near-zero false-positive visual-grounding rates. These events represent visually unsupported affirmative selections under a constrained multi-select format and should not, by themselves, be interpreted as definitive evidence of open-ended generative hallucination.

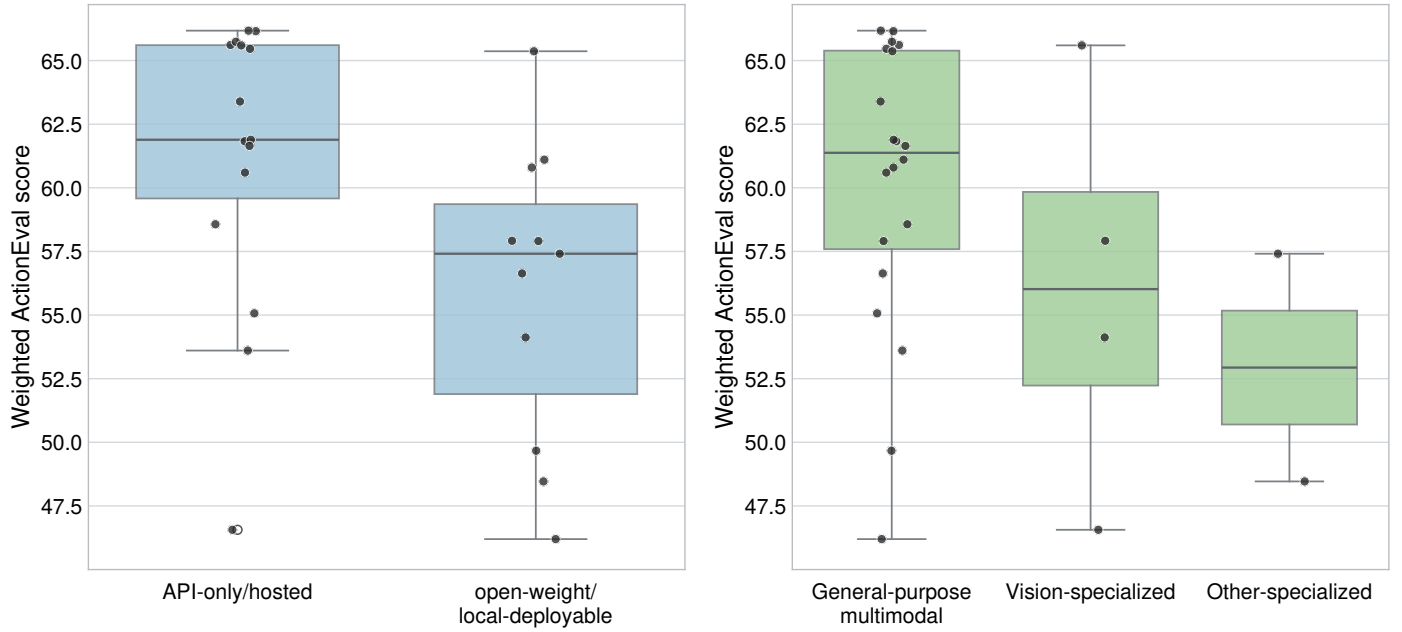


Fig. 9
Subgroup comparison by deployment route and model specialization. **Left** Comparison between API-only/hosted models and open-weight/local-deployable models. **Right** Comparison among general-purpose multimodal models, vision-specialized models, and other specialized models. Boxplots show the distribution of weighted ActionEval scores for each subgroup.

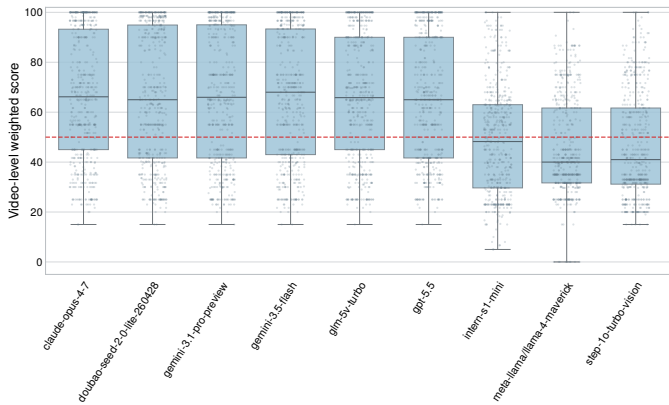


Fig. 10
Video-level score distribution. Boxplots show the descriptive distribution of weighted video-level scores for representative high-performing and lower-performing models across 522 video recordings contributed by 48 participants. Multiple recordings could originate from the same participant; participant-level dependence was accounted for separately in the cluster-robust bootstrap confidence intervals. The red dashed line indicates an exploratory low-score reference cutoff of 50 points rather than a validated clinical usability threshold.

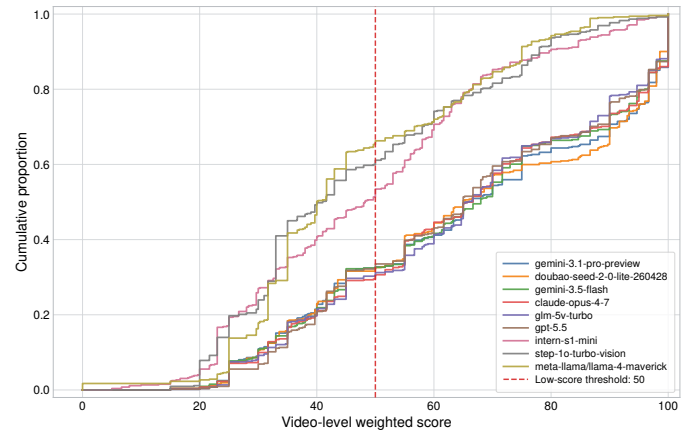


Fig. 11
CDF and downside risk curve. The cumulative distribution function curves show the distribution of video-level weighted scores for representative models. The red dashed vertical line marks an exploratory low-score reference cutoff of 50 points. This cutoff is used for descriptive comparison and does not represent an externally validated clinical usability boundary. Curves shifted toward the lower right indicate lower downside risk and better performance stability.

Table 3

Dataset characteristics, participant-clustering and response-handling structure, and ActionEval-6D weighting scheme.

Component	Assessment focus	Size	Weight (%)	Rationale
Dataset overview	Total de-identified video recordings	522 videos	–	Benchmark source.
Participant-clustering structure	Non-identifying grouping of repeated recordings contributed by the same participant	48 participants	–	Unit used for participant-level cluster bootstrap; 28 clinical and 20 functional participants, with a median of 10 recordings per participant (range, 8–30).
Clinical rehabilitation cohort	Fine-grained clinical neuromotor rehabilitation videos	328 videos; 28 participants	65	Primary clinical cohort.
Functional biomechanics cohort	General functional biomechanical movement videos	194 videos; 20 participants	35	Reference functional cohort.
D1 Semantic action recognition	Macroscopic movement or rehabilitation task category	522 MCQs	10	Baseline visual perception.
D2 Kinematic feature alignment	Joint trajectory, movement amplitude, and posture alignment	522 MCQs	10	Physical kinematic tracking.
D3 Pathological compensation identification	Compensatory strategies and erroneous kinetic-chain patterns	522 MCQs	30	Core clinical reasoning dimension.
D4 Spatiotemporal dynamics analysis	Rhythm, speed variation, pauses, tremor, and smoothness	522 MCQs	10	Temporal motor-control assessment.
D5 Movement quality and clinical decision-making	Functional limitation, risk, and intervention-relevant judgement	522 MCQs	25	Integrated clinical decision-making.
D6 Visual grounding and anti-hallucination	Avoidance of fabricated absent scene, object, or context information	522 MCQs	15	Independent safety endpoint.
Total assessment items	Six ActionEval dimensions per video	3,132 MCQs	–	Primary evaluation task set.
Expected dimension-level response opportunities	3,132 MCQ items evaluated by 26 MLLMs	81,432 response opportunities	–	Complete expected response set.
Successfully returned model responses	Dimension-level responses successfully returned after the predefined retry procedure	81,006 responses	–	Primary returned-response scoring pool; overall response-yield rate of 99.48%.
Missing or system-level failure responses	Response opportunities unavailable because of missing outputs or system-level failures	426 response opportunities	–	Excluded from model-specific primary point estimates and assigned zero only in the complete-denominator sensitivity analysis.
Returned invalid or unparsable responses	Returned responses failing parsing or usability control	75 responses	–	Assigned an item-level score of zero in the primary analysis and excluded in the valid-only sensitivity analysis.
Parsable valid model responses	Parsed and usable responses scored according to their predicted option sets	80,931 responses	–	Valid component of the primary returned-response scoring pool; 99.38% of the complete expected response set.

Table 4

Inter-group reliability and arbitration statistics of the dual-track independent consensus pipeline.

Dimension	MCQs	Cohen’s κ [95% CI]	Discrepant items	Arbitration rate (%)	Pipeline consensus (%)
D1 Semantic action recognition	522	0.896 [0.85, 0.94]	21	4.02	95.98
D2 Kinematic feature alignment	522	0.828 [0.79, 0.86]	75	14.37	85.63
D3 Pathological compensation identification	522	0.650 [0.59, 0.71]	118	22.61	77.39
D4 Spatiotemporal dynamics analysis	522	0.761 [0.72, 0.80]	101	19.35	80.65
D5 Comprehensive decision quality	522	0.706 [0.65, 0.76]	96	18.39	81.61
D6 Visual grounding and anti-hallucination	522	1.000 [1.00, 1.00]	0	0.00	100.00
Overall	3,132	0.807 [0.79, 0.82]	411	13.12	86.88

Notes: Two independent expert groups reviewed the videos and generated group-level consensus labels through within-group majority voting under blinded conditions. Inter-group reliability was assessed before chief-expert arbitration by comparing the group-level consensus labels independently generated by Group A and Group B.

Because each ActionEval item used a multi-label option-set format, each consensus answer was treated as a unique categorical response. For example, $\{A\}$, $\{A, B\}$, $\{B, D\}$, and $\{D\}$ were treated as distinct categorical responses. Cohen’s κ was calculated at the item level by comparing whether the two groups reached the exact same option set, rather than individual options.

The 95% confidence intervals were estimated using item-level bootstrap resampling. Pipeline consensus rate was defined as the proportion of items with complete agreement between the two independent groups before chief-expert arbitration. All Cohen’s κ tests satisfied $P < 0.001$. The pooled κ aggregates all six dimensions and is partly elevated by the perfect agreement observed in D6 ($\kappa = 1.000$); dimension-specific κ estimates and arbitration rates should therefore be considered the more informative indicators of annotation difficulty across ActionEval dimensions.

Table 5

Overall benchmark leaderboard and statistical testing of the top 10 evaluated MLLMs on ActionEval-MedBench.

Rank	Model	Weighted F1-score	EM accuracy (%)	Valid response (%)	Participant-level 95% bootstrap CI	Adjusted P value
1	Gemini-3.1-Pro-Preview	66.17	61.62	100.00	[63.32, 69.15]	0.0003***
2	Doubao-Seed-2.0-Lite-260428	66.15	60.70	100.00	[63.03, 69.25]	0.0005***
3	Gemini-3.5-Flash	65.74	62.48	100.00	[62.94, 68.77]	0.0005***
4	Claude-Opus-4-7	65.61	61.88	100.00	[62.91, 68.57]	0.0007***
5	GLM-5V-Turbo	65.60	62.84	100.00	[63.01, 68.18]	0.0015**
6	GPT-5.5	65.46	61.81	100.00	[63.01, 68.00]	0.0025**
7	Kimi-K2.6	65.37	63.51	100.00	[62.90, 67.89]	0.0003***
8	Doubao-Seed-2.0-Pro-260215	63.39	60.71	100.00	[60.87, 66.03]	0.0772 ^{ns}
9	Gemini-3.1-Flash-Lite-Preview	61.89	59.87	100.00	[59.16, 64.60]	0.3351 ^{ns}
10	Doubao-Seed-2.0-Mini-260428	61.83	57.12	100.00	[59.07, 64.64]	0.5264 ^{ns}

Notes: Only the top 10 models are shown; the full leaderboard is provided in Table C.15. Weighted F1-scores were calculated using the predefined cross-cohort scheme, with 65% weight assigned to the clinical rehabilitation cohort and 35% to the functional biomechanics cohort.

The 95% CIs were estimated using 2,000 participant-level cluster-robust bootstrap resamples. Participants were sampled with replacement separately within the clinical and functional cohorts, and all recordings contributed by the same participant, including left- and right-limb recordings, were retained together within each bootstrap replicate. The 522 recordings were contributed by 48 participants, including 28 clinical and 20 functional participants, with a median of 10 recordings per participant (range, 8–30). The participant-level bootstrap was used only to estimate the confidence intervals and did not alter the model-level point estimates or hypothesis-test results.

Qwen3.6-27B, the median-ranked model in the full leaderboard, was used as the reference model. Pairwise Wilcoxon signed-rank tests were performed on matched video–MCQ item-level F1-scores with Holm–Bonferroni correction. ns, $P \geq 0.05$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

Table 6
Post hoc diagnostic stratification of D3 using adjudicated gold-label status.

D3 subset	Items	Mean F1 (%)	Difference from all-item D3 (pp)
All D3 items	522	45.2	–
Determinate, rationale-supported gold	429	43.0	–2.2
Internally unresolved gold status	93	55.5	+10.3
Positive compensation finding	230	33.8	–11.4
Rationale-confirmed absence	199	53.6	+8.4

Notes: Values are unweighted arithmetic means of the 26 model-level estimates calculated using the primary returned-response scoring rule, under which parsable valid responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned a score of zero, and unavailable responses were excluded from model-specific estimates. The “determinate, rationale-supported gold” subset comprises the 230 items containing at least one expert-labelled pathological-compensation option and the 199 items with a rationale-confirmed absence. “Determinate” in this analysis does not denote six-rater unanimity and should not be equated with the 404 D3 items with exact pre-arbitration inter-group agreement reported in Table 4. The item-level identifiers required to reconstruct the original 404/118 inter-group agreement/arbitration split were not retained in the archived analytical dataset. The determinate and internally unresolved model-level estimates were compared using a two-sided paired Wilcoxon signed-rank test ($P = 0.029$). Comparisons involving empty-ground-truth subsets should be interpreted in light of the predefined scoring convention assigning a score of 1 when both the predicted and ground-truth sets are empty. The complete model-level results are reported in Supplementary Table C.18.

Table 7
Chance-level and empirical human-reference benchmarks across the six ActionEval dimensions.

Dimension	Chance floor, F1 (%)		Group A human reference, F1 (%)				Group A majority vote	Best-observed model
	Random-subset	Select-all	Expert 01	Expert 02	Expert 03	Human ceiling (mean)	F1 (%)	F1 (%)
D1 Semantic action recognition	58.64	84.03	99.05	99.01	98.98	99.01	99.78	95.57
D2 Kinematic feature alignment	43.58	58.52	89.49	89.97	90.53	90.00	95.64	70.17
D3 Pathological compensation identification	16.26	20.63	78.57	78.48	79.18	78.74	88.70	58.30
D4 Spatiotemporal dynamics analysis	28.00	35.62	83.15	83.06	83.22	83.14	90.96	64.17
D5 Movement quality and clinical decision-making	32.70	40.13	81.74	82.06	81.99	81.93	90.80	52.55
D6 Visual grounding and anti-hallucination	32.67	40.00	100.00	100.00	100.00	100.00	100.00	100.00

Notes: All values are reported as dimension-specific set-F1 percentages. For the random-subset baseline, all 16 possible subsets of the four-option universe were assigned equal probability, and the expected score was calculated by exact enumeration. The select-all baseline selected all four options for every item. Expert 01–03 denote the three Group A experts whose independent pre-arbitration annotations were compared with the final adjudicated ground truth. The empirical human ceiling was defined as the arithmetic mean of the three individual-expert F1-scores. The Group A majority-vote score is reported as an additional consensus reference and was not used as the primary human-ceiling estimate. “Best-observed model” denotes the highest dimension-specific set-F1 score among the 26 evaluated models; the best-observed scores across different dimensions may therefore originate from different models. Because the adjudicated ground truth incorporated expert-panel evidence, the human-ceiling values represent empirical agreement with the final reference standard rather than a fully independent theoretical upper bound. Human-reference and chance-level results were not aggregated into composite scores because the archived human-reference summaries did not preserve the cohort-specific estimates required to reproduce the model leaderboard’s prespecified 65% clinical and 35% functional weighting procedure.

Table 8
Statistical summary of clinical reasoning gaps across representative MLLMs.

Overall rank	Model	D1 F1 (%)	D2 F1 (%)	D3 F1 (%)	D4 F1 (%)	D5 F1 (%)	Mean D3–D5 (%)	Clinical gap (pp)	Gap rank
<i>Top-ranked models by overall performance</i>									
1	Gemini-3.1-Pro-Preview	94.25	66.18	57.69	58.80	49.04	55.18	39.07	22
2	Doubao-Seed-2.0-Lite-260428	93.49	67.01	55.43	56.56	52.55	54.85	38.64	23
3	Gemini-3.5-Flash	95.14	70.17	55.56	58.46	47.54	53.85	41.29	19
4	Claude-Opus-4-7	95.57	65.64	56.50	55.38	49.11	53.66	41.90	18
5	GLM-5V-Turbo	94.81	64.18	54.91	57.94	50.64	54.50	40.31	21
<i>Models with the largest reasoning cliffs</i>									
26	Meta-Llama/Llama-4-Maverick	92.57	60.90	21.55	28.37	27.52	25.81	66.76	1
22	Claude-Sonnet-4-6	95.31	61.60	28.26	47.64	39.40	38.43	56.88	2
23	MiniMax-M2.7	89.01	55.00	30.50	34.38	33.80	32.89	56.12	3
21	InternVL3.5-241B-A28B	92.35	59.50	33.68	36.07	43.49	37.75	54.61	5
25	Step-1o-Turbo-Vision	82.08	45.70	31.09	22.78	28.54	27.47	54.61	4

Notes: All dimension-specific F1-scores and clinical reasoning gaps reported in this table follow the primary returned-response scoring rule: parsable valid responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned an item-level score of zero, and unavailable responses were excluded from model-specific estimates. The clinical reasoning gap was calculated as $S_{D1} - (S_{D3} + S_{D4} + S_{D5})/3$ and is reported in percentage points (pp). D2 was not included in the higher-order clinical reasoning mean because it primarily reflects physical kinematic alignment.

Table 9
Statistical summary of cross-cohort performance drops across representative MLLMs.

Overall rank	Model	Deployment	$S_{\text{functional}}$ (%)	S_{clinical} (%)	Performance drop (pp)	S_{total} (%)
9	Gemini-3.1-Flash-Lite-Preview	API-only/hosted	72.88	55.97	16.92	61.89
19	MiniMax-M2.7-Highspeed	Open-weight/local-deployable	66.81	51.15	15.66	56.64
7	Kimi-K2.6	Open-weight/local-deployable	74.94	60.21	14.73	65.37
2	Doubao-Seed-2.0-Lite-260428	API-only/hosted	75.72	61.00	14.71	66.15
8	Doubao-Seed-2.0-Pro-260215	API-only/hosted	72.95	58.24	14.70	63.39
1	Gemini-3.1-Pro-Preview	API-only/hosted	75.62	61.09	14.53	66.17
18	Intern-S1-Pro	Open-weight/local-deployable	66.54	52.49	14.05	57.41
16	Step-3	Open-weight/local-deployable	66.29	53.41	12.88	57.92
4	Claude-Opus-4-7	API-only/hosted	73.82	61.19	12.63	65.61
12	Qwen3.6-35B-A3B	Open-weight/local-deployable	68.81	56.95	11.86	61.11

Notes: The cohort-specific scores and performance drops reported in this table follow the primary returned-response scoring rule: parsable valid returned responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned a score of zero, and unavailable responses were excluded from model-specific estimates. Models are sorted in descending order according to cross-cohort performance drop. Performance drop was calculated as $\text{Performance Drop} = S_{\text{functional}} - S_{\text{clinical}}$ and is reported in percentage points (pp). A larger value indicates greater performance loss when shifting from the functional biomechanics cohort to the clinical fine-grained rehabilitation cohort. The performance drop reflects the combined domain shift between two video cohorts that differ in movement scale, body region, data source, recording context, participant status, and clinical complexity; it should not be interpreted as the isolated causal effect of pathology or pathological compensation difficulty.

Table 10
Summary of subgroup performance and Mann–Whitney U tests by deployment attribute and architectural specialization, with multiplicity-adjusted P values.

Comparison family	Metric	Group A (baseline)	Group B (comparison)	Mean A (%)	Mean B (%)	Mean diff (%)	U	Unadjusted P	BH–FDR P
Deployment	Weighted composite score	API-only/hosted ($n = 15$)	Open-weight/local-deployable ($n = 11$)	61.19	55.96	+5.23	130.00	0.0147	0.0735 ^{ns}
	D3 (pathological compensation)	API-only/hosted	Open-weight/local-deployable	48.07	41.27	+6.80	116.00	0.0868	0.1736 ^{ns}
	D5 (clinical decision-making)	API-only/hosted	Open-weight/local-deployable	44.81	40.78	+4.03	119.00	0.0617	0.1653 ^{ns}
	D6 (anti-hallucination F1)	API-only/hosted	Open-weight/local-deployable	99.59	97.44	+2.15	135.50	0.0045	0.0450 [*]
	FP_{D6} hallucination rate	API-only/hosted	Open-weight/local-deployable	0.14	0.92	-0.78	52.00	0.0661	0.1653 ^{ns}
Specialization	Weighted composite score	General-purpose multimodal ($n = 20$)	Vision-specialized ($n = 4$)	60.17	56.05	+4.12	54.00	0.3091	0.4416 ^{ns}
	D3 (pathological compensation)	General-purpose multimodal	Vision-specialized	46.46	42.49	+3.97	46.00	0.6815	0.7572 ^{ns}
	D5 (clinical decision-making)	General-purpose multimodal	Vision-specialized	43.71	40.81	+2.90	48.00	0.5754	0.7193 ^{ns}
	D6 (anti-hallucination F1)	General-purpose multimodal	Vision-specialized	99.23	99.04	+0.19	41.00	0.9674	0.9674 ^{ns}
	FP_{D6} hallucination rate	General-purpose multimodal	Vision-specialized	0.29	0.00	+0.29	56.00	0.1513	0.2522 ^{ns}

Notes: All statistics were calculated using two-sided Mann–Whitney U tests. The 10 comparisons presented in this table were treated as a single hypothesis family, and the unadjusted P values were corrected using the Benjamini–Hochberg false-discovery-rate (BH–FDR) procedure. Statistical significance and significance markers were determined using the BH–FDR-adjusted P values. After correction, only the deployment-based difference in D6 anti-hallucination F1 remained statistically significant. Holm–Bonferroni and Bonferroni corrections applied to the same 10 comparisons produced the same substantive conclusion. The specialization subgroup analysis should be interpreted as exploratory because the number of vision-specialized models was small ($n = 4$). ns, BH–FDR-adjusted $P \geq 0.05$; *BH–FDR-adjusted $P < 0.05$.

Table 11
Qualitative thematic analysis of clinically interpretable failure patterns in MLLMs.

Failure mode	Thematic label	Operational signature	Interpretive hypothesis	Potential clinical risk
FM-1	Perception–reasoning disconnection	Correct or near-correct D1 action recognition accompanied by substantial performance loss in D3 and D5	Pattern consistent with incomplete linkage between action-level semantic recognition and fine-grained kinematic or clinical interpretation	Overestimation of movement quality or rehabilitation status, potentially leading to inappropriate progression or training intensity.
FM-2	Pathological compensation blind spots	Clinically relevant compensatory movements are misclassified as normal motor variations	Possible insufficient sensitivity to fine-grained kinetic-chain deviations and pathology-specific visual features	Missed detection of abnormal compensation, potentially resulting in inappropriate exercise continuation or increased risk of secondary injury.
FM-3	Language priors overriding visual evidence	Selection or description of scene elements, objects, or signs that are absent from the corresponding video, particularly in D6 negative probes	Apparent over-reliance on language-model priors despite absent or contradictory visual evidence	False-positive observations that may interfere with environmental safety assessment or subsequent clinical decision support.
FM-4	Temporal integration breakdown	Failure to identify temporally localized abnormalities such as tremor, pauses, rhythm disturbance, or freezing of gait	Pattern consistent with insufficient integration of temporally distributed evidence; variation in temporal-input preprocessing may also contribute	Underestimation of dynamic movement abnormalities, potentially delaying recognition of clinically important motor impairment.
FM-5	Kinetic-chain reasoning deficiency	Responses rely primarily on medical keywords or isolated joint descriptions without integrating cross-joint biomechanical relationships	Apparent reliance on surface-level semantic associations rather than structured reasoning over kinetic-chain coordination	Biomechanically unsupported interpretation or intervention suggestions, limiting the reliability of individualized rehabilitation planning.

Notes: The five categories were generated through human-led, Grounded Theory-informed thematic analysis of cases retrieved by the deterministic screening module. The interpretive hypotheses summarize recurrent patterns observed in model outputs and should not be interpreted as direct evidence of the models' internal cognitive or computational mechanisms. Failure-mode categories were used only for qualitative interpretation and did not affect quantitative scores or model rankings.

Table B.14

Model-family-specific API access, visual-input routes, and temporal-sampling configurations.

Model family	Access channel	Visual-input route	Temporal-sampling configuration	Effective sampling rate	Processed frames by cohort	Implementation notes
Gemini family ($n = 3$)	Programmatic API	Native video-file input	Provider-side default video preprocessing	1 frame/s	Clinical: approximately 5; functional: median 15 (range 5–135)	The complete video was submitted through the corresponding API. Frame counts are expected values because the API did not return realized frame indices.
Doubao Seed 2.0 family ($n = 3$)	Programmatic API	Native video-file input	Provider-side default video preprocessing	1 frame/s	Clinical: approximately 5; functional: median 15 (range 5–135)	The complete video was submitted through the documented video-understanding API.
Qwen3.6 family via DashScope ($n = 4$)	Programmatic API	Video-file or provider-mediated frame-sequence input	Provider-documented default sampling, with uniform downsampling if the maximum-frame limit was exceeded	2 frames/s	Clinical: approximately 10; functional: median 30 (range 10–270)	The complete video was submitted through the documented DashScope API. Counts may be reduced by a provider-specific maximum-frame limit.
Kimi-K2.6 via DashScope ($n = 1$)	Programmatic API	Video-file or provider-mediated frame-sequence input	Provider-documented default sampling, with uniform downsampling if the maximum-frame limit was exceeded	2 frames/s	Clinical: approximately 10; functional: median 30 (range 10–270)	The complete video was submitted through the documented DashScope API. Counts may be reduced by a provider-specific maximum-frame limit.
GLM-5V-Turbo ($n = 1$)	Programmatic API	Native video-capable multimodal input	Provider-managed temporal preprocessing	NR	NR	The video input was submitted through the provider API, which did not expose the effective sampling rate or realized frame indices.
Step family ($n = 2$)	Programmatic API	Native video-file input	Provider-managed temporal preprocessing	NR	NR	The complete video was submitted through the provider API; the provider did not publicly report its internal temporal-sampling configuration.
Intern/InternVL family ($n = 3$)	Programmatic API	Native or implementation-level video input	Provider- or implementation-managed temporal preprocessing	NR	NR	The visual input was submitted through the corresponding evaluation API. The effective sampling rate and realized frame indices were not exposed by the interface.
Claude, GPT-5.5, Grok-4.2, and Llama-4 families ($n = 7$)	Programmatic API	Chronologically ordered 4×3 contact-sheet images	Client-side uniform temporal sampling with $k = \min(N, 96)$; frames resized to 320×180 pixels; up to 12 frames per contact sheet	96/ T frames/s	Clinical: 96 frames (8 sheets); functional: 96 frames (8 sheets)	All 522 videos reached the 96-frame limit. Contact sheets were encoded as JPEG at quality 85 and submitted chronologically through the corresponding APIs.
MiniMax-M2.7 family ($n = 2$)	Programmatic API	Chronologically ordered 4×3 contact-sheet images	Client-side uniform temporal sampling with $k = \min(N, 240)$; frames resized to 320×180 pixels; up to 12 frames per contact sheet	30 frames/s when $N \leq 240$; otherwise 240/ T frames/s	Clinical: 150–184 frames (13–16 sheets); functional: 240 frames (20 sheets) for 193/194 clips and 150 frames (13 sheets) for 1/194 clip	All clinical source frames were retained. In the functional cohort, only the 5-s sit-up clip remained below the 240-frame limit. Contact sheets were encoded as JPEG at quality 85 and submitted through the corresponding API.

Notes: All 26 evaluated models were accessed through programmatic API endpoints during benchmark execution. No model was evaluated through a provider-operated web interface or through local/self-hosted inference. The terms “API-only/hosted” and “open-weight/local-deployable” used in the subgroup analysis describe model availability and deployment potential rather than the inference access channel used in this benchmark. Thus, models classified as open-weight/local-deployable were also evaluated through API endpoints to maintain a consistent programmatic inference workflow.

The model-family counts sum to the 26 MLLMs included in the benchmark. The clinical cohort comprised 328 videos with a mean duration of 5.005 s; all were encoded at 30 frames/s with a resolution of 640×960 pixels and contained 150–184 source frames. The functional cohort comprised 194 videos with a mean duration of 15.022 s, a median duration of 15.000 s, and a range of 5.000–134.800 s; all were encoded at 30 frames/s with a resolution of 1280×720 pixels.

For provider-managed native-video routes, processed-frame counts are expected values calculated from source-video duration and the provider-documented default sampling rate. These counts may be reduced when an internal maximum-frame limit is applied. NR denotes not reported, indicating that the provider did not expose the effective sampling rate or realized frame indices.

For client-side contact-sheet routes, N denotes the number of source frames, $K_{\max}$ denotes the model-family-specific maximum frame budget, $k = \min(N, K_{\max})$ denotes the realized number of selected frames, and T denotes source-video duration in seconds. The effective sampling rate was therefore duration-adaptive rather than fixed. Contact-sheet count was calculated as $\lceil k/12 \rceil$. All selected frames were sampled uniformly across the complete temporal axis, resized to 320×180 pixels, arranged chronologically into 4×3 contact sheets containing up to 12 frames each, and encoded as JPEG at quality 85. No question-dependent frame selection, key-frame optimization, retrospective manual adjustment, repeated-output selection, or manual choice among alternative model responses was performed.

Although all models were evaluated through programmatic APIs, some providers did not expose complete backend versioning, decoding settings, realized frame indices, or internal media-processing parameters. The reported results should therefore be interpreted as comparisons of the end-to-end API systems available during the evaluation period rather than as architecture-controlled comparisons under fully identical inference conditions.

Table C.15

Complete model leaderboard and statistical testing of 26 evaluated MLLMs on ActionEval-MedBench, with participant-level cluster-robust confidence intervals.

Rank	Model name	Weighted F1-score	Overall EM accuracy (%)	Valid response (%)	Participant-level 95% bootstrap CI	vs. reference <i>P</i> value	Reference model	<i>n</i> paired videos
1	gemini-3.1-pro-preview	66.17	61.62	100.00	[63.32, 69.15]	0.0003***	qwen3.6-27b	522
2	doubao-seed-2-0-lite-260428	66.15	60.70	100.00	[63.03, 69.25]	0.0005***	qwen3.6-27b	522
3	gemini-3.5-flash	65.74	62.48	100.00	[62.94, 68.77]	0.0005***	qwen3.6-27b	521
4	claude-opus-4-7	65.61	61.88	100.00	[62.91, 68.57]	0.0007***	qwen3.6-27b	522
5	glm-5v-turbo	65.60	62.84	100.00	[63.01, 68.18]	0.0015**	qwen3.6-27b	522
6	gpt-5.5	65.46	61.81	100.00	[63.01, 68.00]	0.0025**	qwen3.6-27b	522
7	kimi-k2.6	65.37	63.51	100.00	[62.90, 67.89]	0.0003***	qwen3.6-27b	522
8	doubao-seed-2-0-pro-260215	63.39	60.71	100.00	[60.87, 66.03]	0.0772 ^{ns}	qwen3.6-27b	520
9	gemini-3.1-flash-lite-preview	61.89	59.87	100.00	[59.16, 64.60]	0.3351 ^{ns}	qwen3.6-27b	522
10	doubao-seed-2-0-mini-260428	61.83	57.12	100.00	[59.07, 64.64]	0.5264 ^{ns}	qwen3.6-27b	522
11	qwen3.6-plus	61.65	58.43	100.00	[59.39, 64.06]	0.1792 ^{ns}	qwen3.6-27b	522
12	qwen3.6-35b-a3b	61.11	56.86	100.00	[59.02, 63.17]	0.4408 ^{ns}	qwen3.6-27b	522
13	qwen3.6-27b	60.80	56.29	100.00	[59.01, 62.61]	1.0000 ^{ns}	qwen3.6-27b	522
14	qwen3.6-flash	60.60	55.78	100.00	[58.76, 62.41]	0.3576 ^{ns}	qwen3.6-27b	522
15	grok-4.2	58.57	52.52	100.00	[56.52, 60.57]	0.0247*	qwen3.6-27b	522
16	step-3	57.92	51.60	99.97	[55.28, 60.41]	0.0020**	qwen3.6-27b	522
17	meta-llama/llama-4-scout	57.91	47.29	100.00	[56.22, 59.56]	0.0052**	qwen3.6-27b	522
18	intern-s1-pro	57.41	52.95	100.00	[54.98, 59.89]	0.0051**	qwen3.6-27b	519
19	MiniMax-M2.7-highspeed	56.64	49.90	100.00	[54.94, 58.46]	0.0046**	qwen3.6-27b	478
20	claude-opus-4-6	55.07	53.70	100.00	[53.14, 57.29]	< 0.0001***	qwen3.6-27b	522
21	internv13.5-241b-a28b	54.12	49.97	100.00	[52.37, 55.92]	< 0.0001***	qwen3.6-27b	522
22	claude-sonnet-4-6	53.61	49.74	100.00	[51.69, 55.42]	< 0.0001***	qwen3.6-27b	522
23	MiniMax-M2.7	49.67	42.75	100.00	[48.03, 51.20]	< 0.0001***	qwen3.6-27b	501
24	intern-s1-mini	48.46	35.86	100.00	[46.78, 50.13]	< 0.0001***	qwen3.6-27b	522
25	step-1o-turbo-vision	46.56	39.05	100.00	[44.25, 48.98]	< 0.0001***	qwen3.6-27b	522
26	meta-llama/llama-4-maverick	46.20	43.23	97.64	[44.06, 48.46]	< 0.0001***	qwen3.6-27b	522

Notes: The 95% CIs were estimated using 2,000 participant-level cluster-robust bootstrap resamples. Participants were sampled with replacement separately within the clinical and functional cohorts, and all recordings contributed by the same participant, including left- and right-limb recordings, were retained together within each bootstrap replicate. The 522 recordings were contributed by 48 participants, including 28 clinical and 20 functional participants, with a median of 10 recordings per participant (range, 8–30). Cohort-specific estimates were recalculated within each replicate and combined using the prespecified 65% clinical and 35% functional cohort weights. The participant-level bootstrap was used only to estimate confidence intervals and did not alter the model-level point estimates or hypothesis-test results. The reference model for the paired Wilcoxon signed-rank tests on matched video–MCQ item-level F1-scores was qwen3.6-27b.

“Valid response (%)” denotes the proportion of parsable dimension-level responses among the outputs successfully returned by each model and therefore reflects response-format validity rather than complete benchmark coverage. The column “*n* paired videos” denotes the number of videos for which matched video-level scores were available for both the evaluated model and the reference model. Consequently, a model may achieve a 100% valid-response rate among its returned outputs while having fewer than 522 paired videos because some requests did not yield model outputs available for paired testing. Conversely, a model may retain 522 paired videos while having a small number of invalid dimension-level responses when a video-level aggregate score remained available. The overall response yield reported in the main analysis (81,006 of 81,432 dimension-level response opportunities) was calculated against the complete expected response set and therefore additionally accounted for missing or system-level failure responses.

P-value stars indicate ****P* < 0.001, ***P* < 0.01, **P* < 0.05, and ^{ns}, *P* ≥ 0.05. Weighted F1-score denotes the cohort-weighted ActionEval composite score. EM, exact match; CI, confidence interval.

Table C.16
Summary of robustness and sensitivity analyses.

Sensitivity analysis	Operational change	n_{models}	Spearman ρ	Top-6 overlap	Max rank shift
Alternative weighting	Equal weighting, low-D6 weighting, and D6-excluded weighting	26	0.964	6/6	5.0
D6-excluded scoring	Removed D6 and re-normalized D1–D5	26	0.999	6/6	1.0
Valid-only response scoring	Excluded the 75 returned invalid or unparsable responses rather than assigning zero	26	0.999	6/6	1.0
Complete-denominator scoring	Retained all 81,432 expected response opportunities and assigned zero to both the 75 returned invalid responses and the 426 unavailable responses	26	0.995	6/6	3.0
Video-level aggregated analysis	Aggregated the six dimension-specific item scores within each video using the prespecified D1–D6 weights, followed by the prespecified 65% clinical and 35% functional cohort weighting	26	1.000	6/6	0.0
Complete-case intersection	Restricted analysis to paired complete-case item intersections	26	0.985	5/6	4.0
Clinical cohort only	Restricted evaluation to the neuromotor rehabilitation domain	26	0.932	6/6	7.0

Notes: Spearman's ρ was calculated between the model ranking under each sensitivity analysis and the primary model ranking. Top-6 overlap indicates the number of models shared with the primary top-performing six-model set. Maximum rank shift denotes the largest absolute change in rank position across all 26 evaluated models.

The primary benchmark scoring used the 81,006 successfully returned responses: 80,931 parsable valid responses were scored according to their predicted option sets, whereas 75 returned invalid or unparsable responses were assigned zero. The 426 response opportunities unavailable because of missing outputs or system-level failures were excluded from model-specific primary point estimates.

The valid-only sensitivity analysis excluded the 75 returned invalid or unparsable responses rather than assigning zero. The complete-denominator sensitivity analysis retained all 81,432 expected response opportunities and assigned zero to both returned invalid responses and unavailable response opportunities. Under complete-denominator scoring, the Top-6 model set remained unchanged; the maximum three-position displacement was observed for MiniMax-M2.7-Highspeed, which shifted from rank 19 to rank 22.

In the video-level aggregated analysis, the six ActionEval item scores associated with each video were first combined using the prespecified dimension weights of 10%, 10%, 30%, 10%, 25%, and 15% for D1–D6, respectively. The resulting video-level scores were averaged separately within the clinical and functional cohorts, after which the cohort-specific estimates were combined using the prespecified 65% clinical and 35% functional weights. All 13,501 observed model–video records contained complete six-dimension score sets after application of the primary response-handling rule. Consequently, video-level aggregation was algebraically equivalent to the primary cohort-specific item-pooled calculation up to numerical precision (maximum absolute difference $< 10^{-14}$). The resulting model ranking was identical to the primary ranking (Spearman's $\rho = 1.000$; Top-6 overlap, 6/6; maximum rank shift, 0), indicating that the principal rank-order conclusions were not driven by treating the six dimension-specific items from the same video as separate ranking units. This sensitivity analysis evaluated ranking robustness and did not replace the participant-level cluster-robust bootstrap used for confidence-interval estimation.

Table C.17
Sensitivity of model rankings to alternative dimension-weighting schemes.

Weighting scheme	D1–D6 weights (%)	Spearman's ρ	Top-5 overlap	Maximum rank change	Interpretation
Primary	10.00/10.00/30.00/10.00/25.00/15.00	Reference	5/5	0	Prespecified primary weighting scheme.
Equal weighting	16.67/16.67/16.67/16.67/16.67/16.67	0.964	4/5	5	The ranking remained strongly correlated with the primary ranking, although limited changes occurred among individual models.
Lower-D6 weighting	11.18/11.18/33.53/11.18/27.94/5.00	0.999	5/5	1	The leading model set and overall rank order were essentially unchanged.
D6 excluded	11.76/11.76/35.29/11.76/29.41/0.00	0.999	5/5	1	Excluding D6 had negligible influence on the leading model set and overall rank order.

Notes: Spearman's ρ was calculated between each alternative ranking and the ranking obtained using the prespecified primary weighting scheme. Top-5 overlap indicates the number of models shared with the primary top-five set. Maximum rank change denotes the largest absolute change in rank position across all evaluated models. For the lower-D6 and D6-excluded schemes, the remaining weight was proportionally redistributed across D1–D5 while preserving their relative ratios. Weight totals may differ marginally from 100% because of rounding.

Table C.18
Model-level post hoc D3 performance stratified by adjudicated gold-label status.

Model	All D3 F1 (%)	Determinate gold		Internally unresolved		Positive- compensation F1 (%)	Confirmed- absence F1 (%)
		Valid <i>n</i>	F1 (%)	Valid <i>n</i>	F1 (%)		
Gemini-3.1-Pro-Preview	57.7	429	54.3	93	73.1	23.1	90.5
Doubao-Seed-2.0-Lite-260428	55.4	429	46.5	93	96.8	9.3	89.4
Gemini-3.5-Flash	55.6	429	47.4	92	93.5	15.0	84.9
Claude-Opus-4-7	56.5	429	53.6	93	69.9	34.8	75.4
GLM-5V-Turbo	54.9	429	47.9	93	87.1	19.0	81.4
GPT-5.5	56.3	429	50.5	93	82.8	25.1	79.9
Kimi-K2.6	58.3	429	52.5	93	84.9	21.0	88.9
Doubao-Seed-2.0-Pro-260215	54.5	427	45.0	93	97.8	11.1	84.3
Gemini-3.1-Flash-Lite-Preview	50.0	429	49.2	93	53.8	34.3	66.3
Doubao-Seed-2.0-Mini-260428	49.1	429	40.4	93	89.2	20.1	63.8
Qwen3.6-Plus	47.4	429	49.8	93	36.6	48.0	51.8
Qwen3.6-35B-A3B	49.3	429	47.4	93	58.1	33.7	63.3
Qwen3.6-27B	43.9	429	45.0	93	38.7	43.1	47.2
Qwen3.6-Flash	47.1	429	45.9	93	52.7	34.7	58.8
Grok-4.2	45.4	429	43.8	93	52.7	40.8	47.2
Step-3	50.3	428	45.8	93	71.0	21.5	73.9
Meta-Llama/Llama-4-Scout	45.7	429	42.3	93	61.3	36.4	49.2
Intern-S1-Pro	44.2	427	37.1	92	77.2	17.2	60.1
MiniMax-M2.7-Highspeed	44.9	400	45.6	78	41.0	40.8	51.1
Claude-Opus-4-6	31.9	429	36.7	93	9.7	58.5	11.6
InternVL3.5-241B-A28B	33.7	429	31.0	93	46.2	39.1	21.6
Claude-Sonnet-4-6	28.3	429	32.1	93	10.8	56.7	3.5
MiniMax-M2.7	30.5	415	33.2	86	17.4	43.3	21.6
Intern-S1-Mini	31.7	429	31.3	93	33.3	37.5	24.1
Step-1o-Turbo-Vision	31.1	429	36.4	93	6.5	65.8	2.5
Meta-Llama/Llama-4-Maverick	21.6	417	26.2	91	0.0	48.0	1.0
Cross-model mean	45.2	–	43.0	–	55.5	33.8	53.6

Notes: All F1 values reported in this table follow the primary returned-response scoring rule. Parsable valid returned responses were scored according to their predicted option sets, returned invalid or unparsable responses were assigned an item-level score of zero, and response opportunities unavailable because of missing outputs or system-level failures were excluded from model-specific estimates.

The 522 D3 items were stratified using the archived adjudicated gold set and its accompanying rationale. The determinate, rationale-supported stratum contained 429 items, comprising 230 items with at least one expert-labelled pathological-compensation option and 199 items with a rationale-confirmed absence. The internally unresolved stratum contained 93 items. These strata were derived from `data_package/d3_item_stratum.csv` using the variables `two_group` and `gold_rationale_class`.

“Determinate” does not denote six-rater unanimity and should not be equated with the 404 D3 items with exact pre-arbitration inter-group agreement or the 118 arbitrated items reported in Table 4. Item-level pre-arbitration Group A and Group B consensus labels were not retained; therefore, the original 404/118 agreement/arbitration split was not reconstructed.

“Parsable valid *n*” denotes the number of parsable valid model responses observed within the corresponding stratum and is reported as a response-coverage diagnostic. These counts should not be interpreted as the complete denominator of the primary F1 estimate because returned invalid or unparsable responses were retained in primary scoring and assigned a score of zero, whereas unavailable response opportunities were excluded. Values below the nominal stratum sizes of 429 determinate items and 93 internally unresolved items may therefore reflect returned invalid or unparsable responses, unavailable response opportunities, or both; they do not indicate changes in the item definitions.

For models with complete response coverage, the positive-compensation and confirmed-absence subsets contained 230 and 199 items, respectively. The positive-compensation and confirmed-absence columns report subset-specific F1-scores calculated using the same primary returned-response scoring rule. Their model-specific parsable-valid counts are not displayed to preserve table readability.

The cross-model mean is the unweighted arithmetic mean of the 26 model-level primary-scoring estimates. Dashes in the cross-model mean row indicate that model-specific parsable-valid response counts were not averaged. The determinate and internally unresolved F1 estimates differed in a two-sided paired Wilcoxon signed-rank test across the 26 model-level pairs ($P = 0.029$). This post hoc diagnostic analysis did not affect the primary leaderboard, composite scores, or model rankings.

In contrast, Supplementary Table C.19 reports the separate valid-response-only decomposition of predicted-set size, precision, recall, set-F1, and empty-prediction frequency. Its Panel B is restricted to positive-ground-truth D3 items.

Table C.19

Valid-response-only decomposition of item-level set-F1 by model and ActionEval dimension, with a D3 positive-ground-truth diagnostic.

Dim.	Model	Valid n	Mean P	Mean G	Precision (%)	Recall (%)	Set-F1 (%)	Empty prediction (%)
Panel A. Valid-response-only decomposition by model and ActionEval dimension								
<i>D1. Semantic action recognition</i>								
D1	Gemini-3.1-Pro-Preview	522	2.852	2.948	96.63	93.20	94.25	0.00
D1	Doubao-Seed-2.0-Lite-260428	522	3.017	2.948	93.36	95.00	93.49	0.57
D1	Gemini-3.5-Flash	521	2.929	2.948	95.97	95.31	95.14	0.38
D1	Claude-Opus-4-7	522	2.933	2.948	96.55	95.71	95.57	0.00
D1	GLM-5V-Turbo	522	2.906	2.948	96.23	94.60	94.81	0.19
D1	GPT-5.5	522	2.918	2.948	96.10	95.10	95.09	0.38
D1	Kimi-K2.6	522	2.770	2.948	93.36	90.44	91.21	3.45
D1	Doubao-Seed-2.0-Pro-260215	520	2.794	2.946	91.09	90.82	90.50	5.19
D1	Gemini-3.1-Flash-Lite-Preview	522	3.038	2.948	94.73	96.90	95.19	0.00
D1	Doubao-Seed-2.0-Mini-260428	522	2.877	2.948	95.32	92.88	93.22	0.19
D1	Qwen3.6-Plus	522	2.893	2.948	93.95	93.06	92.76	0.96
D1	Qwen3.6-35B-A3B	522	2.830	2.948	95.29	91.19	92.05	0.19
D1	Qwen3.6-27B	522	3.033	2.948	93.17	95.15	93.39	0.38
D1	Qwen3.6-Flash	522	2.837	2.948	94.03	90.33	91.11	0.77
D1	Grok-4.2	522	2.837	2.948	94.60	90.33	91.13	0.19
D1	Step-3	522	2.613	2.948	93.44	85.04	87.56	2.87
D1	Meta-Llama/Llama-4-Scout	522	2.464	2.948	97.62	80.80	86.69	0.00
D1	Intern-S1-Pro	519	2.362	2.948	84.23	76.16	78.88	12.52
D1	MiniMax-M2.7-Highspeed	478	2.826	2.962	91.75	90.25	90.22	3.56
D1	Claude-Opus-4-6	522	2.914	2.948	96.39	94.83	95.09	0.00
D1	InternVL3.5-241B-A28B	522	2.883	2.948	94.70	91.86	92.35	0.00
D1	Claude-Sonnet-4-6	522	3.017	2.948	95.08	96.71	95.31	0.00
D1	MiniMax-M2.7	501	2.834	2.952	90.97	89.17	89.01	3.19
D1	Intern-S1-Mini	522	2.228	2.948	94.11	71.04	79.18	0.00
D1	Step-1o-Turbo-Vision	522	2.278	2.948	97.35	74.79	82.08	0.57
D1	Meta-Llama/Llama-4-Maverick	508	3.002	2.949	95.28	96.23	95.12	0.00
D1	Cross-model mean	–	2.803	2.949	94.28	90.27	91.17	1.37
<i>D2. Kinematic feature alignment</i>								
D2	Gemini-3.1-Pro-Preview	522	2.153	1.789	64.37	73.61	66.18	1.53
D2	Doubao-Seed-2.0-Lite-260428	522	2.192	1.789	65.74	74.23	67.01	1.92
D2	Gemini-3.5-Flash	521	2.144	1.789	67.29	78.31	70.17	2.11
D2	Claude-Opus-4-7	522	1.816	1.789	67.86	67.85	65.64	1.92
D2	GLM-5V-Turbo	522	1.628	1.789	66.76	65.34	64.18	10.73
D2	GPT-5.5	522	1.611	1.789	69.97	65.96	65.81	9.39
D2	Kimi-K2.6	522	1.718	1.789	64.73	66.60	63.54	13.41
D2	Doubao-Seed-2.0-Pro-260215	520	1.838	1.785	61.59	66.88	61.90	16.35
D2	Gemini-3.1-Flash-Lite-Preview	522	1.596	1.789	62.58	56.48	57.97	0.00
D2	Doubao-Seed-2.0-Mini-260428	522	1.768	1.789	57.41	57.15	54.62	4.98
D2	Qwen3.6-Plus	522	1.784	1.789	69.96	69.48	67.60	2.87
D2	Qwen3.6-35B-A3B	522	1.877	1.789	62.08	64.24	61.25	3.64
D2	Qwen3.6-27B	522	1.998	1.789	64.34	72.11	65.65	3.26
D2	Qwen3.6-Flash	522	1.864	1.789	62.69	65.66	61.95	4.21
D2	Grok-4.2	522	1.948	1.789	61.70	64.72	60.80	0.96
D2	Step-3	522	1.301	1.789	54.93	46.89	47.94	17.43
D2	Meta-Llama/Llama-4-Scout	522	1.966	1.789	61.13	62.21	59.15	0.00
D2	Intern-S1-Pro	519	1.551	1.794	54.30	54.08	52.21	21.00
D2	MiniMax-M2.7-Highspeed	478	2.121	1.824	57.86	67.71	60.09	5.44
D2	Claude-Opus-4-6	522	1.898	1.789	66.83	67.59	65.33	0.00
D2	InternVL3.5-241B-A28B	522	1.864	1.789	61.94	61.77	59.50	0.96
D2	Claude-Sonnet-4-6	522	1.979	1.789	63.35	65.49	61.62	0.00
D2	MiniMax-M2.7	501	2.064	1.800	53.64	61.14	55.00	5.59
D2	Intern-S1-Mini	522	1.649	1.789	55.00	49.62	49.44	0.38
D2	Step-1o-Turbo-Vision	522	1.182	1.789	53.67	41.94	45.71	12.07
D2	Meta-Llama/Llama-4-Maverick	508	2.301	1.754	59.35	70.96	62.62	0.00
D2	Cross-model mean	–	1.839	1.789	61.96	63.77	60.50	5.39
<i>D3. Pathological compensation identification</i>								

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Supplementary Table C.19 continued

Dim.	Model	Valid n	Mean P	Mean G	Precision (%)	Recall (%)	Set-F1 (%)	Empty prediction (%)
D3	Gemini-3.1-Pro-Preview	522	0.324	0.556	57.82	58.52	57.69	77.01
D3	Doubao-Seed-2.0-Lite-260428	522	0.130	0.556	55.36	55.84	55.43	89.27
D3	Gemini-3.5-Flash	521	0.207	0.557	55.85	55.76	55.56	82.73
D3	Claude-Opus-4-7	522	0.483	0.556	56.70	57.60	56.50	63.98
D3	GLM-5V-Turbo	522	0.259	0.556	55.36	54.82	54.91	76.82
D3	GPT-5.5	522	0.375	0.556	55.91	57.82	56.29	73.75
D3	Kimi-K2.6	522	0.255	0.556	58.43	58.88	58.30	79.12
D3	Doubao-Seed-2.0-Pro-260215	520	0.144	0.556	55.00	54.23	54.49	85.58
D3	Gemini-3.1-Flash-Lite-Preview	522	0.477	0.556	51.92	49.11	50.00	53.83
D3	Doubao-Seed-2.0-Mini-260428	522	0.310	0.556	49.81	49.01	49.11	71.26
D3	Qwen3.6-Plus	522	0.858	0.556	47.13	50.06	47.41	38.89
D3	Qwen3.6-35B-A3B	522	0.565	0.556	49.65	50.06	49.32	53.64
D3	Qwen3.6-27B	522	0.870	0.556	44.16	45.82	43.91	38.70
D3	Qwen3.6-Flash	522	0.596	0.556	47.19	48.15	47.09	51.92
D3	Grok-4.2	522	0.674	0.556	46.74	45.59	45.36	43.68
D3	Step-3	521	0.340	0.557	50.77	50.67	50.38	70.06
D3	Meta-Llama/Llama-4-Scout	522	0.638	0.556	47.06	45.56	45.72	50.38
D3	Intern-S1-Pro	519	0.355	0.557	44.99	43.99	44.22	68.21
D3	MiniMax-M2.7-Highspeed	478	1.289	0.561	42.28	51.50	44.86	41.21
D3	Claude-Opus-4-6	522	1.103	0.556	33.72	32.28	31.92	9.20
D3	InternVL3.5-241B-A28B	522	0.807	0.556	35.06	33.81	33.68	31.23
D3	Claude-Sonnet-4-6	522	1.297	0.556	28.29	31.26	28.26	4.41
D3	MiniMax-M2.7	501	1.685	0.553	28.11	36.76	30.50	20.76
D3	Intern-S1-Mini	522	0.948	0.556	32.85	32.25	31.66	28.16
D3	Step-1o-Turbo-Vision	522	1.054	0.556	34.00	30.20	31.09	3.45
D3	Meta-Llama/Llama-4-Maverick	508	1.530	0.553	21.59	26.51	22.15	1.18
D3	Cross-model mean	–	0.676	0.556	45.61	46.39	45.22	50.32

D4. Spatiotemporal dynamics analysis

D4	Gemini-3.1-Pro-Preview	522	1.192	0.967	58.33	62.02	58.80	6.13
D4	Doubao-Seed-2.0-Lite-260428	522	1.310	0.967	56.03	61.05	56.56	2.68
D4	Gemini-3.5-Flash	521	1.219	0.967	57.77	62.35	58.46	6.53
D4	Claude-Opus-4-7	522	1.276	0.967	54.37	59.55	55.38	6.32
D4	GLM-5V-Turbo	522	0.808	0.967	58.05	60.36	57.94	38.89
D4	GPT-5.5	522	1.178	0.967	56.83	62.20	57.86	11.49
D4	Kimi-K2.6	522	0.845	0.967	65.52	64.16	64.17	26.63
D4	Doubao-Seed-2.0-Pro-260215	520	1.100	0.967	60.71	65.64	61.74	18.27
D4	Gemini-3.1-Flash-Lite-Preview	522	1.343	0.967	51.05	53.94	51.22	0.19
D4	Doubao-Seed-2.0-Mini-260428	522	0.925	0.967	58.14	55.68	55.72	17.24
D4	Qwen3.6-Plus	522	1.025	0.967	56.07	56.88	55.09	18.39
D4	Qwen3.6-35B-A3B	522	1.090	0.967	53.35	55.11	52.98	13.98
D4	Qwen3.6-27B	522	1.128	0.967	53.90	56.90	53.86	13.41
D4	Qwen3.6-Flash	522	1.148	0.967	55.46	58.01	55.34	12.07
D4	Grok-4.2	522	1.341	0.967	45.37	47.29	45.04	2.68
D4	Step-3	522	0.734	0.967	46.93	46.07	45.68	42.34
D4	Meta-Llama/Llama-4-Scout	522	1.255	0.967	35.44	38.14	34.99	13.98
D4	Intern-S1-Pro	519	0.917	0.971	58.57	59.28	57.85	26.78
D4	MiniMax-M2.7-Highspeed	478	1.502	0.990	43.38	50.87	45.01	9.00
D4	Claude-Opus-4-6	522	1.533	0.967	50.51	57.98	52.49	0.00
D4	InternVL3.5-241B-A28B	522	1.232	0.967	38.41	36.41	36.07	1.34
D4	Claude-Sonnet-4-6	522	1.525	0.967	46.14	52.75	47.64	0.77
D4	MiniMax-M2.7	501	1.705	0.972	32.67	40.59	34.38	7.58
D4	Intern-S1-Mini	522	1.251	0.967	33.76	35.36	32.83	4.79
D4	Step-1o-Turbo-Vision	522	1.167	0.967	25.29	22.45	22.78	3.26
D4	Meta-Llama/Llama-4-Maverick	508	1.892	0.955	27.43	34.76	29.15	0.00
D4	Cross-model mean	–	1.217	0.968	49.21	52.15	49.19	11.72

D5. Movement quality and clinical decision-making

D5	Gemini-3.1-Pro-Preview	522	0.998	1.008	49.43	48.85	49.04	0.19
D5	Doubao-Seed-2.0-Lite-260428	522	0.987	1.008	52.87	52.39	52.55	1.34
D5	Gemini-3.5-Flash	521	0.990	1.008	47.79	47.41	47.54	0.96
D5	Claude-Opus-4-7	522	1.000	1.008	49.43	48.95	49.11	0.00

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Supplementary Table C.19 continued

Dim.	Model	Valid n	Mean P	Mean G	Precision (%)	Recall (%)	Set-F1 (%)	Empty prediction (%)
D5	GLM-5V-Turbo	522	0.996	1.008	50.96	50.48	50.64	0.38
D5	GPT-5.5	522	0.998	1.008	47.70	47.22	47.38	0.19
D5	Kimi-K2.6	522	0.902	1.008	45.40	45.11	45.21	9.77
D5	Doubao-Seed-2.0-Pro-260215	520	0.873	1.008	43.85	43.56	43.65	12.69
D5	Gemini-3.1-Flash-Lite-Preview	522	1.000	1.008	47.70	47.22	47.38	0.00
D5	Doubao-Seed-2.0-Mini-260428	522	0.985	1.008	49.23	48.85	48.98	1.53
D5	Qwen3.6-Plus	522	0.989	1.008	44.64	44.16	44.32	1.15
D5	Qwen3.6-35B-A3B	522	0.994	1.008	44.06	43.77	43.87	0.77
D5	Qwen3.6-27B	522	0.996	1.008	46.55	45.98	46.17	0.38
D5	Qwen3.6-Flash	522	0.992	1.008	43.68	43.30	43.42	0.77
D5	Grok-4.2	522	0.998	1.008	42.53	42.15	42.27	0.19
D5	Step-3	522	0.923	1.008	40.80	40.42	40.55	7.66
D5	Meta-Llama/Llama-4-Scout	522	1.000	1.008	45.02	44.44	44.64	0.00
D5	Intern-S1-Pro	519	0.881	1.008	43.16	42.97	43.03	11.95
D5	MiniMax-M2.7-Highspeed	478	0.935	1.006	40.17	39.85	39.96	6.49
D5	Claude-Opus-4-6	522	1.000	1.008	38.12	37.84	37.93	0.00
D5	InternVL3.5-241B-A28B	522	1.000	1.008	43.68	43.39	43.49	0.00
D5	Claude-Sonnet-4-6	522	1.000	1.008	39.66	39.27	39.40	0.00
D5	MiniMax-M2.7	501	0.950	1.008	34.13	33.63	33.80	5.39
D5	Intern-S1-Mini	522	0.998	1.008	40.42	40.33	40.36	0.19
D5	Step-1o-Turbo-Vision	522	1.000	1.008	28.74	28.45	28.54	0.00
D5	Meta-Llama/Llama-4-Maverick	513	1.000	1.008	28.27	27.88	28.01	0.00
D5	Cross-model mean	–	0.976	1.008	43.38	43.00	43.12	2.38

D6. Visual grounding and anti-hallucination

D6	Gemini-3.1-Pro-Preview	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Doubao-Seed-2.0-Lite-260428	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Gemini-3.5-Flash	521	1.000	1.000	100.00	100.00	100.00	0.00
D6	Claude-Opus-4-7	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	GLM-5V-Turbo	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	GPT-5.5	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Kimi-K2.6	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Doubao-Seed-2.0-Pro-260215	520	1.000	1.000	100.00	100.00	100.00	0.00
D6	Gemini-3.1-Flash-Lite-Preview	522	1.002	1.000	99.81	99.81	99.81	0.00
D6	Doubao-Seed-2.0-Mini-260428	522	0.979	1.000	97.89	97.89	97.89	2.11
D6	Qwen3.6-Plus	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Qwen3.6-35B-A3B	522	1.000	1.000	99.81	99.81	99.81	0.00
D6	Qwen3.6-27B	522	1.000	1.000	99.81	99.81	99.81	0.00
D6	Qwen3.6-Flash	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Grok-4.2	522	1.000	1.000	98.28	98.28	98.28	0.00
D6	Step-3	522	0.990	1.000	99.04	99.04	99.04	0.96
D6	Meta-Llama/Llama-4-Scout	522	0.998	1.000	99.81	99.81	99.81	0.19
D6	Intern-S1-Pro	519	0.988	1.000	98.84	98.84	98.84	1.16
D6	MiniMax-M2.7-Highspeed	478	0.979	1.000	96.65	96.65	96.65	2.09
D6	Claude-Opus-4-6	522	0.992	1.000	98.66	98.66	98.66	1.15
D6	InternVL3.5-241B-A28B	522	0.971	1.000	97.13	97.13	97.13	2.87
D6	Claude-Sonnet-4-6	522	0.992	1.000	99.23	99.23	99.23	0.77
D6	MiniMax-M2.7	501	0.984	1.000	97.01	97.01	97.01	1.60
D6	Intern-S1-Mini	522	0.933	1.000	86.02	86.02	86.02	7.47
D6	Step-1o-Turbo-Vision	522	1.000	1.000	100.00	100.00	100.00	0.00
D6	Meta-Llama/Llama-4-Maverick	513	1.000	1.000	99.42	99.42	99.42	0.00
D6	Cross-model mean	–	0.993	1.000	98.75	98.75	98.75	0.78

Supplementary Table C.19 continued

Dim.	Model	Valid n	Mean $ P $	Mean $ G $	Precision (%)	Recall (%)	Set-F1 (%)	Empty prediction (%)
Panel B. D3 positive-ground-truth diagnostic ($G > 0$)								
D3+	Gemini-3.1-Pro-Preview	230	0.465	1.261	23.41	25.00	23.10	66.96
D3+	Doubao-Seed-2.0-Lite-260428	230	0.174	1.261	9.13	10.22	9.28	86.09
D3+	Gemini-3.5-Flash	230	0.287	1.261	15.65	15.43	14.99	76.52
D3+	Claude-Opus-4-7	230	0.657	1.261	35.22	37.25	34.75	51.74
D3+	GLM-5V-Turbo	230	0.352	1.261	20.00	18.77	18.97	68.70
D3+	GPT-5.5	230	0.526	1.261	24.28	28.62	25.14	64.78
D3+	Kimi-K2.6	230	0.391	1.261	21.30	22.32	21.01	68.26
D3+	Doubao-Seed-2.0-Pro-260215	229	0.183	1.262	12.23	10.48	11.06	81.66
D3+	Gemini-3.1-Flash-Lite-Preview	230	0.578	1.261	38.70	32.32	34.35	43.04
D3+	Doubao-Seed-2.0-Mini-260428	230	0.326	1.261	21.74	19.93	20.14	70.43
D3+	Qwen3.6-Plus	230	1.074	1.261	47.39	54.06	48.04	28.70
D3+	Qwen3.6-35B-A3B	230	0.700	1.261	34.42	35.36	33.67	43.48
D3+	Qwen3.6-27B	230	0.974	1.261	43.70	47.46	43.14	31.30
D3+	Qwen3.6-Flash	230	0.722	1.261	34.93	37.10	34.71	45.65
D3+	Grok-4.2	230	0.752	1.261	43.91	41.30	40.77	36.96
D3+	Step-3	230	0.400	1.261	22.39	22.17	21.52	66.09
D3+	Meta-Llama/Llama-4-Scout	230	0.622	1.261	39.42	36.01	36.36	46.96
D3+	Intern-S1-Pro	229	0.306	1.262	19.00	16.74	17.25	71.62
D3+	MiniMax-M2.7-Highspeed	212	1.443	1.264	34.94	55.74	40.77	32.55
D3+	Claude-Opus-4-6	230	1.109	1.261	62.61	59.35	58.52	6.96
D3+	InternVL3.5-241B-A28B	230	0.765	1.261	42.17	39.35	39.06	33.48
D3+	Claude-Sonnet-4-6	230	1.387	1.261	56.81	63.55	56.74	2.61
D3+	MiniMax-M2.7	221	1.597	1.253	37.93	57.54	43.35	21.27
D3+	Intern-S1-Mini	230	0.904	1.261	40.22	38.84	37.51	29.57
D3+	Step-1o-Turbo-Vision	230	1.035	1.261	72.39	63.77	65.78	3.04
D3+	Meta-Llama/Llama-4-Maverick	222	1.572	1.266	48.50	59.76	49.77	1.80
D3+	Cross-model mean	–	0.742	1.261	34.71	36.48	33.84	45.39

Notes: P denotes the model-predicted option set and G denotes the adjudicated expert ground-truth option set. Values were macro-averaged across parsable valid responses at the item level. When $P = G = \emptyset$, precision, recall, and set-F1 were defined as 1; when exactly one of the two sets was empty, all three metrics were defined as 0.

Panel A reports the valid-response-only decomposition across all six ActionEval dimensions. Panel B is restricted to the 230 D3 items with $|G| > 0$ and directly quantifies sensitivity to expert-labelled pathological-compensation options. “Empty prediction” denotes $|P| = 0$. Cross-model means are unweighted arithmetic means of the 26 model-level estimates.

Returned invalid or unparsable responses and response opportunities unavailable because of missing outputs or system-level failures were excluded from this table because predicted-set size, precision, recall, and empty-prediction frequency could not be meaningfully defined for unusable outputs. Consequently, the Set-F1 values reported in this table are valid-response-only diagnostic estimates and may differ slightly from the primary dimension-level scores, in which returned invalid or unparsable responses were assigned zero while unavailable responses were excluded.

The Panel B diagnostic was descriptive and did not affect the primary leaderboard or model rankings. Complete-denominator response-handling sensitivity results are reported separately in Supplementary Table C.16.

Appendix D. Thematic analysis of model failure modes

Appendix D.1. Summary of failure-mode categories

The deterministic rule-based retrieval module identified 1,232 candidate high-conflict items across the 26 evaluated MLLMs using the prespecified screening criteria described in Section 2.5. The automated module served only to retrieve and organize candidate error records; it did not generate qualitative codes, assign FM-1–FM-5 labels, or interpret the clinical mechanisms underlying the errors.

Because the same candidate item could produce errors across multiple models, the retrieved dataset contained 12,091 rule-triggered model–item error instances. The numbers of error instances associated with the five quantitative retrieval patterns were 3,459 for the perception–reasoning discrepancy pattern, 2,267 for the pathological-compensation error pattern, 63 for the language-prior or D6 false-positive visual-grounding pattern, 4,038 for the temporal-integration error pattern, and 2,264 for the kinetic-chain error pattern. These values represent automated retrieval-event counts rather than human-coded failure-mode frequencies or mutually exclusive sample counts.

Clinical and algorithmic reviewers subsequently examined the retrieved cases using a Grounded Theory-informed constant-comparison approach. Through iterative coding, cross-case comparison, and consensus consolidation, the reviewers identified five clinically interpretable thematic categories: perception–reasoning disconnection (FM-1), pathological compensation blind spots (FM-2), language priors overriding visual evidence (FM-3), temporal integration breakdown (FM-4), and kinetic-chain reasoning deficiency (FM-5). These categories were used to support qualitative interpretation and model-error analysis and did not affect the quantitative scores or model rankings.

Appendix D.2. Attribution table of failure-mode characteristics

Table D.21 summarizes the clinical manifestations, hypothesized cognitive mechanisms, potential clinical risks, and optimization targets of the five failure modes.

Appendix D.3. Representative failure-mode case analyses

Case 1: FM-1 Perception–reasoning disconnect

Sample information: Clinical / left arm touching the right shoulder followed by extension.

Observed failure: The model performed well in D1 semantic action recognition and accurately described the action process, including crossing the midline, elbow flexion, and touching the contralateral shoulder. However, it failed completely in D3 pathological compensation and D5 movement-quality assessment.

Clinical interpretation: This case illustrates a “knowing what but not knowing why” limitation. The model could complete low-level action classification but lacked deeper biomechanical reasoning about movement quality and pathological features. It directly equated movement completion with good movement quality and failed to identify compensatory cues in the video.

Case 2: FM-2 Pathological blindness

Sample information: Functional / seated dumbbell lateral raise.

Observed failure: For a sample expert-labelled as containing an abnormal pattern in D3, the model not only judged it as normal but also actively produced an affirmative description such as “good movement control” and “no compensation observed.”

Clinical interpretation: The model showed a strong normalization bias. This item had an extremely high error rate across models, indicating that this particular mechanical deviation pattern was a common blind spot among current MLLMs. Models tended to identify the semantic label of “lateral raise” and automatically ignore subtle shoulder or trunk compensatory behavior.

Case 3: FM-3 Language-prior hallucination

Sample information: Functional / push-up.

Observed failure: In an indoor push-up video performed on a hard floor without auxiliary equipment, the model hallucinated a blue yoga mat and even described its material in detail.

Clinical interpretation: This case represents a factual hallucination. The high-frequency association between “push-up” and “yoga mat” in pretraining data overwhelmed the actual visual evidence in the current video. This type of hallucination interferes with environmental safety assessment in D6 and exposes a

deeper limitation of current models: they may fail to suppress language priors even in simple real-world physical scenes.

The above qualitative findings support the main conclusion that current MLLMs face a substantial bottleneck when moving from low-level visual perception to higher-order clinical reasoning. Future research should focus on explicit feature-guided reasoning and negative-grounding strategies to mitigate these failure modes.

Table D.21
Qualitative thematic analysis of clinical reasoning failures in MLLMs.

Failure mode	Clinical manifestation	Hypothesized mechanism	Potential clinical risk	Optimization target
FM-1 Perception–reasoning disconnect	D1 semantic recognition is correct, but D3/D5 fails	Visual recognition is disconnected from biomechanical reasoning	Overestimation of functional quality and missed compensation	Construct explicit perception-to-reasoning chains
FM-2 Pathological blindness	High-conflict abnormal samples are judged as normal	Lack of visual-clinical mapping for pathological movement patterns	Pathological movement is misclassified as normal variation	Introduce pathology-specific visual anchors
FM-3 Language-prior hallucination	Visual evidence is overwritten by language priors	Language priors dominate over visual evidence	Interference with safety assessment and factual judgement	Strengthen negative probes and hallucination penalties
FM-4 Temporal integration breakdown	Temporal abnormalities are missed or denied	Short-frame or weak temporal attention is insufficient	Missed tremor, delay, and rhythm disturbance	Improve long-context modeling and temporal supervision
FM-5 Kinetic-chain reasoning deficiency	Normal judgement is given despite erroneous kinetic-chain patterns	Shallow keyword matching replaces cross-joint reasoning	Misleading conclusions under compensatory movement	Strengthen causal kinetic-chain template modeling